Open19 Brick Mechanical Specifications

Linux Foundation Sustainable Scalable Infrastructure Alliance

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Chapter 1. Change log

Author	Date	Version	Change Details
My Truong	November 2023	2.0	Public Spec release

Definitions

ASHRAE

American Society of Heating, Refrigerating and Air-Conditioning Engineers

ASHRAE W27

ASHRAE specification for water temperature range from 2 °C to 27 °C

OCP

Open Compute Project

SFP

SNIA SFF-8402 SFP+ 1X Pluggable Transceiver Solution

SNIA

Storage Networking Industry Association

QSFP

SNIA SFF-8679 QSFP+ 4X Pluggable Transceiver Solution

QSFP-DD

QSFP-DD MSA QSFP DOUBLE DENSITY 8X AND QSFP 4X PLUGGABLE TRANSCEIVERS

Chapter 2. Open19 Bricks

Four brick designs are the basis of Open19 V2. They are:

- SHHW: Single-High Half-Width, "Brick"
- DHHW: Double-High Half-Width
- SHDW: Single-High Double-Width, 1RU server equivalent
- DHDW: Double-High Double-Width, 2RU server equivalent

Chapter 3. Brick Mechanical Specification [SHHW]

3.1. SHHW Brick

The SHHW brick form-factor is a half-wide 1RU commonly used as the basis of 2U 4-node designs.

3.2. Brick Dimensions

The brick is designed with the overall dimensions shown.

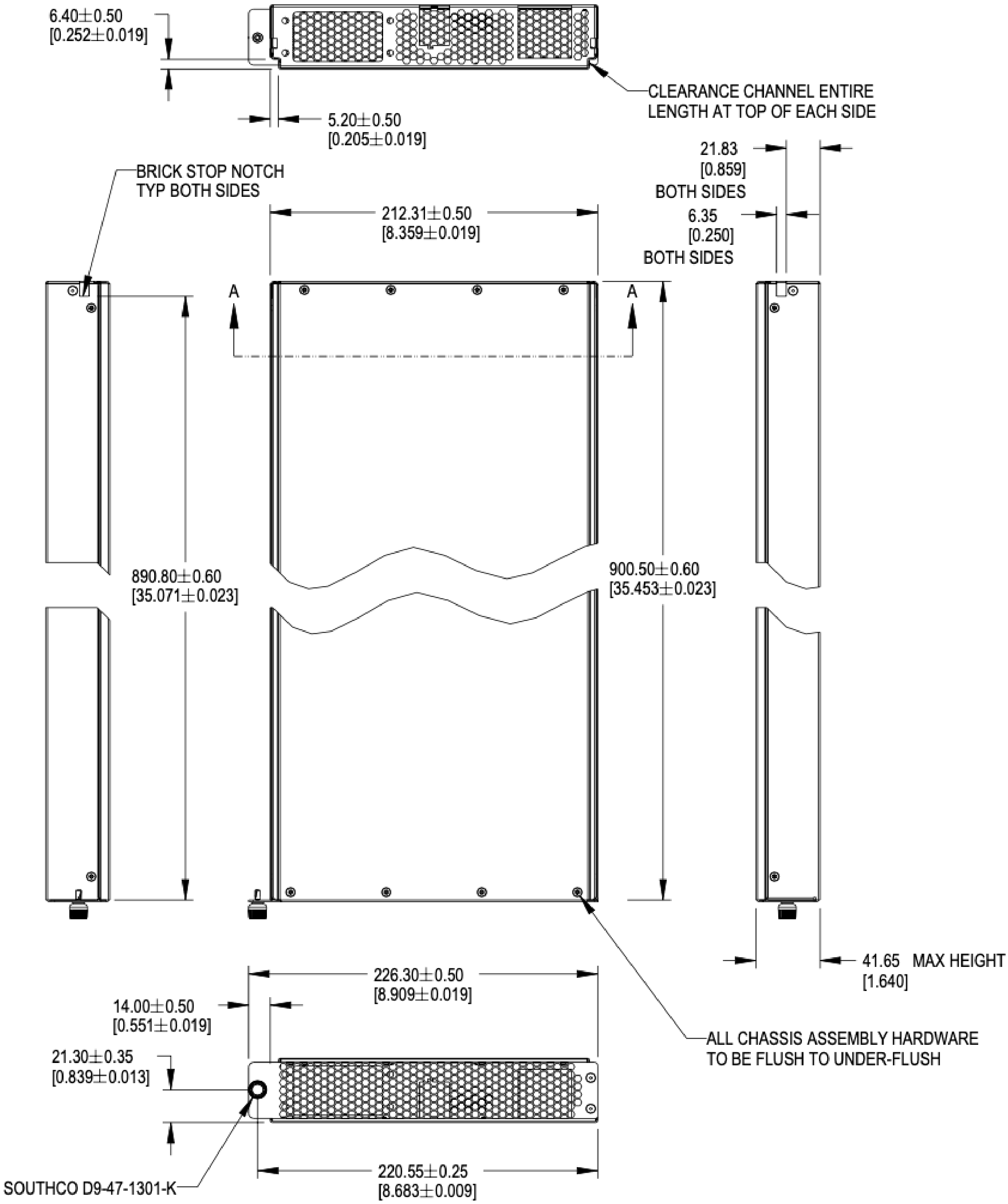


Figure 1. Brick dimensions

Please refer to the drawings, and 3D files for further details.

The V1 specification for bricks defined 901.7mm as the overall length of the brick, including the thickness of the front panel.

The dimension is modified to reference to the inside of the front panel to the rear of the brick. This dimension is 900.50mm +/-0.60 from the rear face of the front panel (mating surface to the cage) to the outside of the rear panel.

For half-width designs using the OCP DC-MHS M-DNO board specification, note the tolerance guidance in order to fit boards in a half-width chassis. This requires setting the nominal width at 209.55mm or smaller.

3.3. Mechanical features

3.3.1. Brick retention

The brick is secured into the cage with a single Southco D9-47-1302-K quarter-turn fastener, located on the left side of the faceplate. This is a change from V1. The change aligns the retention screw with high force areas of the optional liquid cooling interface.

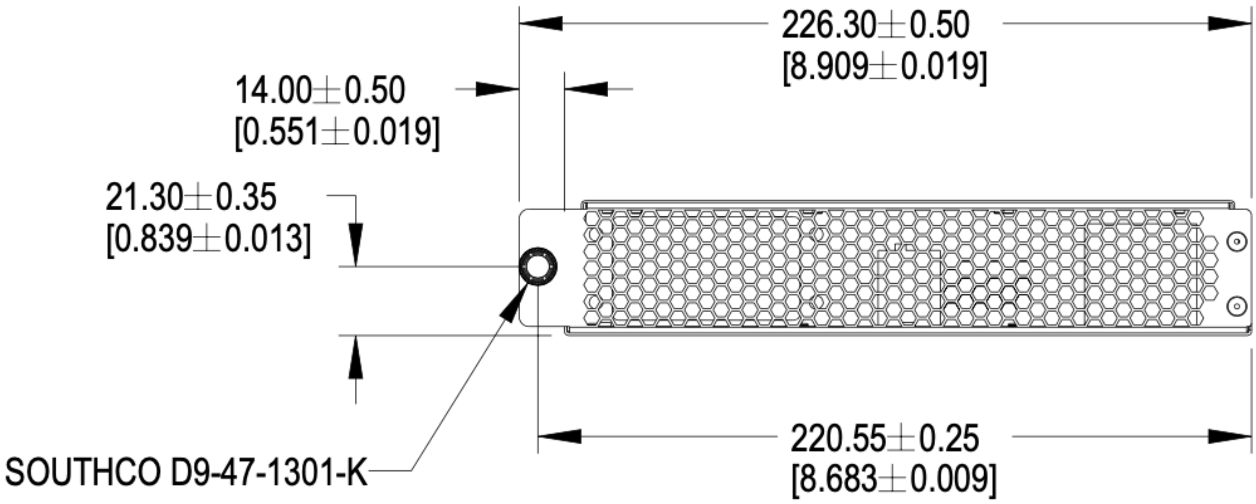


Figure 2. Brick retention

3.3.2. Brick mechanical guide

Clearance channels are required along the top edges of the brick, extending forward from the rear panel to the back of the faceplate.

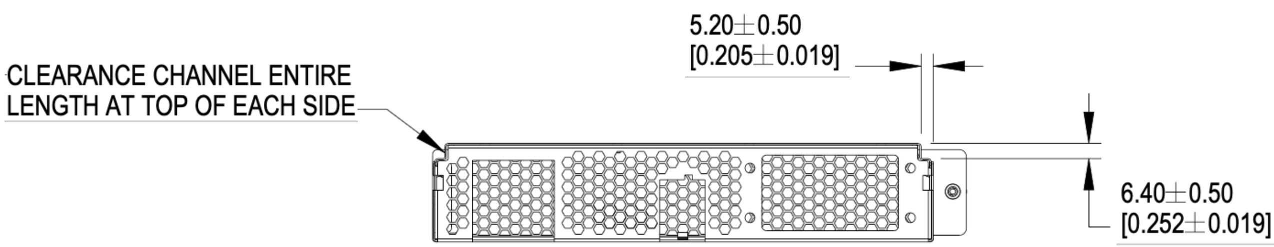


Figure 3. Mechanical guide at top edge of brick

Brick insertion and retention assistance for liquid cooling

For liquid cooling applications, two different mechanical assist for insertion and retention are available in the cage. Bricks can implement features to make use either/both features.

The features are:

- Brick retention standoff, available on both sides in the mechanical guide channel
- Slots, available on both sides of the cage wall

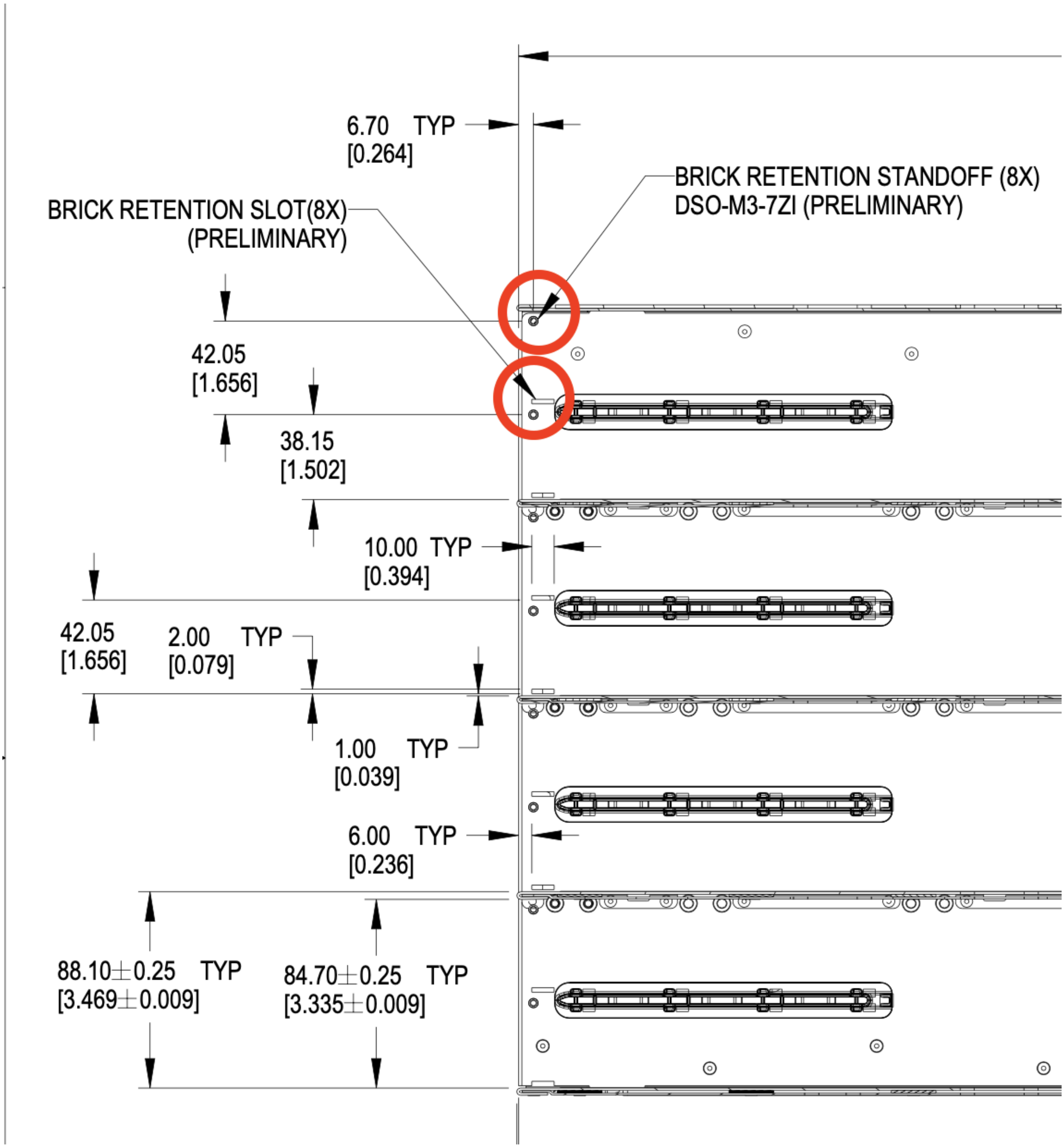


Figure 4. Brick mechanical assistance features in cage

Each feature shall support 100 lbf (445N) force or greater, using the recommended 1.3mm SGCC material.

3.3.3. Brick Mechanical Keying

Two keying notches are positioned in the left and right corners of the sheet metal enclosure as shown below. All dimensions below are critical.

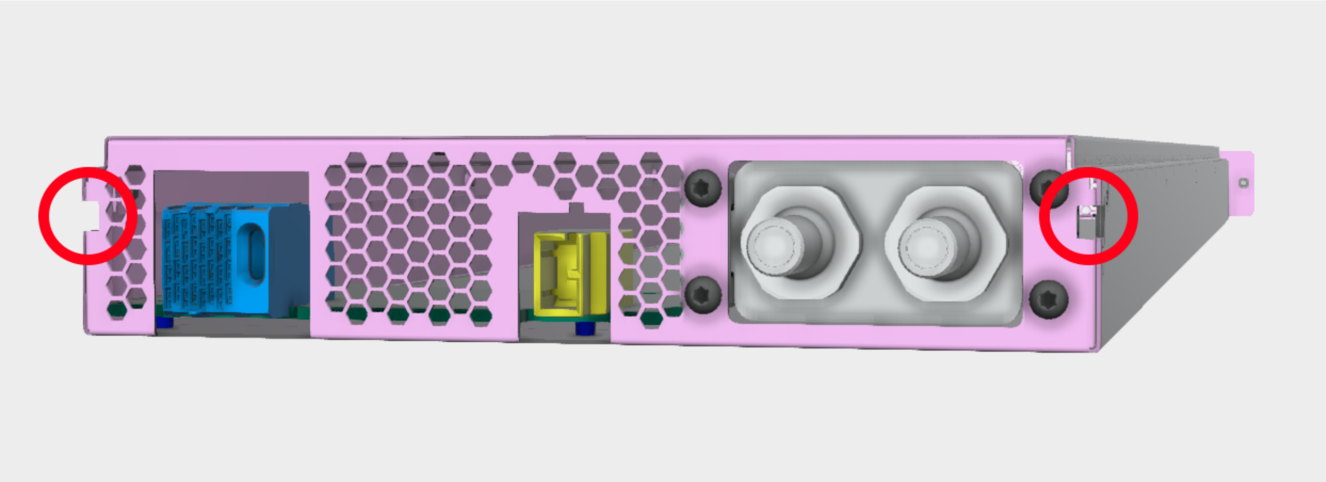


Figure 5. Brick keying

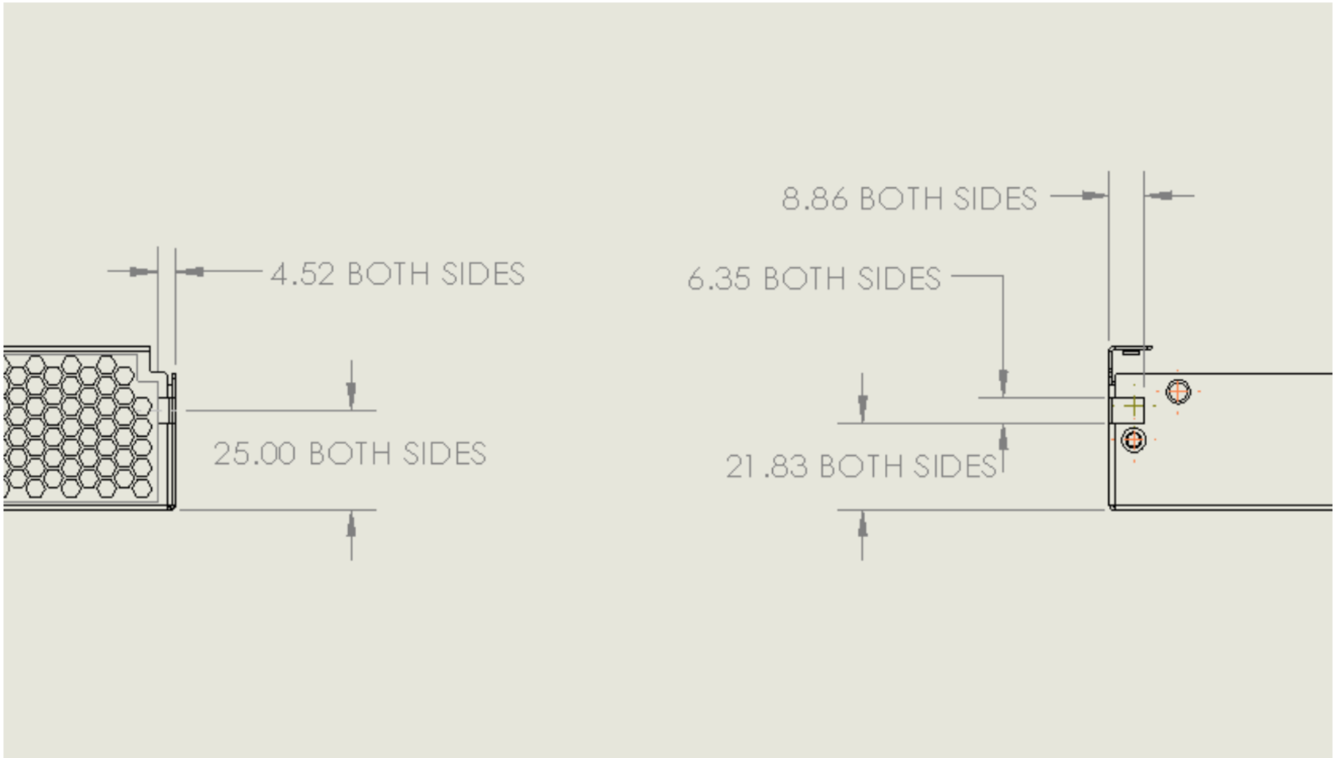


Figure 6. Brick keying dimensions

3.3.4. PCBA

The rear PCBA (Printed Circuit Board Assembly), also referenced as the interposer, provides the +48V DC input power and a data connections. The top of the PCBA is the reference surface, located 6.5mm +/- 0.6mm above the bottom surface of the sheet metal chassis.

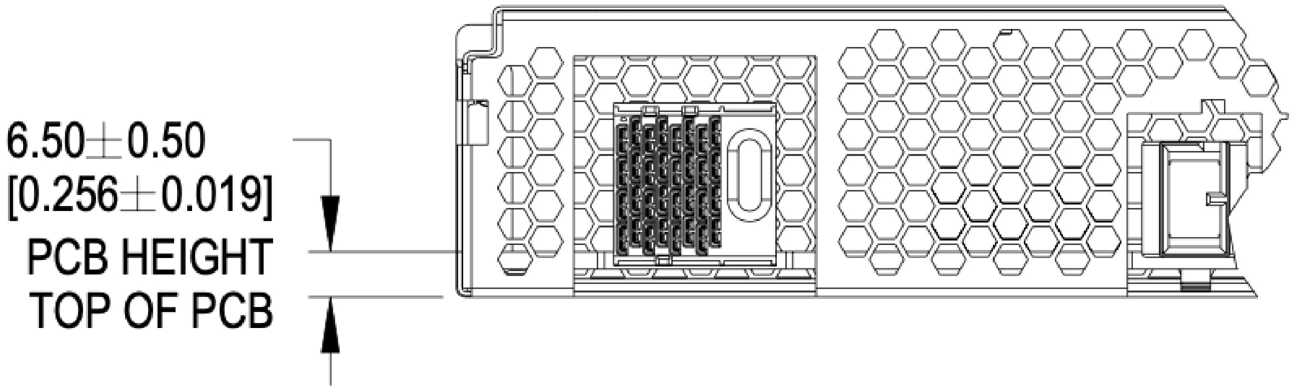


Figure 7. PCBA References

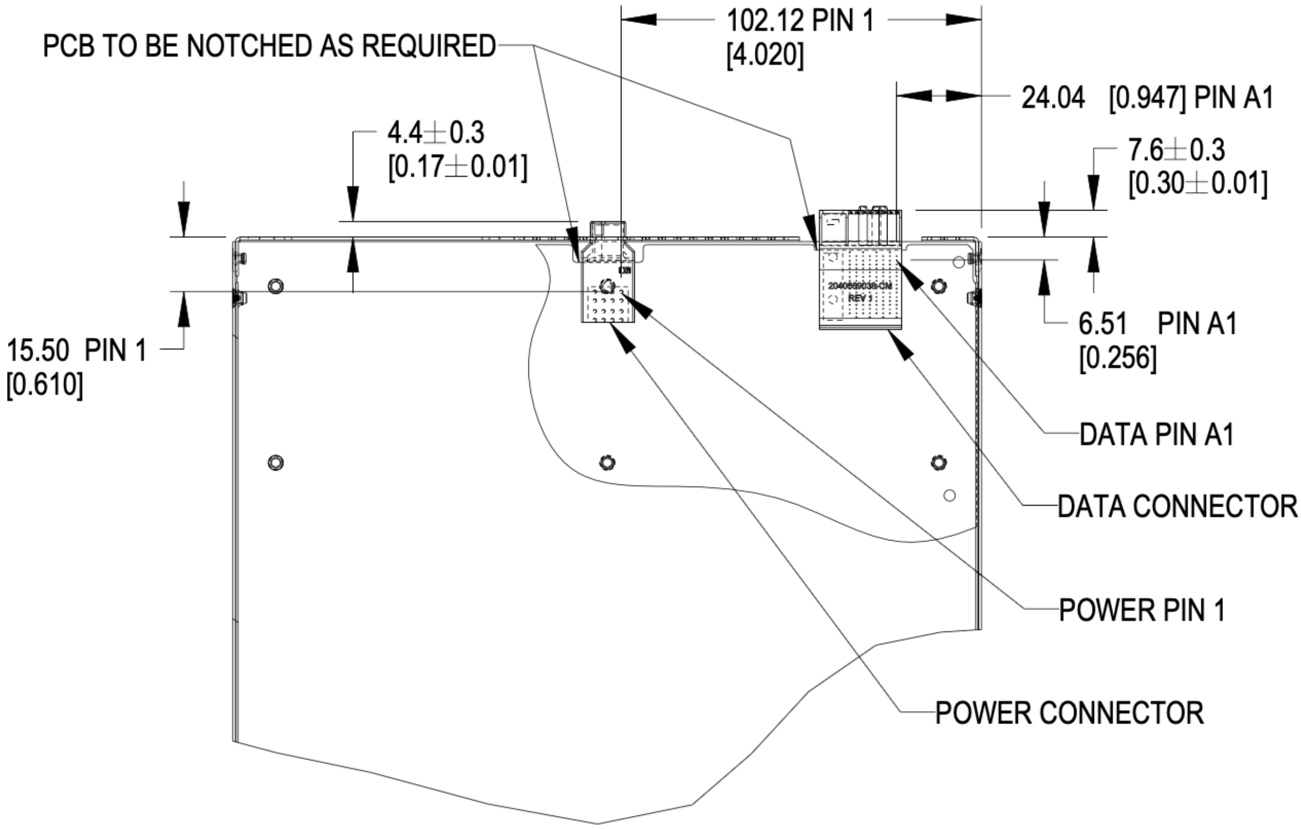


Figure 8. Brick PCB dimensions

3.3.5. Materials

The brick enclosure is constructed of sheet metal with adequate strength and rigidity. Suggested material is 0.8mm SGCC for the chassis base and 0.6mm SGCC for the top cover.

3.3.6. Blind mate interfaces

A brick is equipped with at least one set of blind mate connectors for receiving +48V DC power and transferring data with an external device, typically a network switch. Introduced in this version is are blind mate connections to facilitate liquid cooling. The connections mate with cable and manifold projections from the rear wall of the cage.

- Power connector: Molex Ten60 PN: 220370-0001 (35A) 22037-0000 (70A), refer to Molex document PSD-220370-0000
- Data connector: Molex 4x8 Impel with guide receptacle PN: 204066-9038 (PCB thickness 2.2-4.1mm), 204066-9138 (PCB thickness 1.0-1.9mm), refer to Molex document PSD-204066-9001

The power connector has two ratings:

- Power: Not-To-Exceed (NTE) 70A at +48V DC
- Power: Not-To-Exceed (NTE) 35A at +48V DC

The ratings shall be coordinated with the power shelf. 35A bricks shall be enabled in 70A capable infrastructure. 70A bricks are keyed to prevent unsafe operation in 35A infrastructure.

3.3.7. Weight

The maximum weight of the brick, when equipped with all drives and/or accessories, is 30lbs (13.6kg). This is an increase of 5lbs from V1.

3.4. Connector Pinout

3.4.1. Brick power connector

The brick blind-mate power connector for receiving 48V DC (51V nominal) power from the power shelf. The connector mates with the power cable connector projected from the brick cage wall.

Two variations of the brick connector exist, with different Not-to-exceed (NTE) ratings:

- 35A Power Connector: Molex 2203700001 (NTE 35 AMPS) No Key
- 70A Power Connector: Molex 2203700000 (NTE 70 AMPS) Keyed

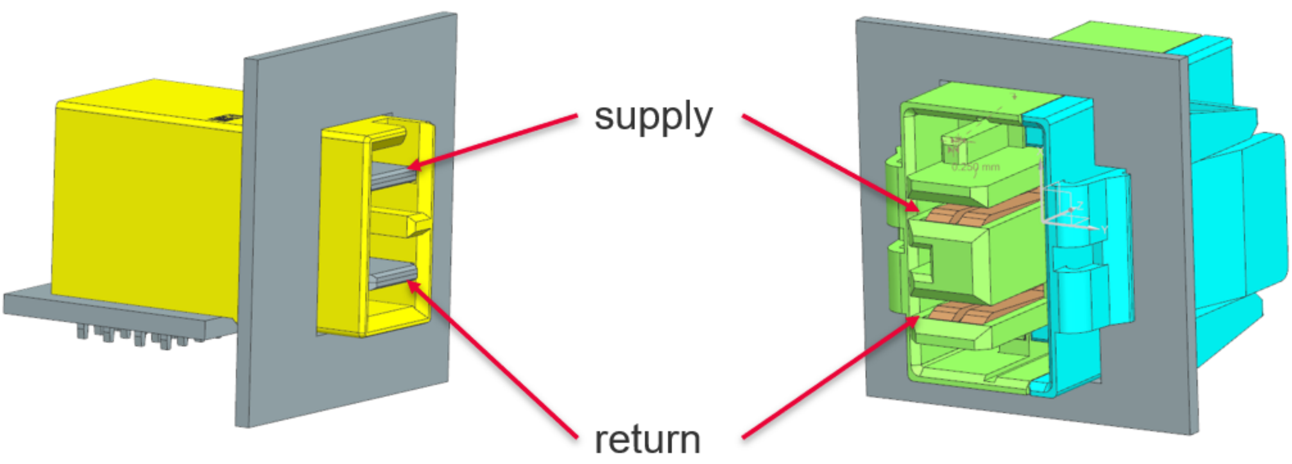


Figure 9. Power connector supply/return definition

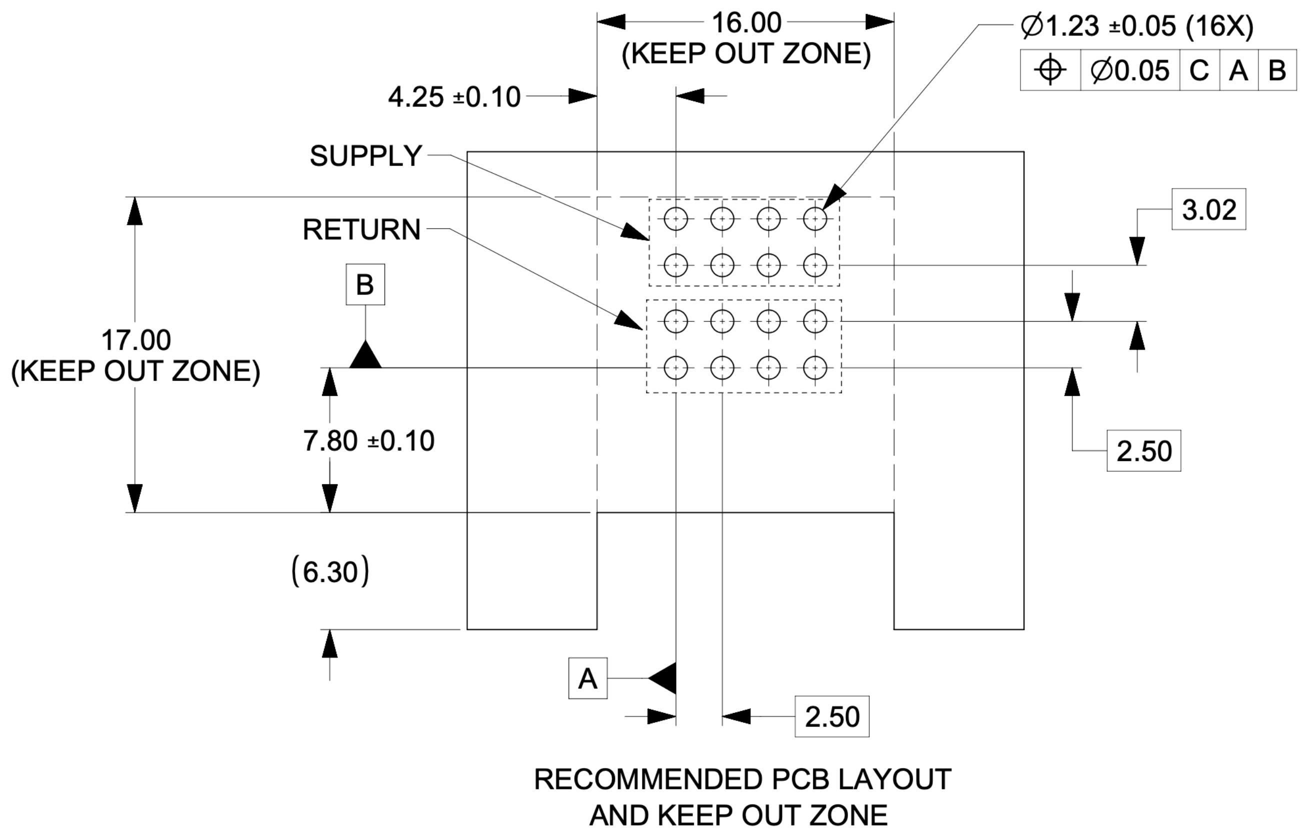


Figure 10. Power connector footprint definition

Connector drawings are available: Molex Brick power connector

3.4.2. Brick critical dimensions

The brick power connector has a critical dimension for the projection from the brick panel. Spacing between panels is 6.23mm. The power connector shall be proud of the panel 4.40mm.

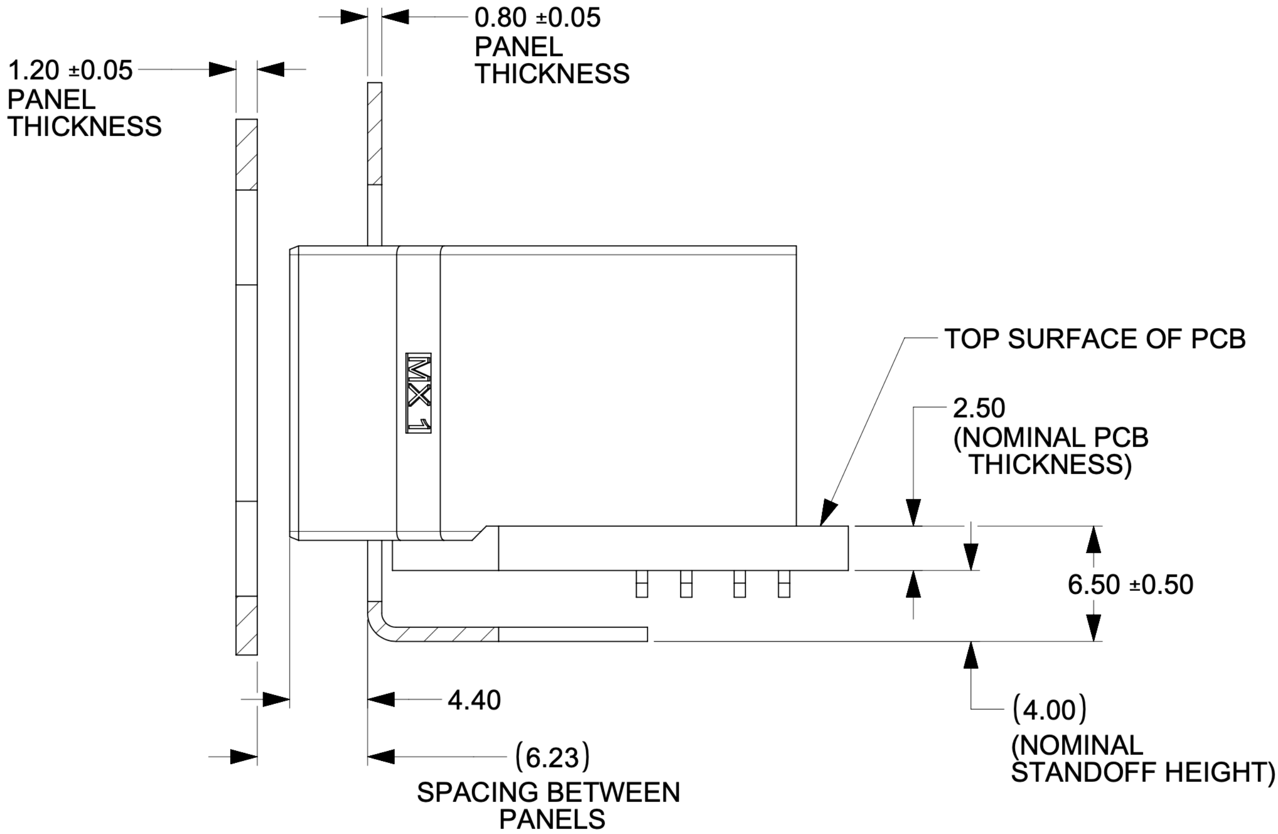


Figure 11. Power connector critical dimensions

3.4.3. Brick voltage

Open19 V2 power shares Open Compute ORV3 power characteristics. Open19 bricks shall safely operate in the DC range 48-52V and shall remain functional when exposed to 0-54V. The power rail shall not exceed 52.5V for more than 200mS. DC voltage shall never exceed 54V.

- Operating range: 48-52V
- NTE voltage: 54V
- Max voltage: ⇐52.5V for <200mS

3.4.4. Data Connector

The data connector is rated for 56G PAM4 with an air dielectric. Variations of cables deliver different lane counts, data rates, and physical interfaces. In the reference Ethernet implementation, the data connector pinout explicitly supports a dedicated 10/100/1000Base-T management port, four 25G NRZ/50G PAM4 lanes. Optionally, a single QSFP-DD or two QSFP interfaces are supported.

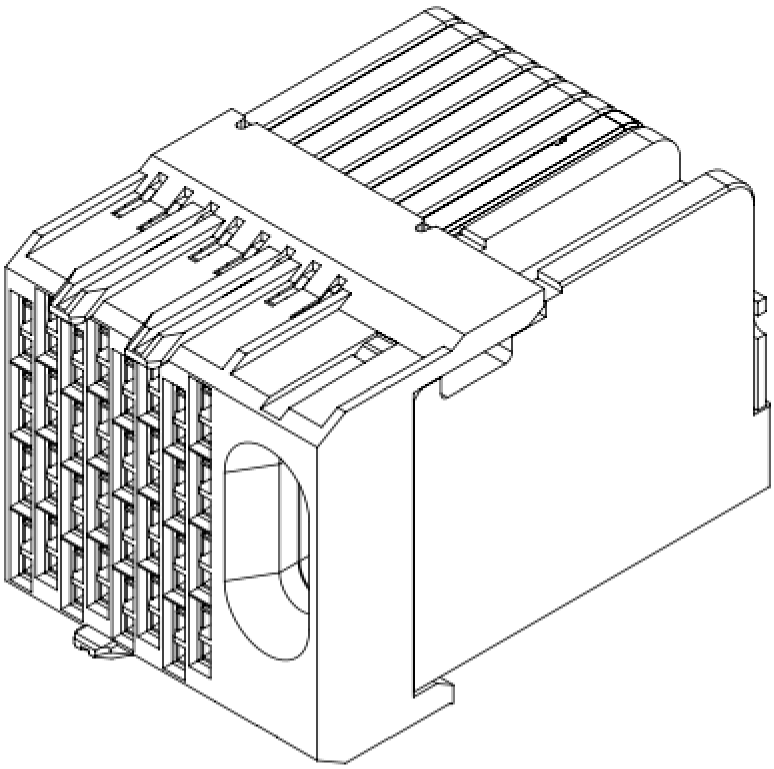


Figure 12. Impel Connector with guide feature

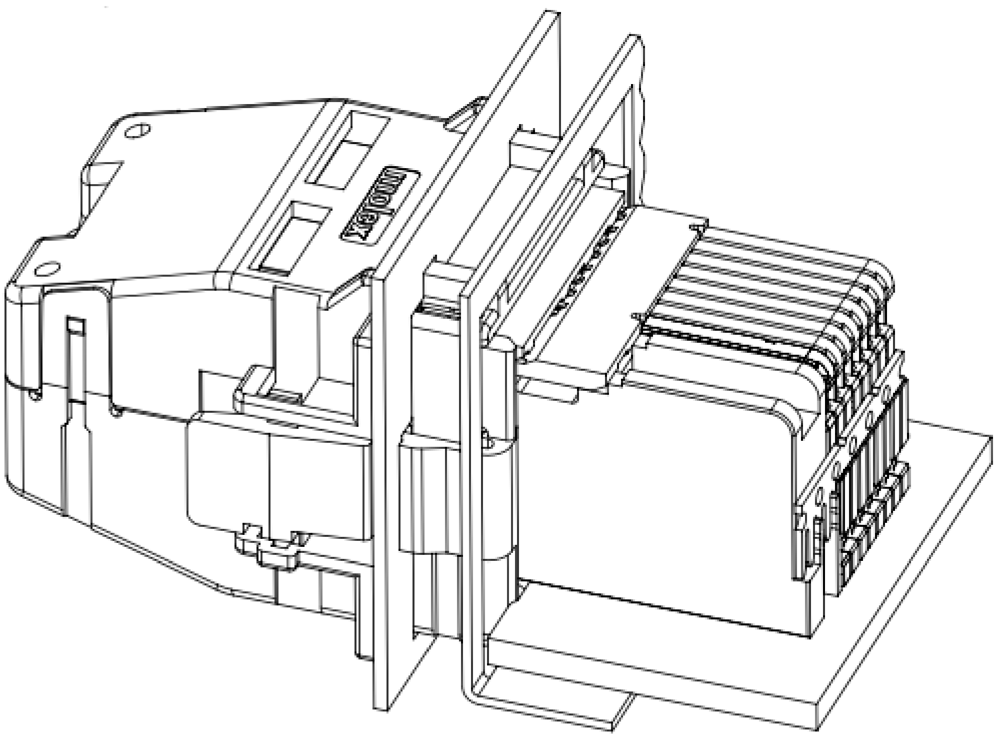


Figure 13. Impel Mated connector with cable backshell

The data connector pinout is defined in the [Appendix]

There is no attempt to support 50G (2x25G NRZ) nor 50G (1x50G NRZ).

3.4.5. Ethernet interface inter-operability

The inter-operability of bricks at deployment with Ethernet interfaces is critical. To support 25G NRZ, 50G PAM4, 100G (4x25G NRZ), 200G (4x50G PAM4), and 400G (8x50G PAM4), the following table:

Port Count	Data Rate	Implementation Interface and order
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5		SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1
6		SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1, QSFP2 LN 1
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5	100G (4x25G) 200G (4x50G PAM4)	SFP1, SPF2, SFP3, SPF4, QSFP1
6		SFP1, SPF2, SFP3, SPF4, QSFP1, QSFP2
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5	400G (8x50G PAM4)	QSFP-DD (QSFP1+QSFP2) 200G (4x50G PAM4)
QSFP (SFP1+SFP2+SFP3+SFP4)	2	QSFP, QSFP1

3.5. Liquid cooling

3.5.1. Brick liquid cooling

Brick liquid cooling is divided into two broad categories, single-phase and two phase. They are mutually incompatible and the designs are mechanically incompatible by design. All bricks inside a single cage are expected to be the same liquid cooling technology.

Unified cutout

Open19 V2 liquid cooling implements a unified liquid cooling cutout. This allows for a consistent brick design to implement both liquid cooling solutions. Implementing the unified cutout is not required, provided all the requirements of a single solution are met.

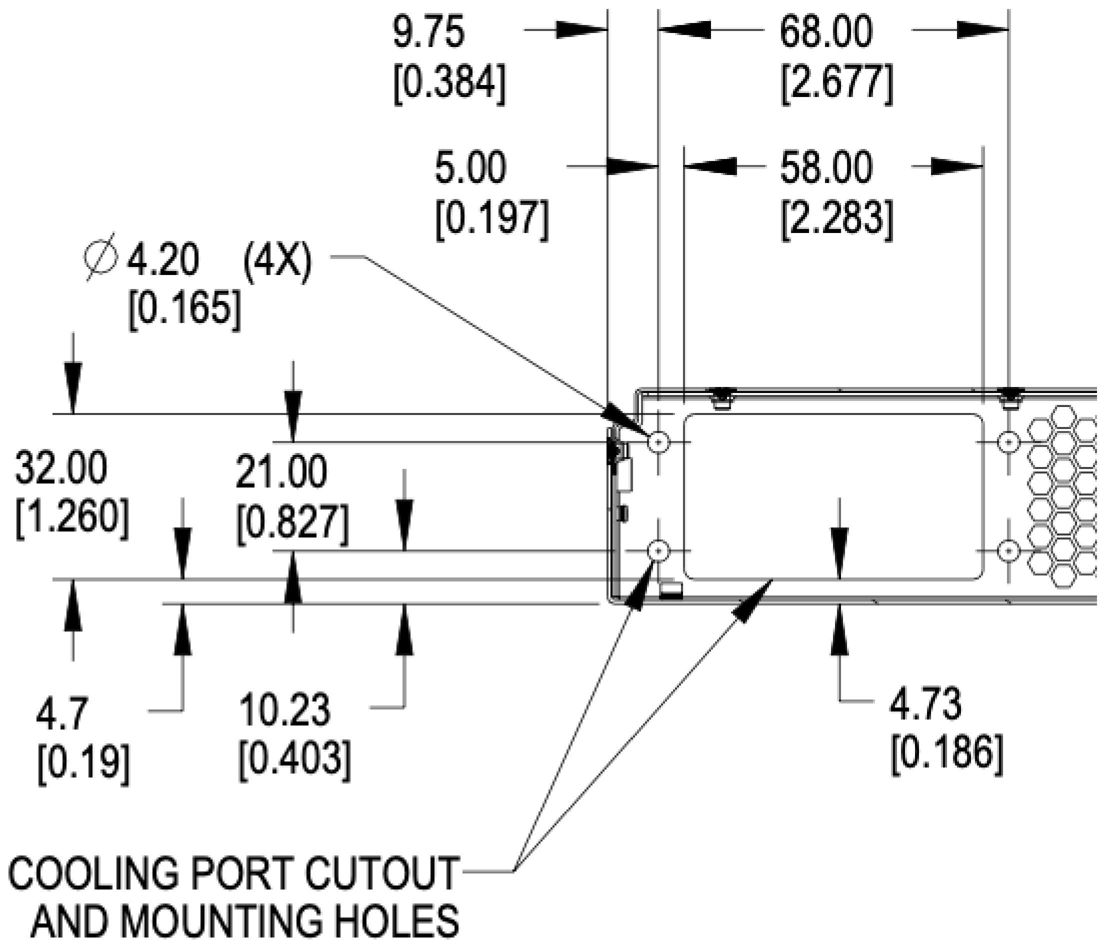


Figure 14. Brick unified cooling cutout

3.5.2. Single phase

Single-phase Connector selection

Single-phase cooling solutions shall use the CPC BLQ4 family of interfaces.

Figure 15. BLQ4 body for manifold

Figure 16. BLQ4 insert for bricks

BLQ4_SpecSheet.pdf

The BLQ4 connectors are blind mate. Both sides of the connector have G 3/8" threads which thread into manifold (body) and brick (insert), respectively.

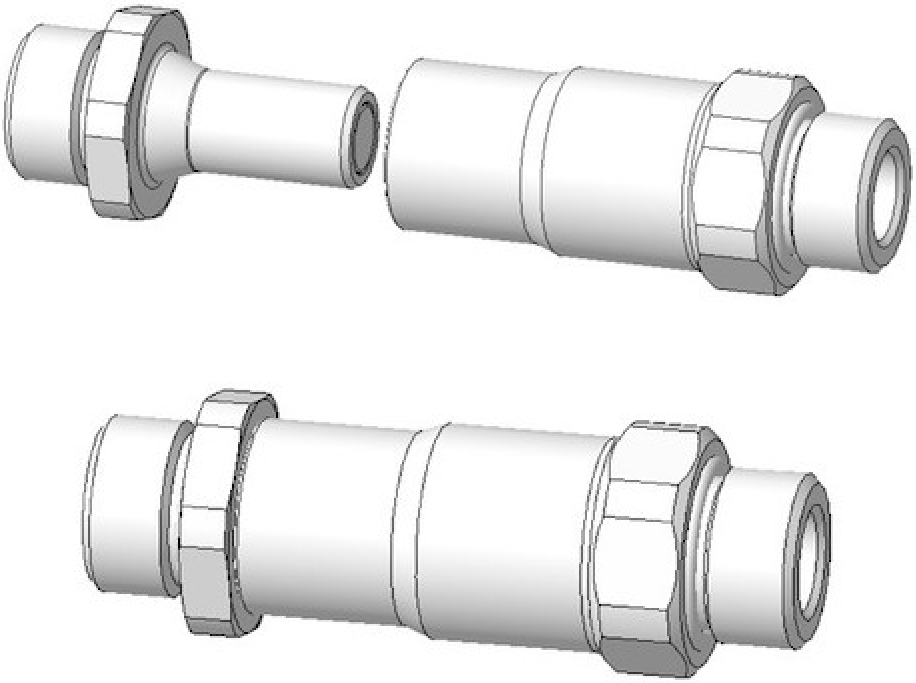


Figure 17. BLQ4 pair

Single-phase connector mounting and polarization

Bricks shall have male blind-mate connectors (inserts), BLQ4D47006 or compatible part.

When viewed from the rear of the brick, in reference to the server: Liquid return shall be located to the left. Liquid supply shall be located in the to the right.

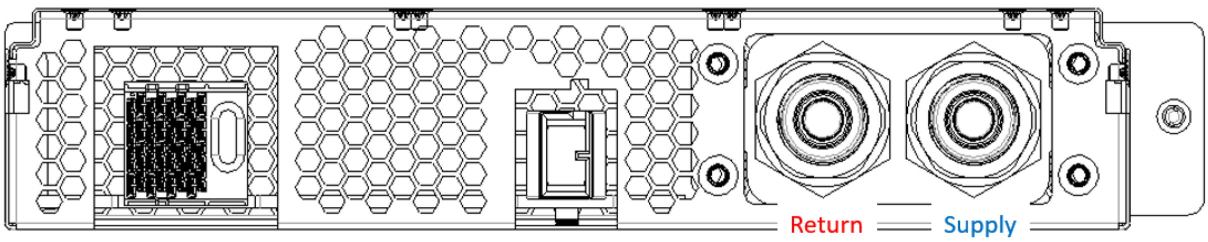


Figure 18. Brick side supply and return

Single-phase brick side connector locations

The supply connector should be 29.5mm from the outer side of the brick (opposite the thumbscrew) in the x-direction and 20.73mm from the bottom of face of the brick in the y-direction. THe two

connectors should be lined up in the horizontal plane and spaced 28.5mm from each other.

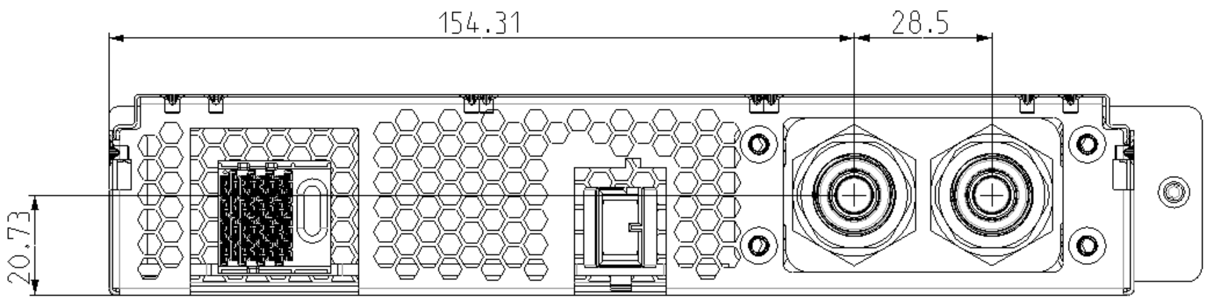


Figure 19. Brick side liquid connector locations for single-phase

Single-phase offset depth

The mating surface of the brick side connector sits xx mm behind (outside) the outer edge of the back plane of the brick. Specifying this dimension ensures the server manufacturer and manifold/cage side implementor have a consistent location where the connectors must meet to ensure full mating and thereby full flow.

The reference mating surfaces of each connector are highlighted below along with a drawing showing the specified dimension.

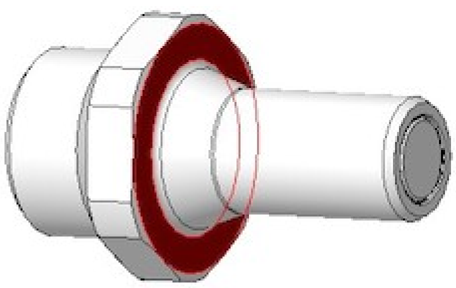


Figure 20. Mating surface BLQ4 insert

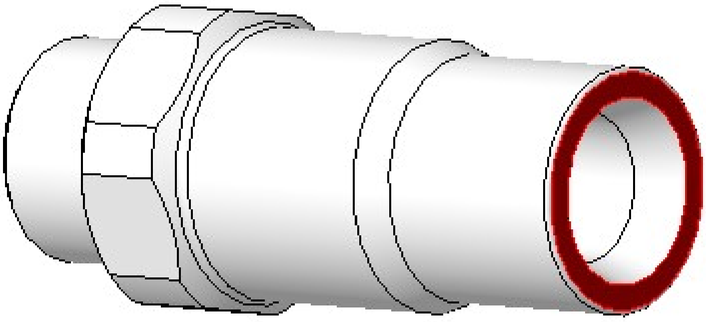


Figure 21. Mating surface BLQ4 body

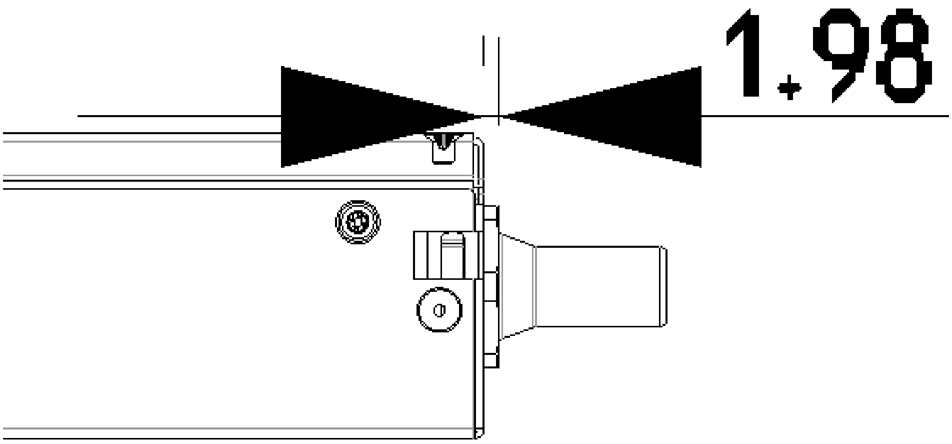


Figure 22. Offset from cage back panel to connector mating interface

The BLQ4 connectors must be within 3mm of this fully engaged state to achieve full flow.

3.5.3. Two phase

Two-phase connector selection

The two-phase cooling solution shall use the CPC BLQ family of interfaces.

On brick side

Figure 23. Liquid - CPC BLQ4 series insert P/N BLQ4D47004PROTO (chrome plated brass) or P/N 4649800PROTO (Aluminum)

Figure 24. Vapor - CPC BLQ6 series insert P/N 4650000PROTO (chrome plated brass)

On the Manifold side

Figure 25. Liquid - CPC BLQ4 series body P/N BLQ4D31004PROTO (chrome plated brass) or P/N 4649900PROTO (Aluminum)

Figure 26. Vapor - CPC BLQ6 series body P/N 4650200PROTO (chrome plated brass)

Two-phase connector mounting polarization

Cage manifolds shall have body (female) blind mate connectors. Bricks shall have insert (male) blind-mate connectors. When viewed from rear of the brick, vapor connection shall be located at the right position. Liquid connection shall be located at the left position.

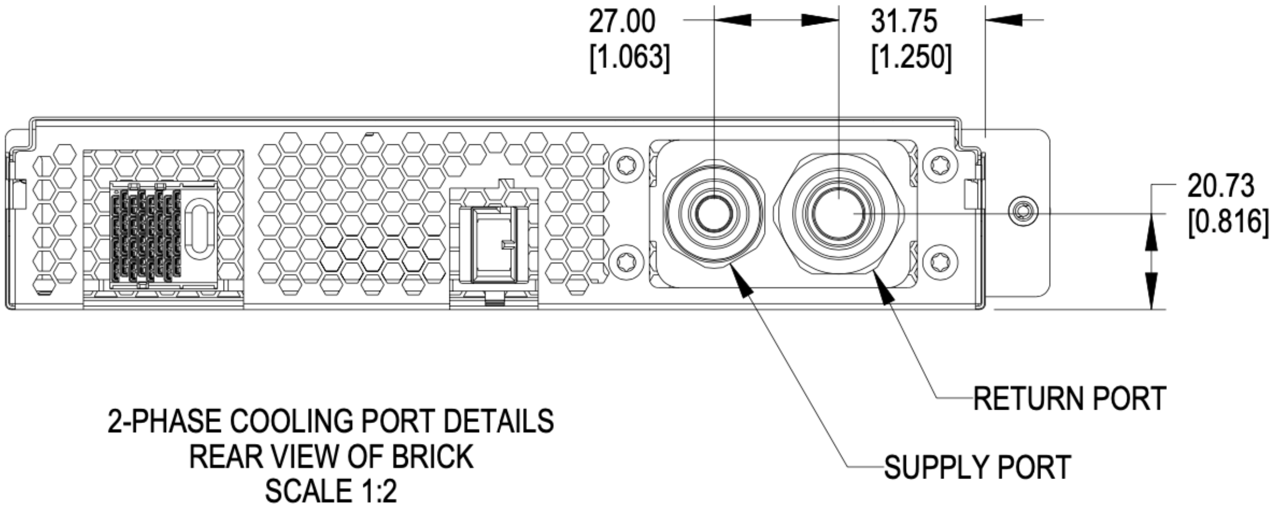


Figure 27. Brick liquid supply and Vapor return

Two-phase brick side connector locations

Using the right-side panel of the brick as a reference, the Vapor return connector on the brick should be 31.85mm in the x direction and 20.73 upwards in the y direction.

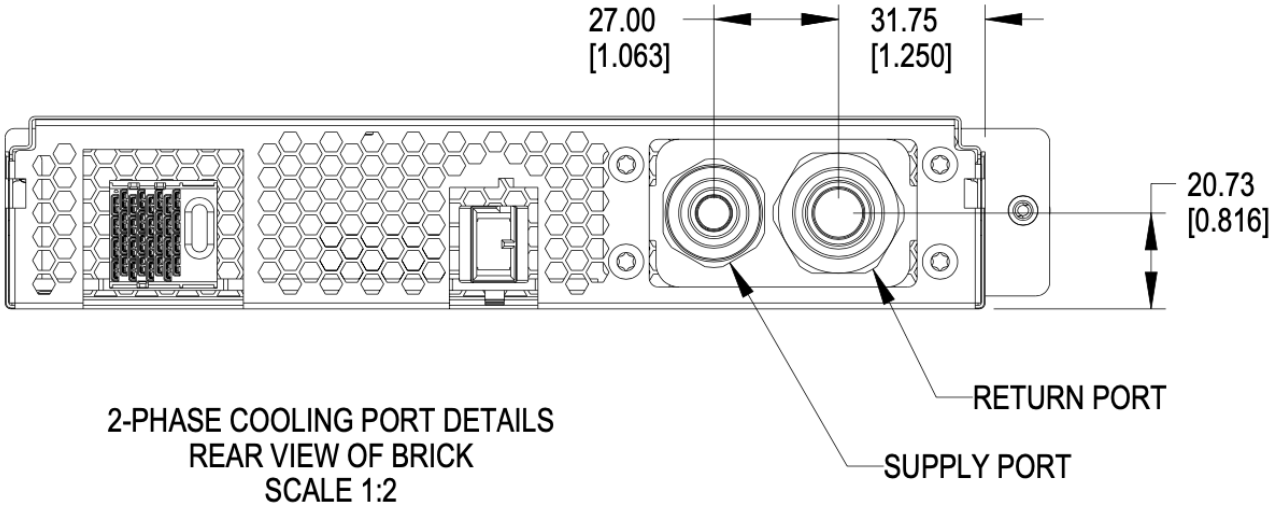


Figure 28. Brick side connectors locations

Two-phase offset depth

The mating surface of the brick side connector sits xx mm behind (outside) the outer edge of the back plane of the brick. Specifying this dimension ensures the server manufacturer and manifold/cage side implementor have a consistent location where the connectors must meet to ensure full mating and thereby full flow.

The reference mating surfaces of each connector are highlighted below along with a drawing showing the specified dimension.

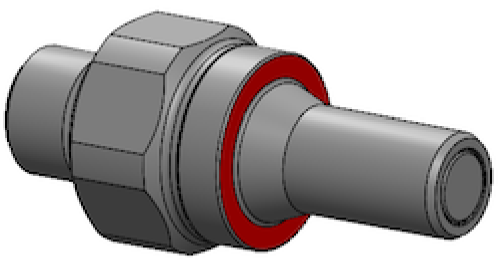


Figure 29. Mating surface BLQ4 insert

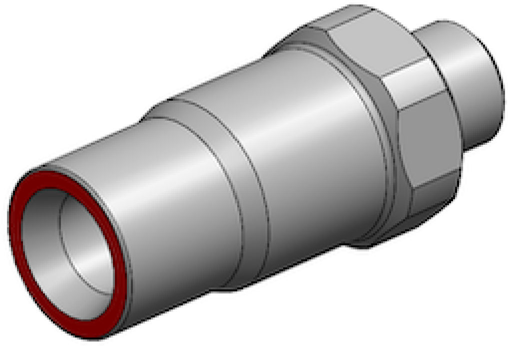


Figure 30. Mating surface BLQ4 body

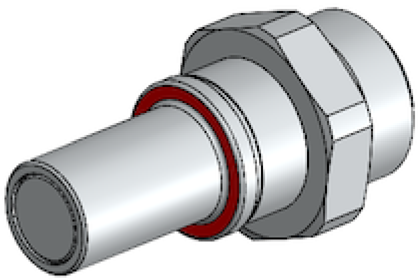


Figure 31. Mating surface BLQ6 insert

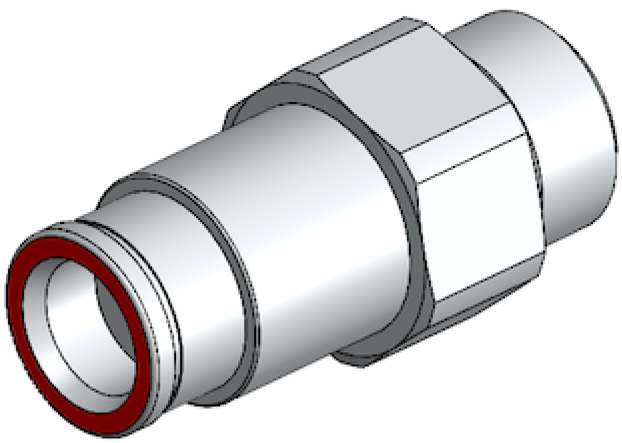


Figure 32. Mating surface BLQ6 body

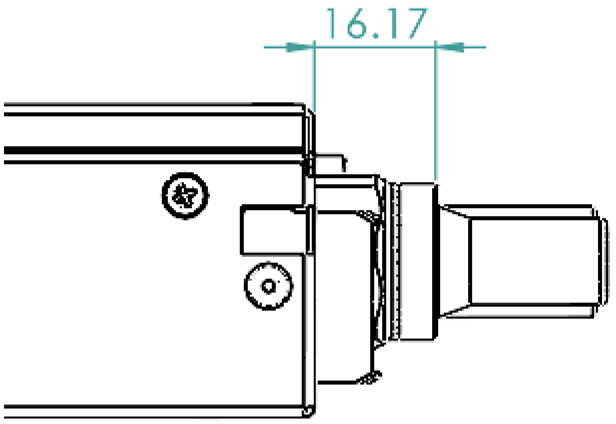


Figure 33. Offset from cage back panel to BLQ4 connector mating interface

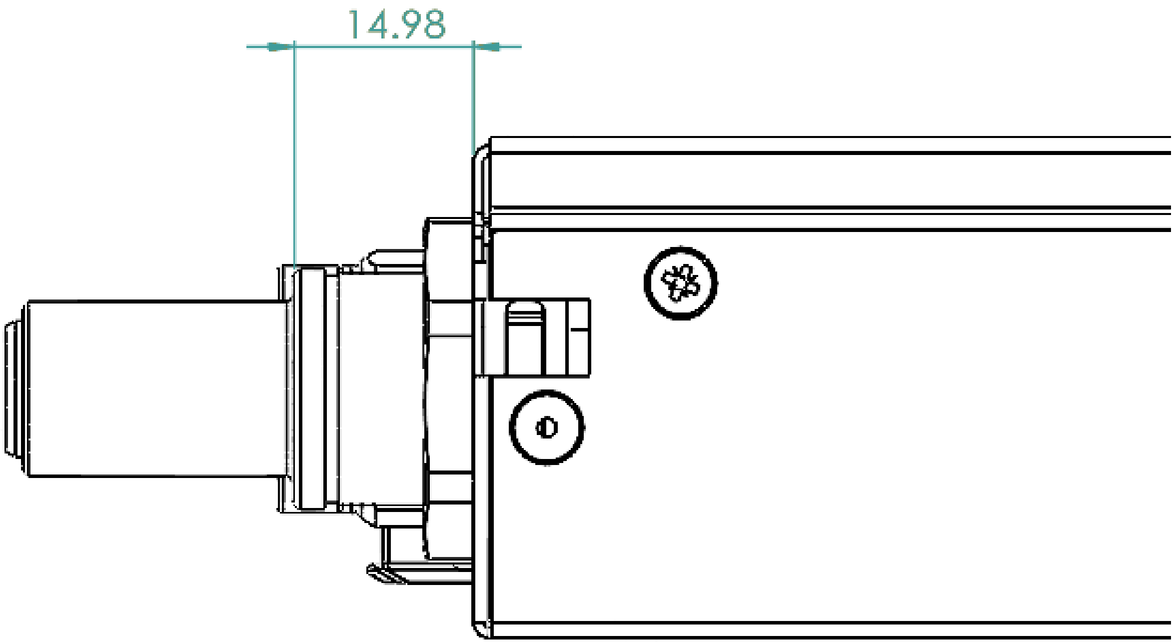


Figure 34. Offset from cage back panel to BLQ6 connector mating interface

The BLQ4 & BLQ6 connectors must be within ±1.5mm of this engaged state to achieve full flow.

3.6. Design guidance

Design guidance is not a specification requirement. It is simple design guidance that can be implemented to simplify future needs, to meet certification requirements, or satisfy general fit & finish.

3.6.1. Brick faceplate guidance

The faceplate is notched above and below the fastener (See Figure 2, “Brick retention”). Note that the faceplate can accommodate any type of ports or drives as needed and is not required to be fully perforated.

Screws and/or rivets used to secure the top cover are flush to be under-flush. The vertical edges on the rear of the brick chassis base are coined to break sharp edges.

3.6.2. Back panel safety perforation

See below for typical dimensions of the airflow perforation. All dimensions below are critical.

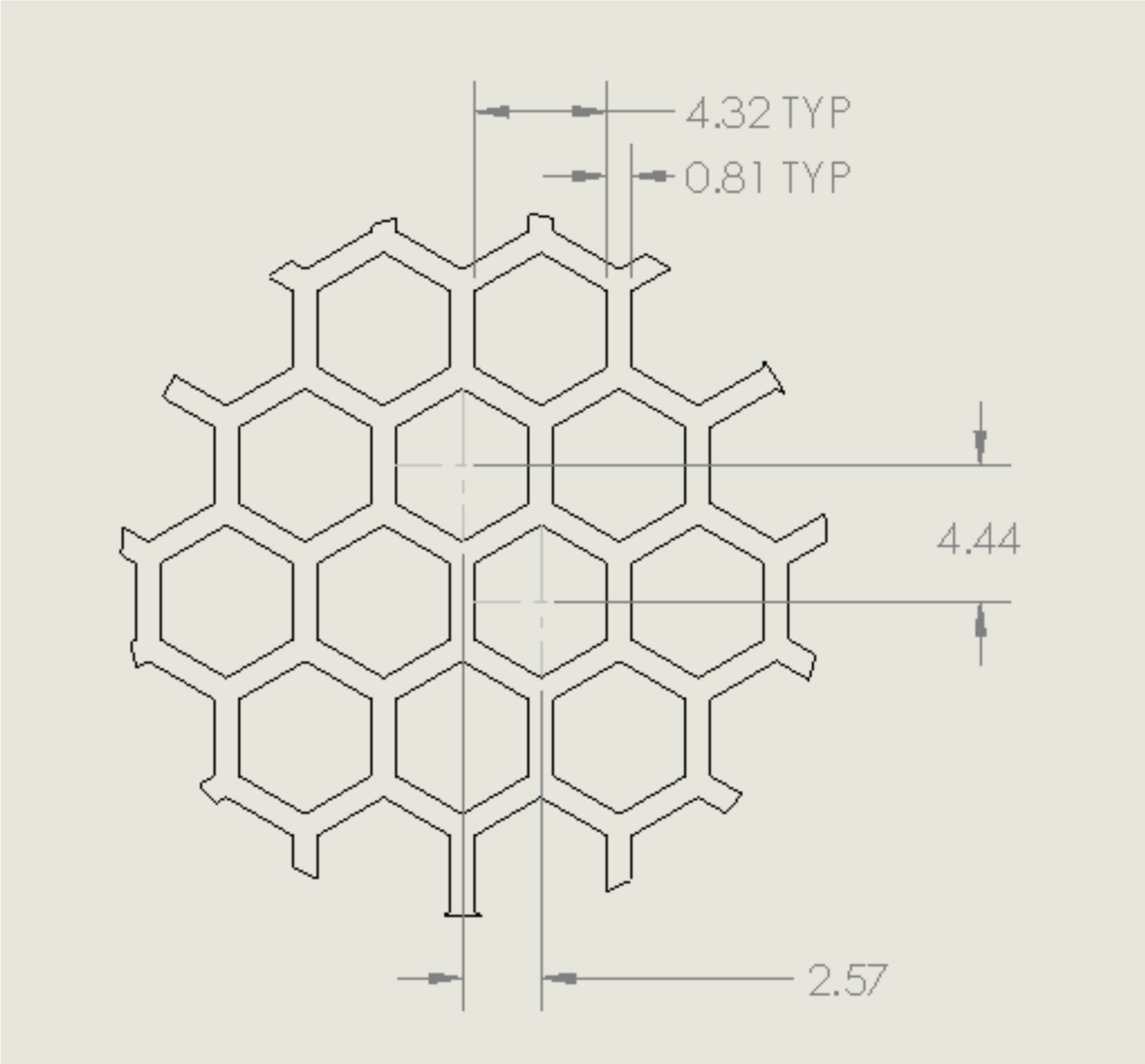


Figure 35. Brick panel perforation

Keep-out zones

Keep-out zones are defined in the brick surrounding the power connector and liquid cooling interfaces. The liquid cooling keep-out spaces "fly" above the PCB, allowing components to be populated below the connections.

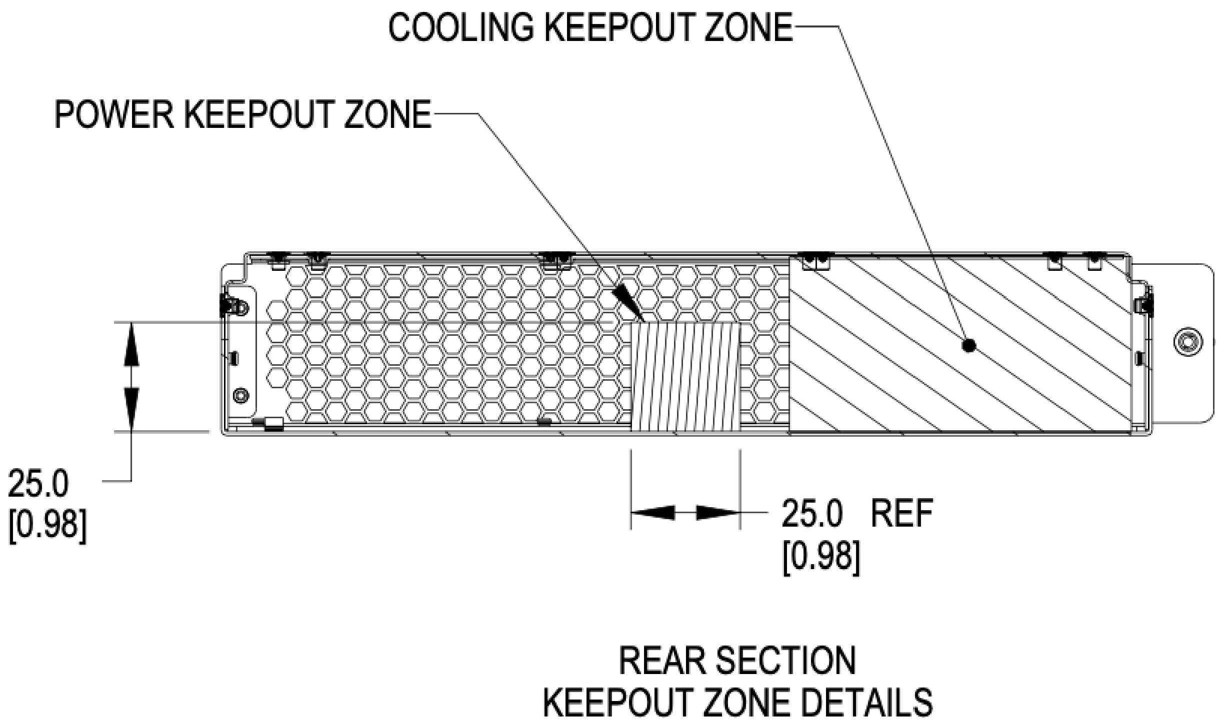


Figure 36. Brick keep-out zone

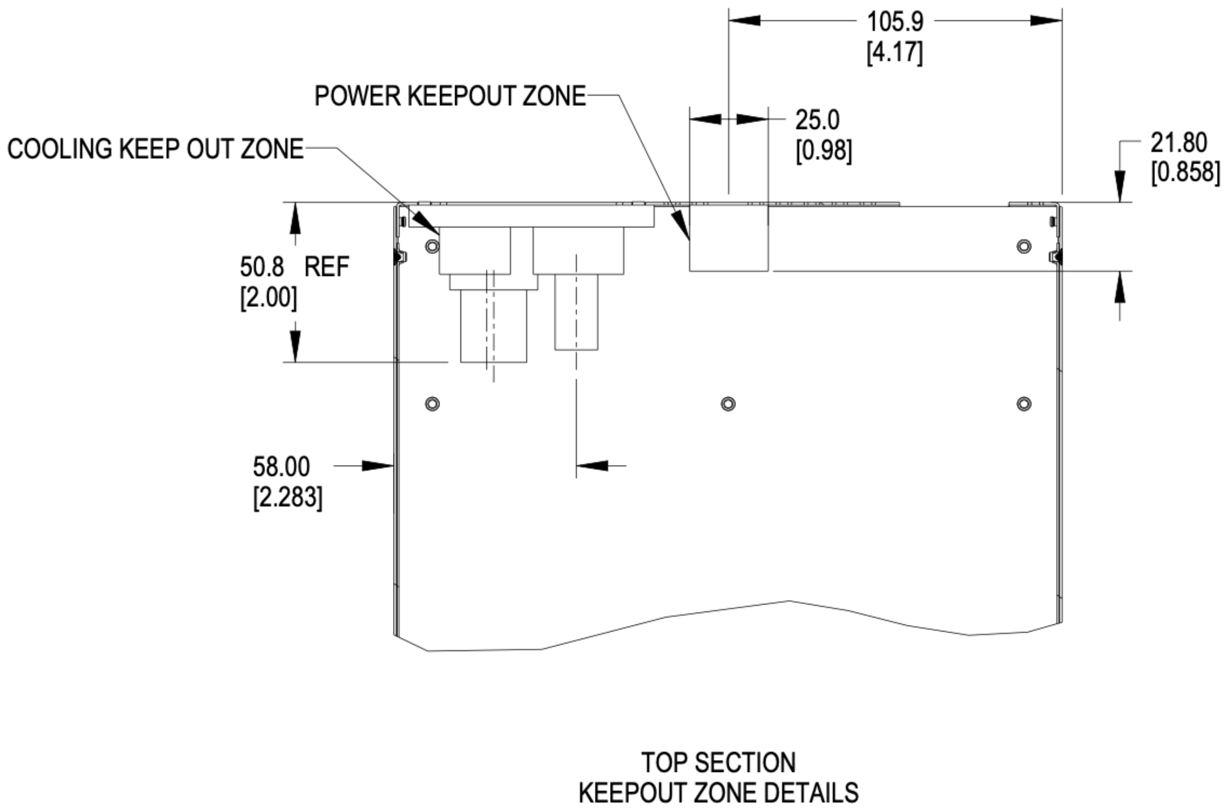


Figure 37. Brick keep-out zone top view

3.7. Brick performance

3.7.1. Brick electrical startup behavior

In order to safely operate the power shelf, we define the following characteristics for brick startup behavior.

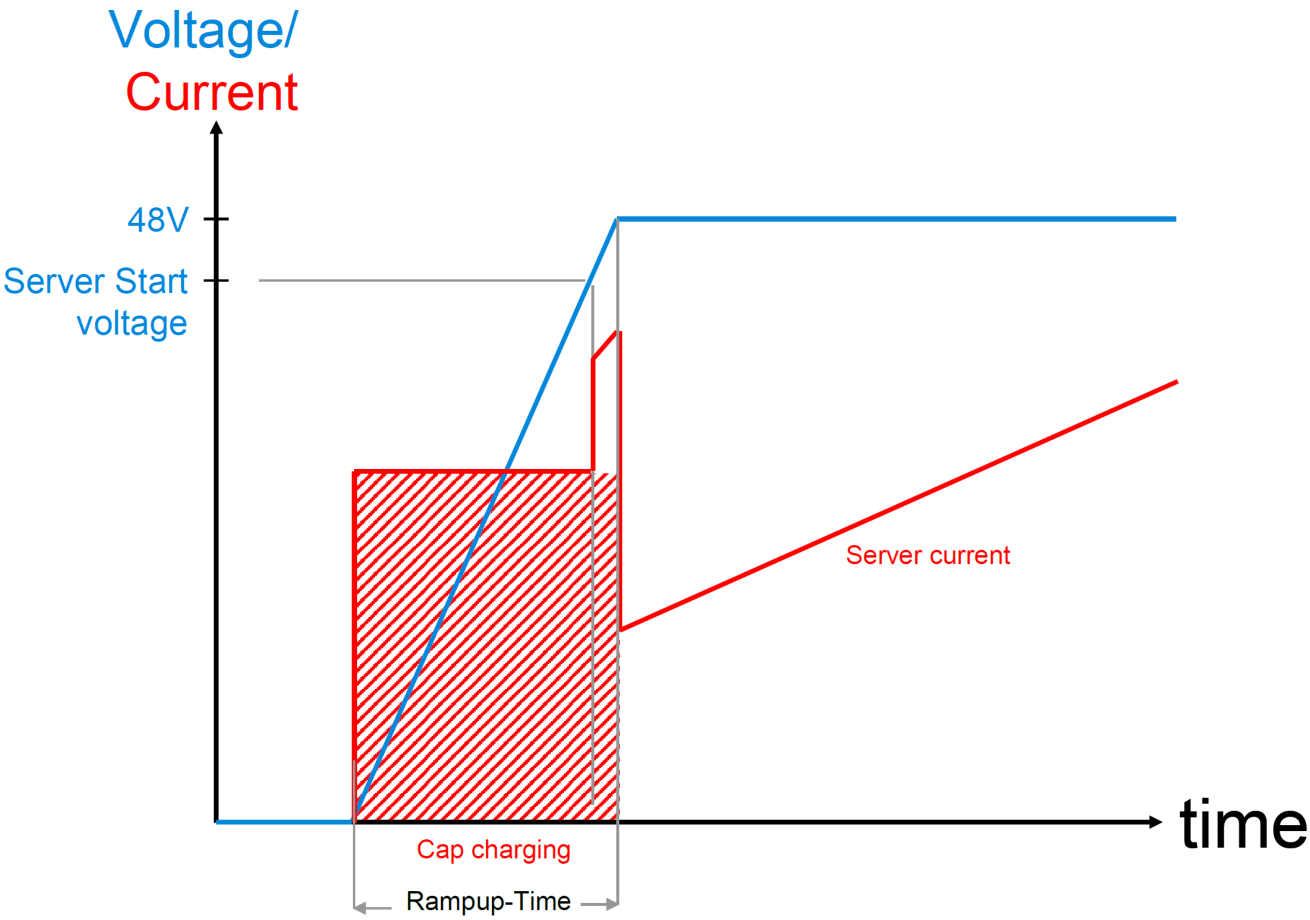


Figure 38. Server startup electrical characteristics

These values are per connector feed from a power shelf.

- Min start-up voltage: +44V DC
- Max input capacitance per brick feed: ⇐1000uF
- Max server current at start-up voltage: 70A
- Ramp time: Follow Intel CRPS spec: ramp time 10-70msec, slew rate 0.5V/msec

https://www.intel.com/content/dam/www/public/us/en/documents/guides/power-supply-design-guide-june.pdf

Chapter 4. Brick Mechanical Specification [DHHW]

4.1. DHHW Brick

The DHHW brick is commonly used as a 2U 2N server form-factor.

4.2. Brick Dimensions

The brick is designed with the overall dimensions shown.

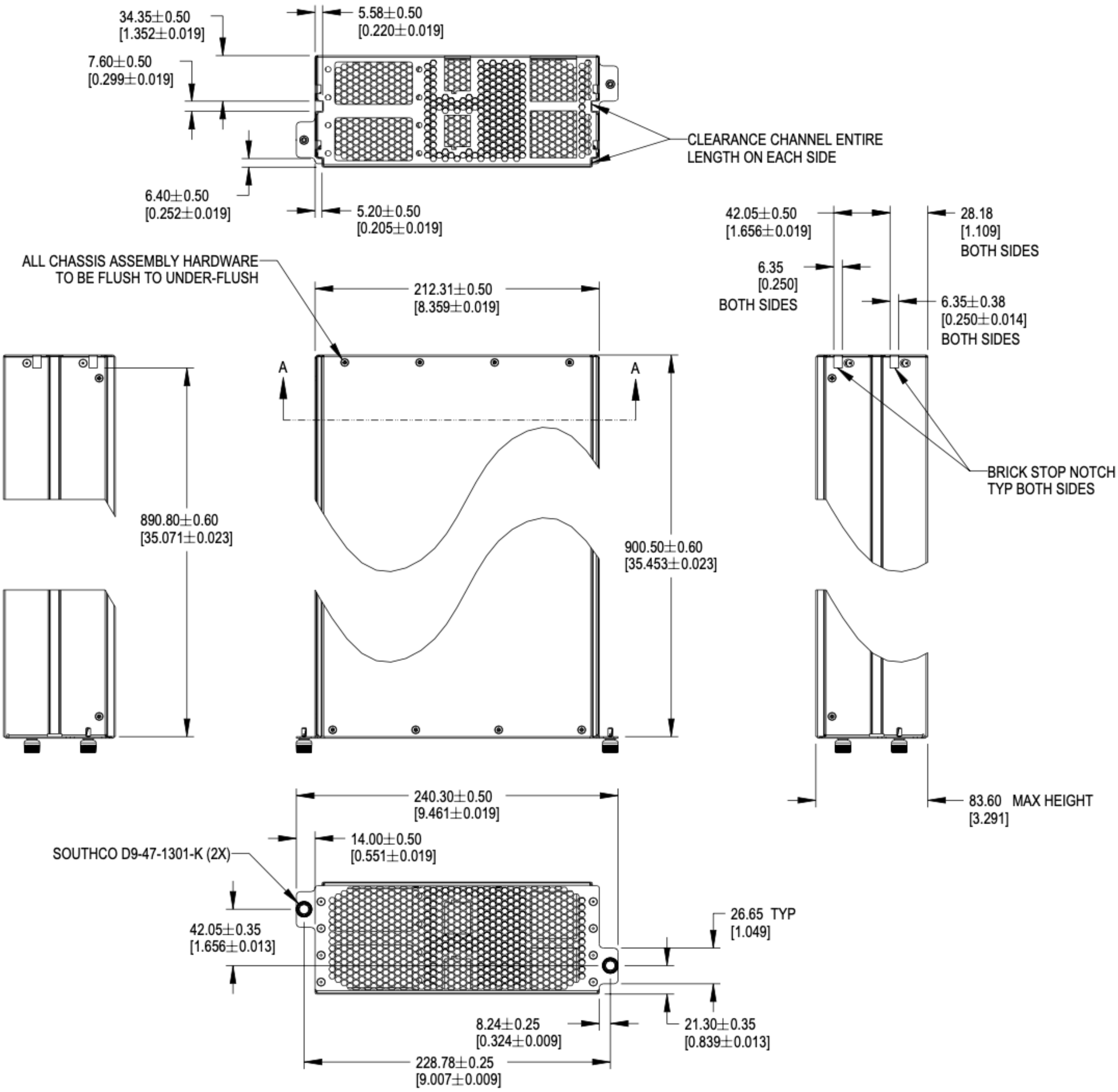


Figure 39. Brick dimensions

Please refer to the drawings, and 3D files for further details.

The V1 specification for bricks defined 901.7mm as the overall length of the brick, including the thickness of the front panel.

The dimension is modified to reference to the inside of the front panel to the rear of the brick. This dimension is 900.50mm +/-0.60 from the rear face of the front panel (mating surface to the cage) to the outside of the rear panel.

For half-width designs using the OCP DC-MHS M-DNO board specification, note the tolerance guidance in order to fit boards in a half-width chassis. This requires setting the nominal width at 209.55mm or smaller.

4.3. Mechanical features

4.3.1. Brick retention

Brick are secured into the cage with Southco D9-47-1302-K quarter-turn fasteners.

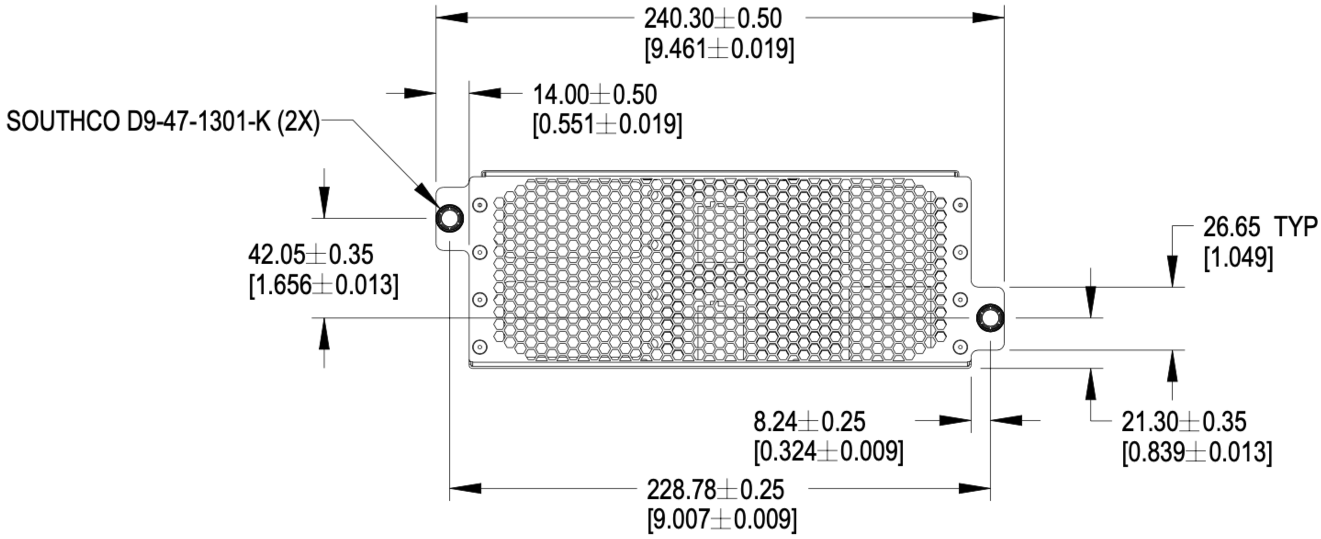


Figure 40. Brick retention

4.3.2. Brick mechanical guide

Clearance channels are required along the top edges of the brick and the sidewalls, extending forward from the rear panel to the back of the faceplate.

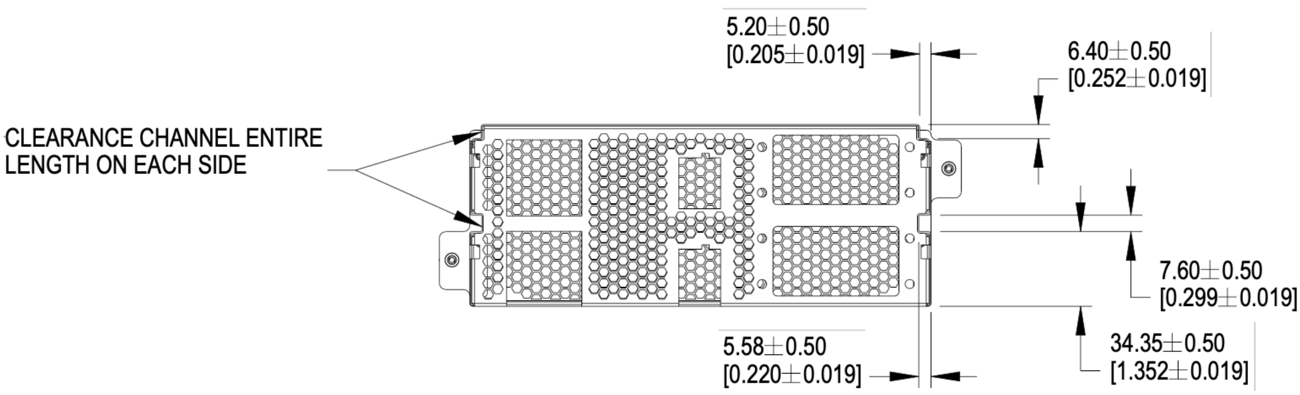


Figure 41. Mechanical guide at top edge and side walls of brick

Brick insertion and retention assistance for liquid cooling

For liquid cooling applications, two different mechanical assist for insertion and retention are available in the cage. Bricks can implement features to make use either/both features.

The features are:

- Brick retention standoff, available on both sides in the mechanical guide channel
- Slots, available on both sides of the cage wall

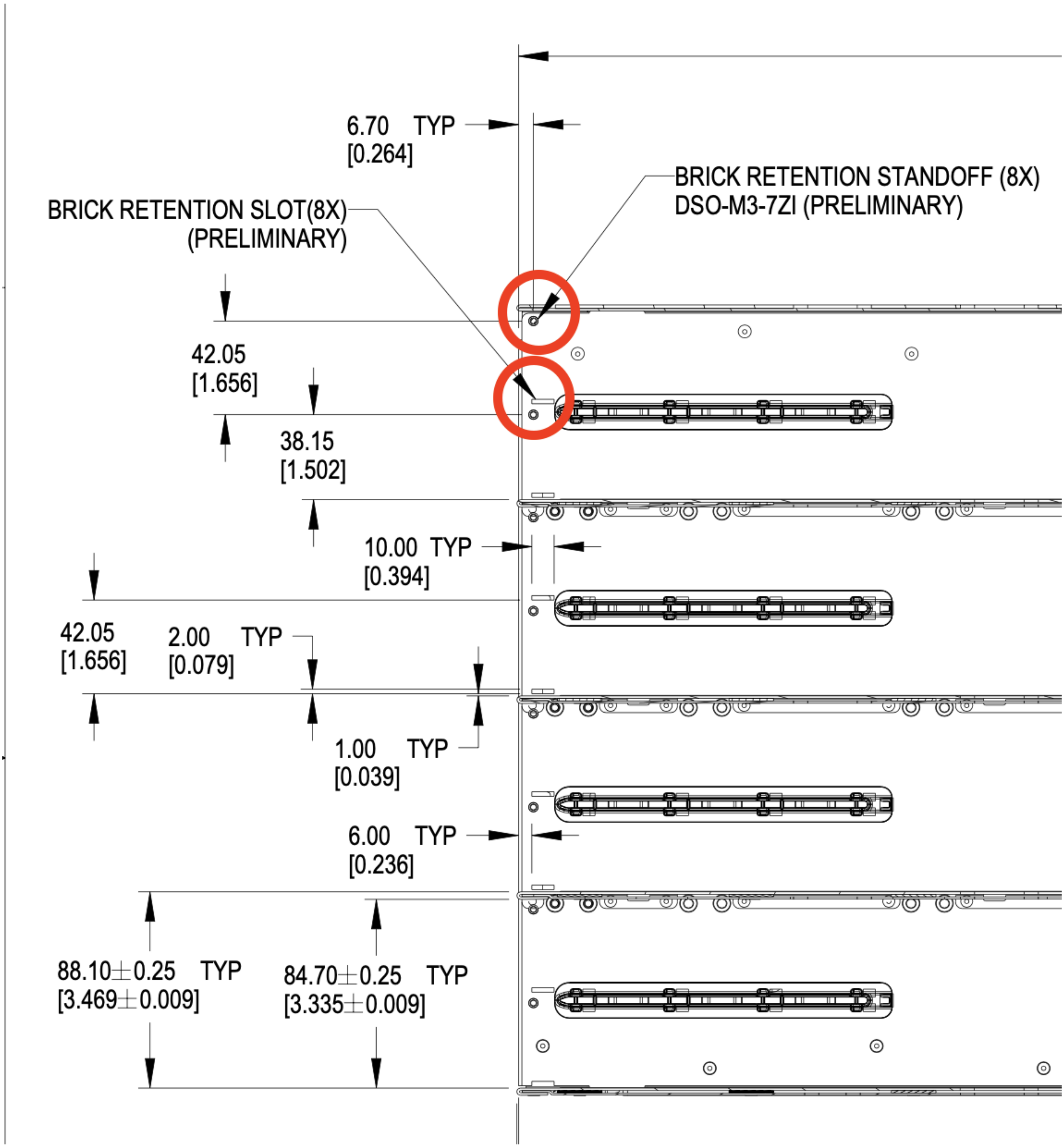


Figure 42. Brick mechanical assistance features in cage

Each feature shall support 100 lbf (445N) force or greater, using the recommended 1.3mm SGCC material.

4.3.3. Brick Mechanical Keying

Two keying notches are positioned in the left and right corners of the sheet metal enclosure as shown below. All dimensions below are critical.

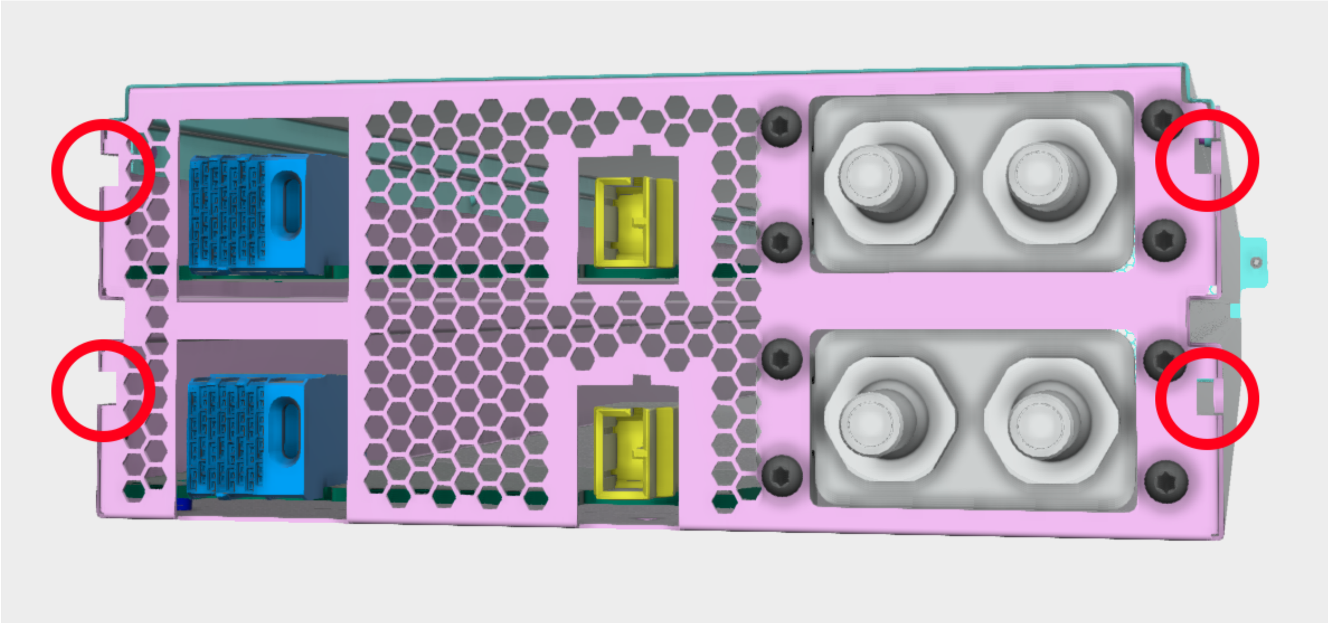


Figure 43. Brick keying

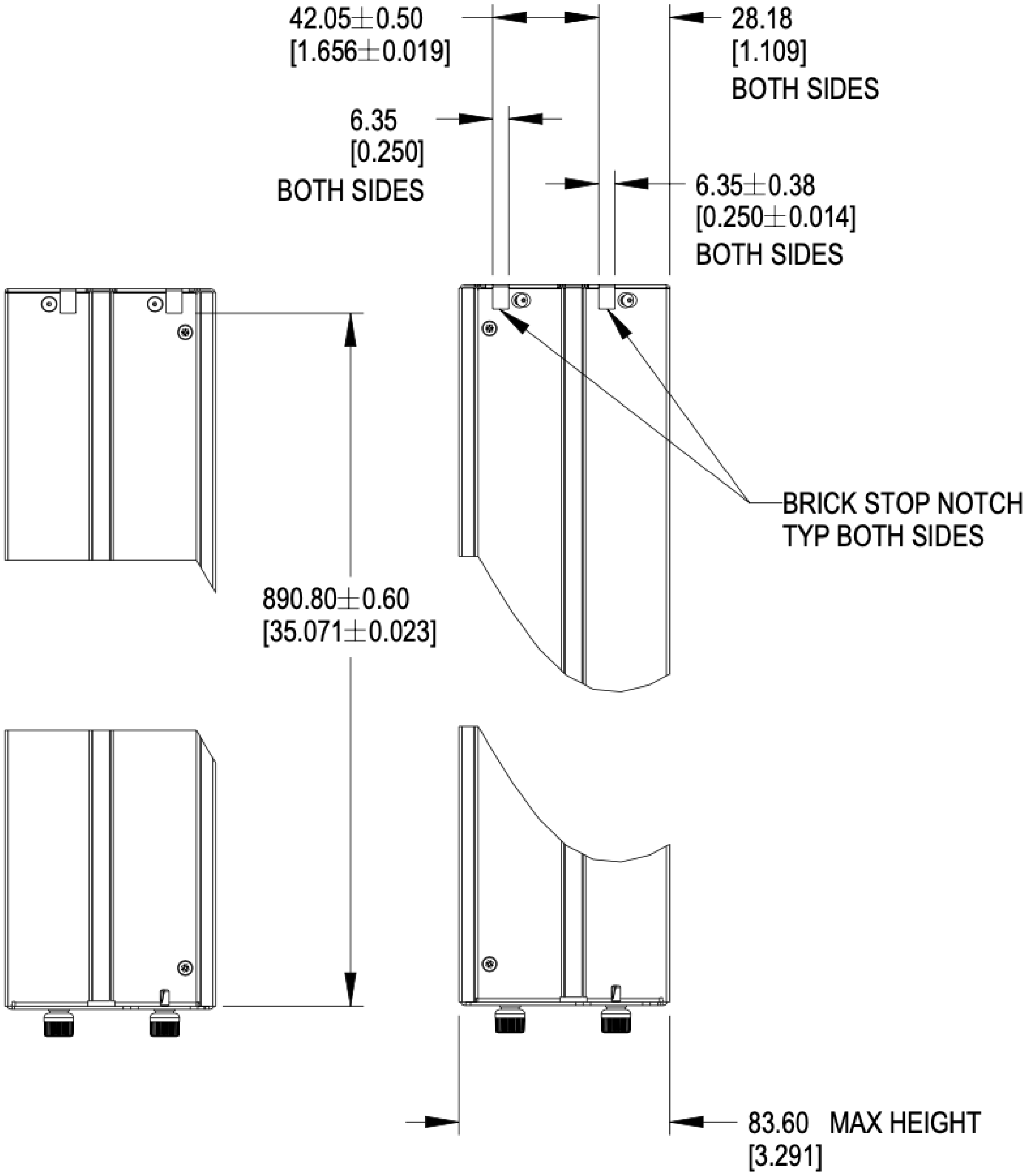


Figure 44. Brick keying dimensions

4.3.4. PCBA

The rear PCBA (Printed Circuit Board Assembly), also referenced as the interposer, provides the +48V DC input power and a data connections. The top of the PCBA is the reference surface, located 6.5mm +/- 0.6mm above the bottom surface of the sheet metal chassis.

A second PCBA may be installed at 48.55mm +/- 0.5mm above the bottom surface of the chassis.

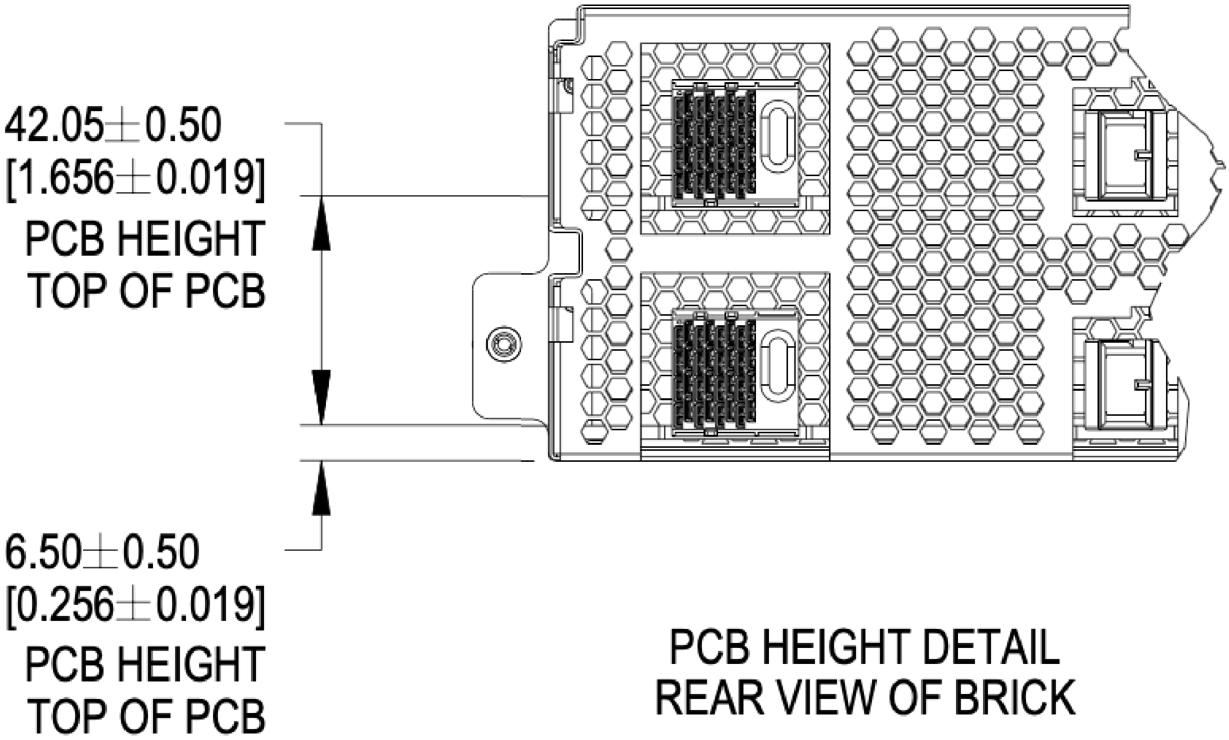


Figure 45. PCBA References

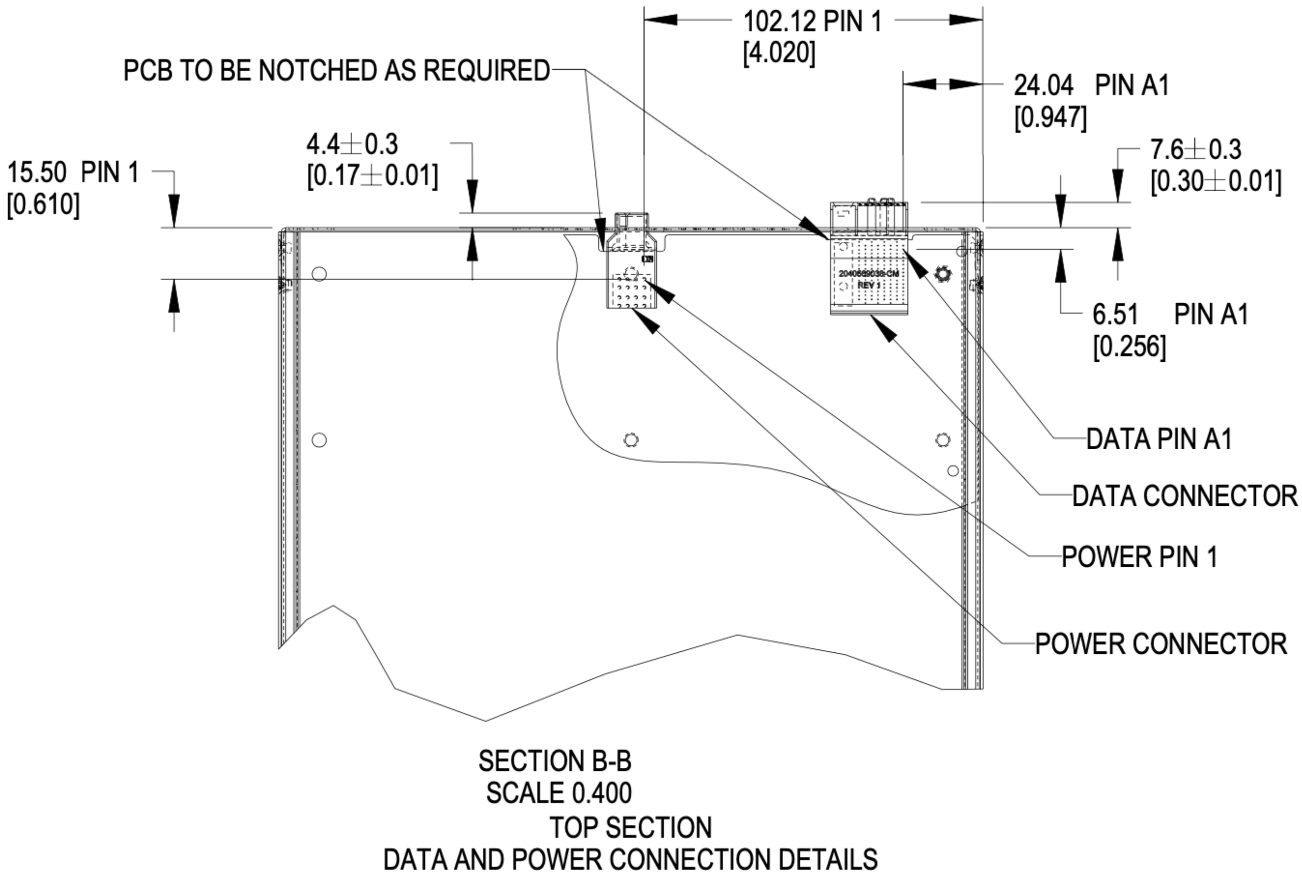


Figure 46. Brick PCB dimensions

One set of connectors are shown in the image. Two sets are available in a brick.

4.3.5. Materials

The brick enclosure is constructed of sheet metal with adequate strength and rigidity. Suggested material is 0.8mm SGCC for the chassis base and 0.6mm SGCC for the top cover.

4.3.6. Blind mate interfaces

A brick is equipped with at least one set of blind mate connectors for receiving +48V DC power and transferring data with an external device, typically a network switch. Introduced in this version is are blind mate connections to facilitate liquid cooling. The connections mate with cable and manifold projections from the rear wall of the cage.

- Power connector: Molex Ten60 PN: 220370-0001 (35A) 22037-0000 (70A), refer to Molex document PSD-220370-0000
- Data connector: Molex 4x8 Impel with guide receptacle PN: 204066-9038 (PCB thickness 2.2-4.1mm), 204066-9138 (PCB thickness 1.0-1.9mm), refer to Molex document PSD-204066-9001

The power connector has two ratings:

- Power: Not-To-Exceed (NTE) 70A at +48V DC
- Power: Not-To-Exceed (NTE) 35A at +48V DC

The ratings shall be coordinated with the power shelf. 35A bricks shall be enabled in 70A capable infrastructure. 70A bricks are keyed to prevent unsafe operation in 35A infrastructure.

The brick may populate a second set of connectors, as required.

4.4. Connector Pinout

4.4.1. Brick power connector

The brick blind-mate power connector for receiving 48V DC (51V nominal) power from the power shelf. The connector mates with the power cable connector projected from the brick cage wall.

Two variations of the brick connector exist, with different Not-to-exceed (NTE) ratings:

- 35A Power Connector: Molex 2203700001 (NTE 35 AMPS) No Key
- 70A Power Connector: Molex 2203700000 (NTE 70 AMPS) Keyed

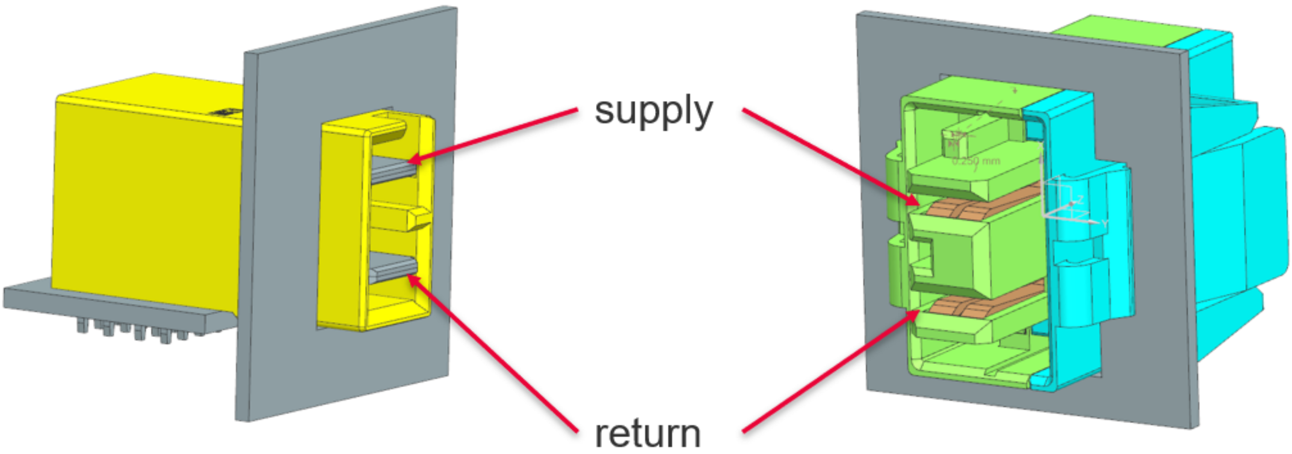


Figure 47. Power connector supply/return definition

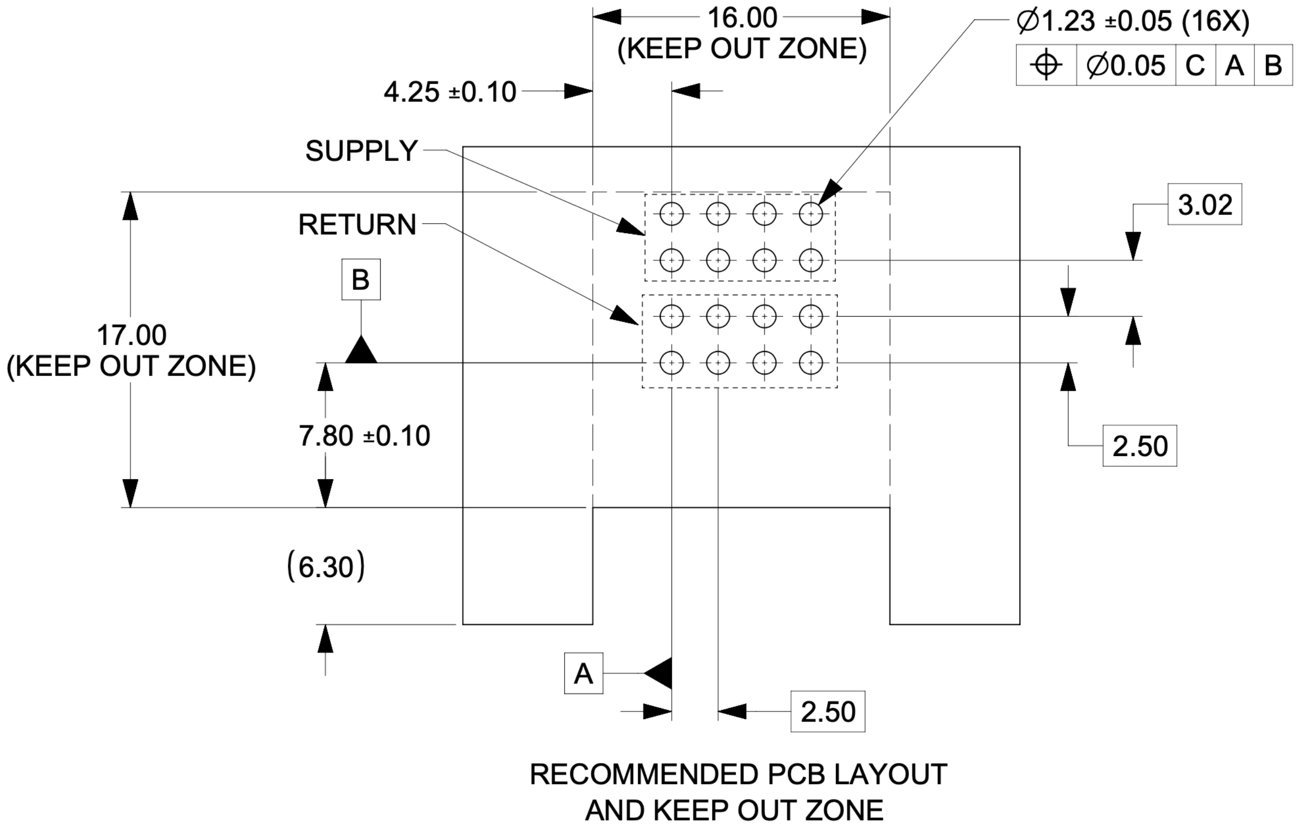


Figure 48. Power connector footprint definition

Connector drawings are available: Molex Brick power connector

4.4.2. Brick critical dimensions

The brick power connector has a critical dimension for the projection from the brick panel. Spacing between panels is 6.23mm. The power connector shall be proud of the panel 4.40mm.

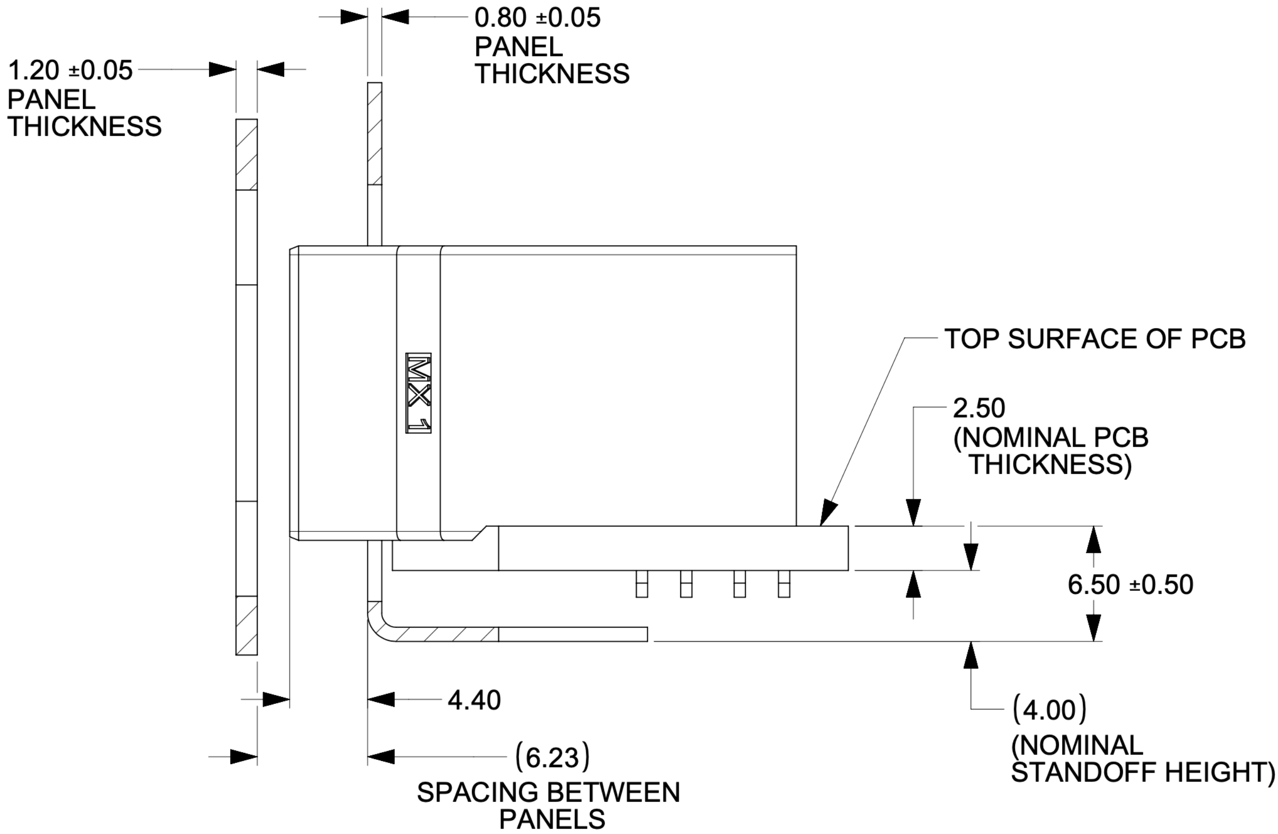


Figure 49. Power connector critical dimensions

4.4.3. Brick voltage

Open19 V2 power shares Open Compute ORV3 power characteristics. Open19 bricks shall safely operate in the DC range 48-52V and shall remain functional when exposed to 0-54V. The power rail shall not exceed 52.5V for more than 200mS. DC voltage shall never exceed 54V.

- Operating range: 48-52V
- NTE voltage: 54V
- Max voltage: ⇐52.5V for <200mS

4.4.4. Data Connector

The data connector is rated for 56G PAM4 with an air dielectric. Variations of cables deliver different lane counts, data rates, and physical interfaces. In the reference Ethernet implementation, the data connector pinout explicitly supports a dedicated 10/100/1000Base-T management port, four 25G NRZ/50G PAM4 lanes. Optionally, a single QSFP-DD or two QSFP interfaces are supported.

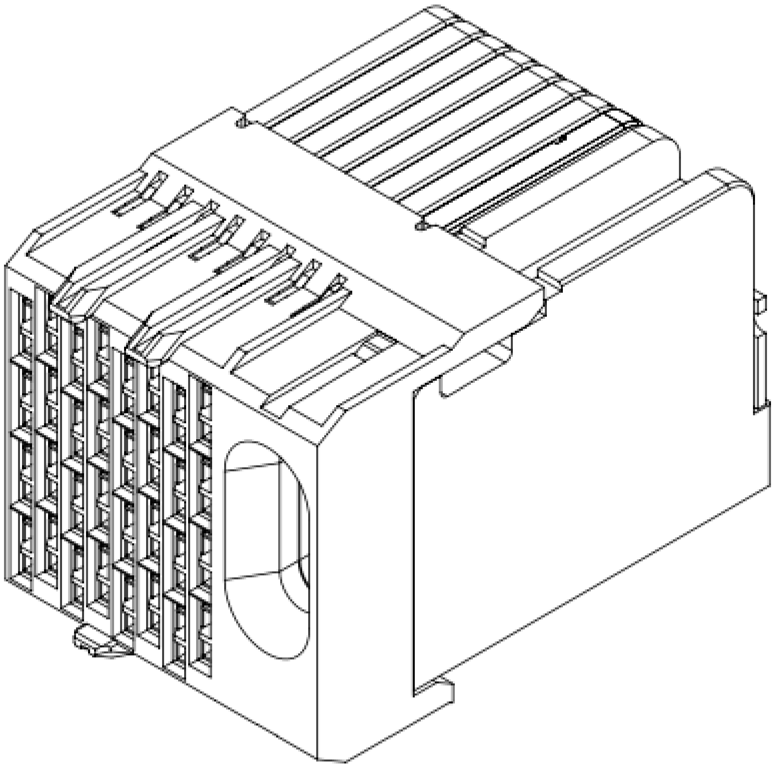


Figure 50. Impel Connector with guide feature

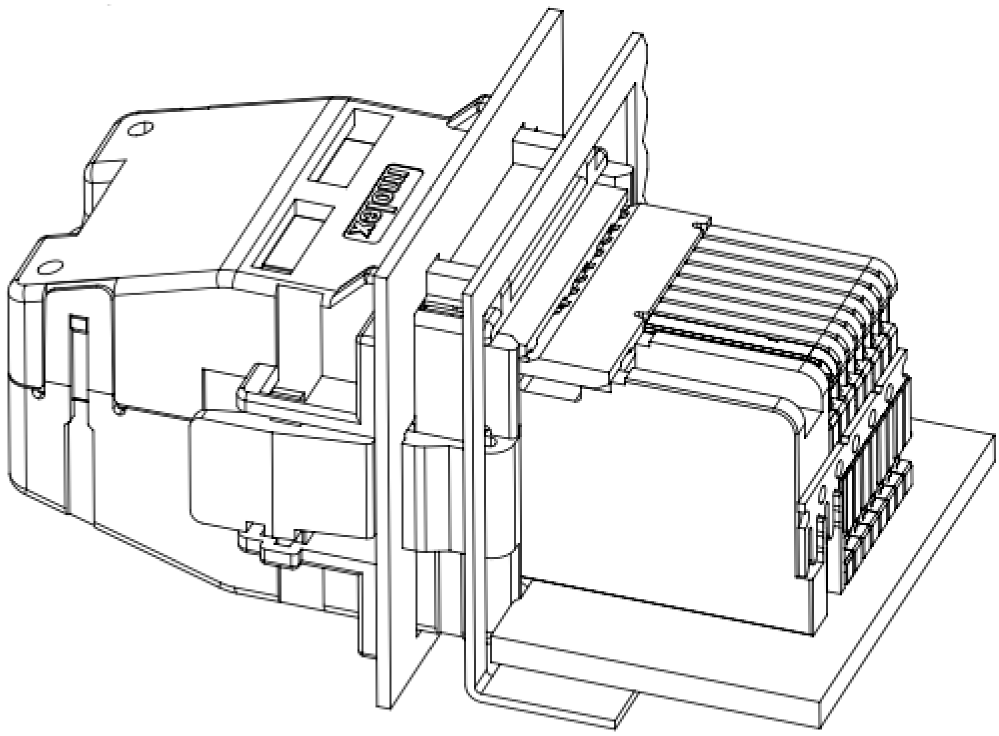


Figure 51. Impel Mated connector with cable backshell

The data connector pinout is defined in the [Appendix]

There is no attempt to support 50G (2x25G NRZ) nor 50G (1x50G NRZ).

4.4.5. Ethernet interface inter-operability

The inter-operability of bricks at deployment with Ethernet interfaces is critical. To support 25G NRZ, 50G PAM4, 100G (4x25G NRZ), 200G (4x50G PAM4), and 400G (8x50G PAM4), the following table:

Port Count	Data Rate	Implementation Interface and order
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5		SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1
6		SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1, QSFP2 LN 1
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5	100G (4x25G) 200G (4x50G PAM4)	SFP1, SPF2, SFP3, SPF4, QSFP1
6		SFP1, SPF2, SFP3, SPF4, QSFP1, QSFP2
1	25G NRZ 50G PAM4	SFP1
2		SFP1, SFP2
3		SFP1, SFP2, SFP3
4		SFP1, SPF2, SFP3, SPF4
5	400G (8x50G PAM4)	QSFP-DD (QSFP1+QSFP2) 200G (4x50G PAM4)
QSFP (SFP1+SFP2+SFP3+ SFP4)	2	QSFP, QSFP1

4.5. Liquid cooling

4.5.1. Brick liquid cooling

Brick liquid cooling is divided into two broad categories, single-phase and two phase. They are mutually incompatible and the designs are mechanically incompatible by design. All bricks inside a single cage are expected to be the same liquid cooling technology.

Unified cutout

Open19 V2 liquid cooling implements a unified liquid cooling cutout. This allows for a consistent brick design to implement both liquid cooling solutions. Implementing the unified cutout is not required, provided all the requirements of a single solution are met.

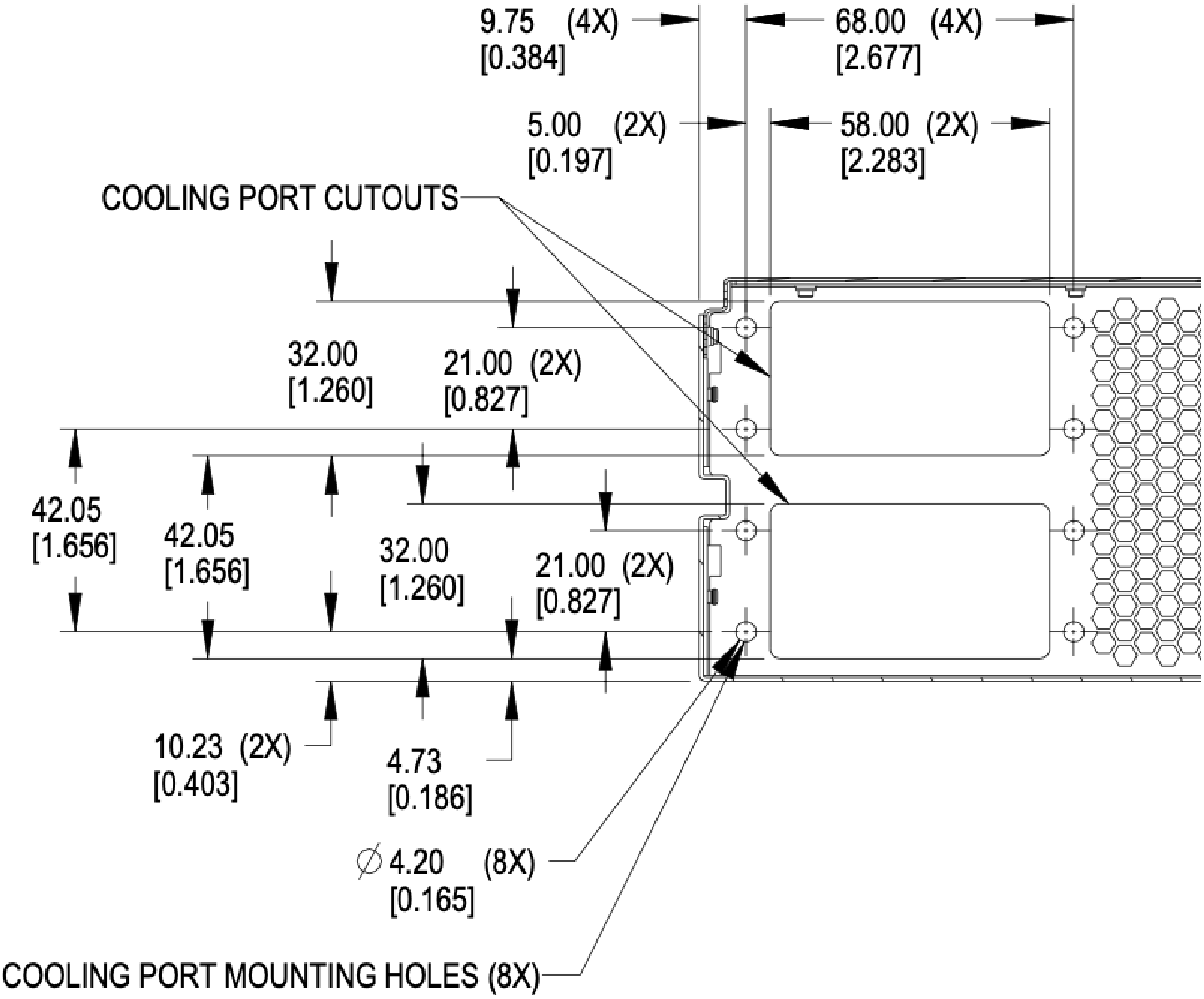


Figure 52. Brick unified cooling cutout

4.5.2. Single phase

Single-phase Connector selection

Single-phase cooling solutions shall use the CPC BLQ4 family of interfaces.

Figure 53. BLQ4 body for manifold

Figure 54. BLQ4 insert for bricks

BLQ4_SpecSheet.pdf

The BLQ4 connectors are blind mate. Both sides of the connector have G 3/8" threads which thread into manifold (body) and brick (insert), respectively.

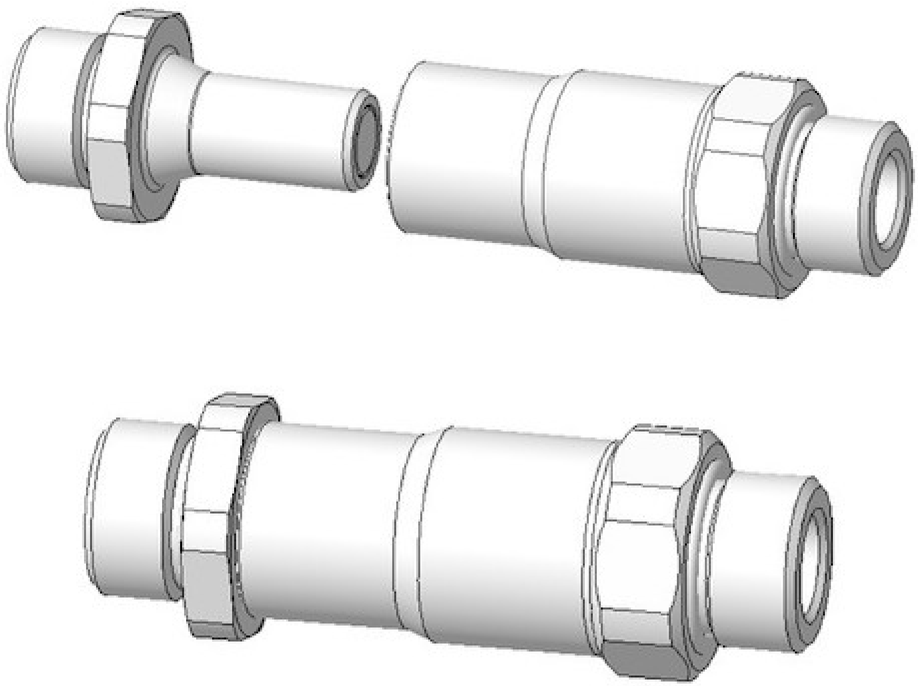


Figure 55. BLQ4 pair

Single-phase connector mounting and polarization

Bricks shall have male blind-mate connectors (inserts), BLQ4D47006 or compatible part.

When viewed from the rear of the brick, in reference to the server: Liquid return shall be located to the left. Liquid supply shall be located in the to the right.

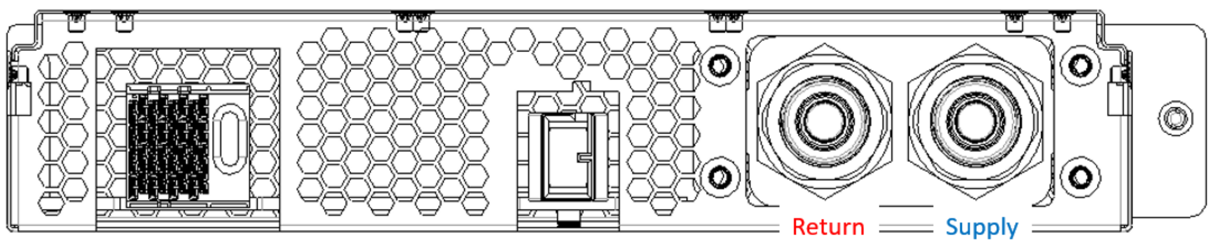


Figure 56. Brick side supply and return

Single-phase brick side connector locations

The supply connector should be 29.5mm from the outer side of the brick (opposite the thumbscrew) in the x-direction and 20.73mm from the bottom of face of the brick in the y-direction. THe two

connectors should be lined up in the horizontal plane and spaced 28.5mm from each other.

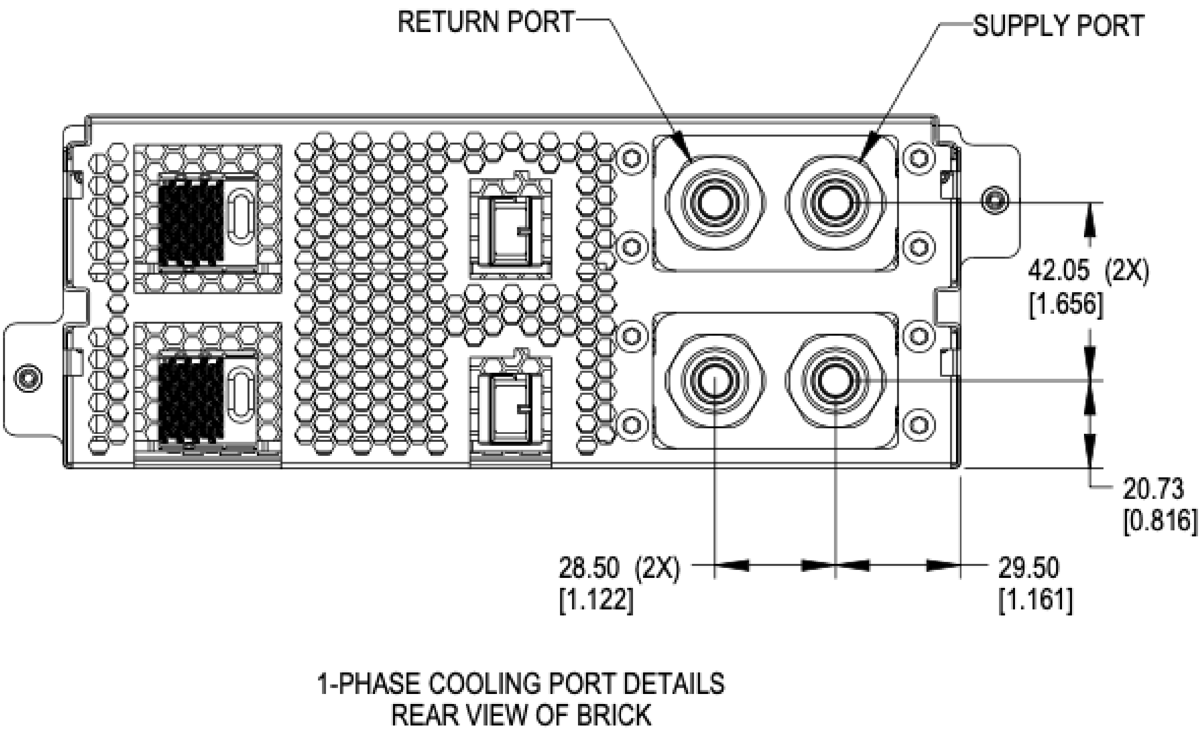


Figure 57. Brick side liquid connector locations for single-phase

Single-phase offset depth

The mating surface of the brick side connector sits xx mm behind (outside) the outer edge of the back plane of the brick. Specifying this dimension ensures the server manufacturer and manifold/cage side implementor have a consistent location where the connectors must meet to ensure full mating and thereby full flow.

The reference mating surfaces of each connector are highlighted below along with a drawing showing the specified dimension.

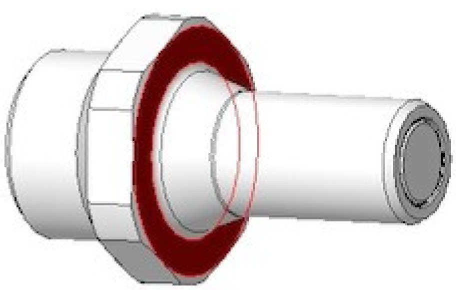


Figure 58. Mating surface BLQ4 insert

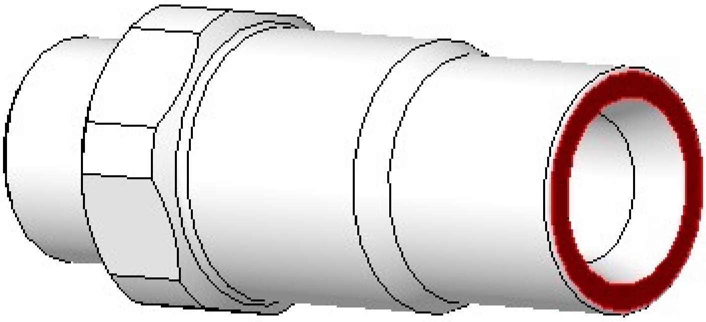


Figure 59. Mating surface BLQ4 body

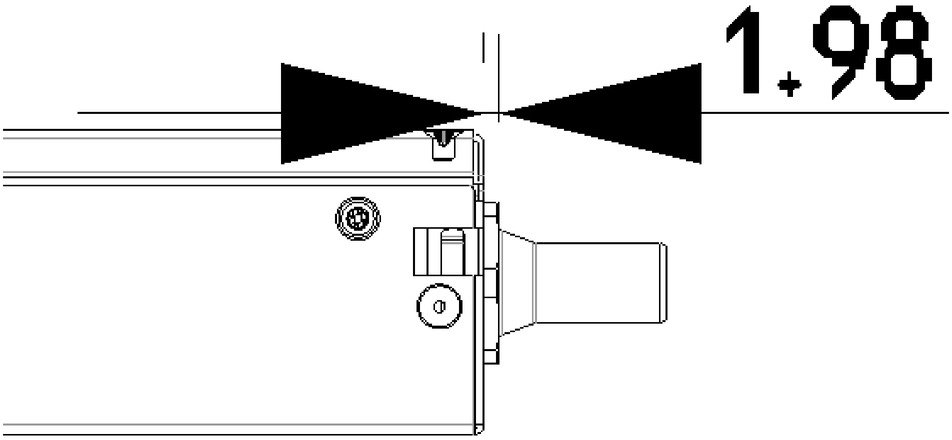


Figure 60. Offset from cage back panel to connector mating interface

The BLQ4 connectors must be within 3mm of this fully engaged state to achieve full flow.

4.5.3. Two phase

Two-phase connector selection

The two-phase cooling solution shall use the CPC BLQ family of interfaces.

On brick side

Figure 61. Liquid - CPC BLQ4 series insert P/N BLQ4D47004PROTO (chrome plated brass) or P/N 4649800PROTO (Aluminum)

Figure 62. Vapor - CPC BLQ6 series insert P/N 4650000PROTO (chrome plated brass)

On the Manifold side

Figure 63. Liquid - CPC BLQ4 series body P/N BLQ4D31004PROTO (chrome plated brass) or P/N 4649900PROTO (Aluminum)

Figure 64. Vapor - CPC BLQ6 series body P/N 4650200PROTO (chrome plated brass)

Two-phase connector mounting polarization

Cage manifolds shall have body (female) blind mate connectors. Bricks shall have insert (male) blind-mate connectors. When viewed from rear of the brick, vapor connection shall be located at the right position. Liquid connection shall be located at the left position.

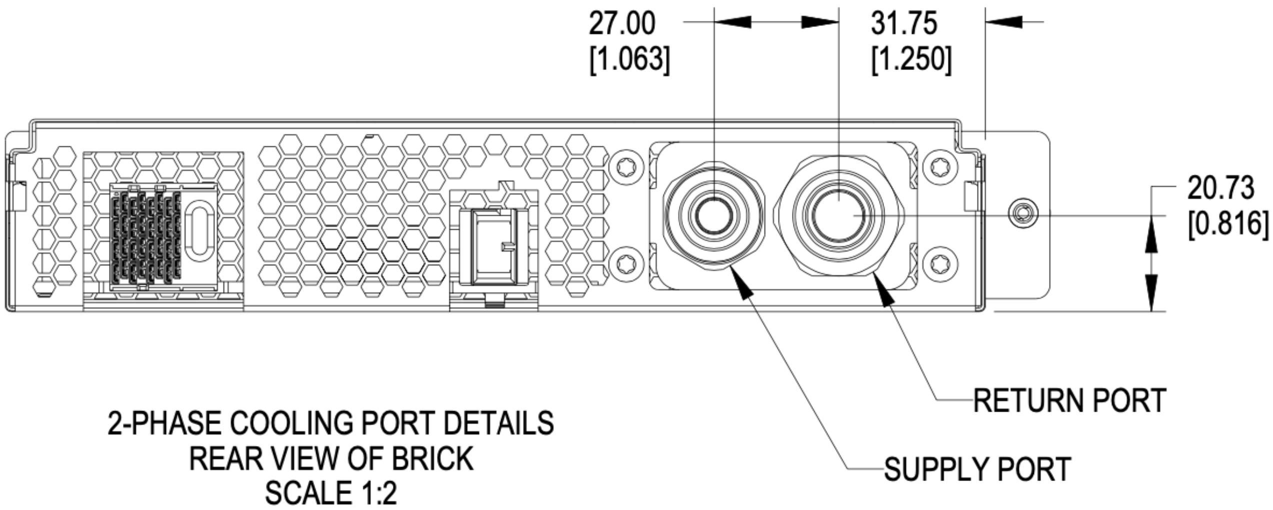


Figure 65. Brick liquid supply and Vapor return

Two-phase brick side connector locations

Using the right-side panel of the brick as a reference, the Vapor return connector on the brick should be 31.85mm in the x direction and 20.73 upwards in the y direction.

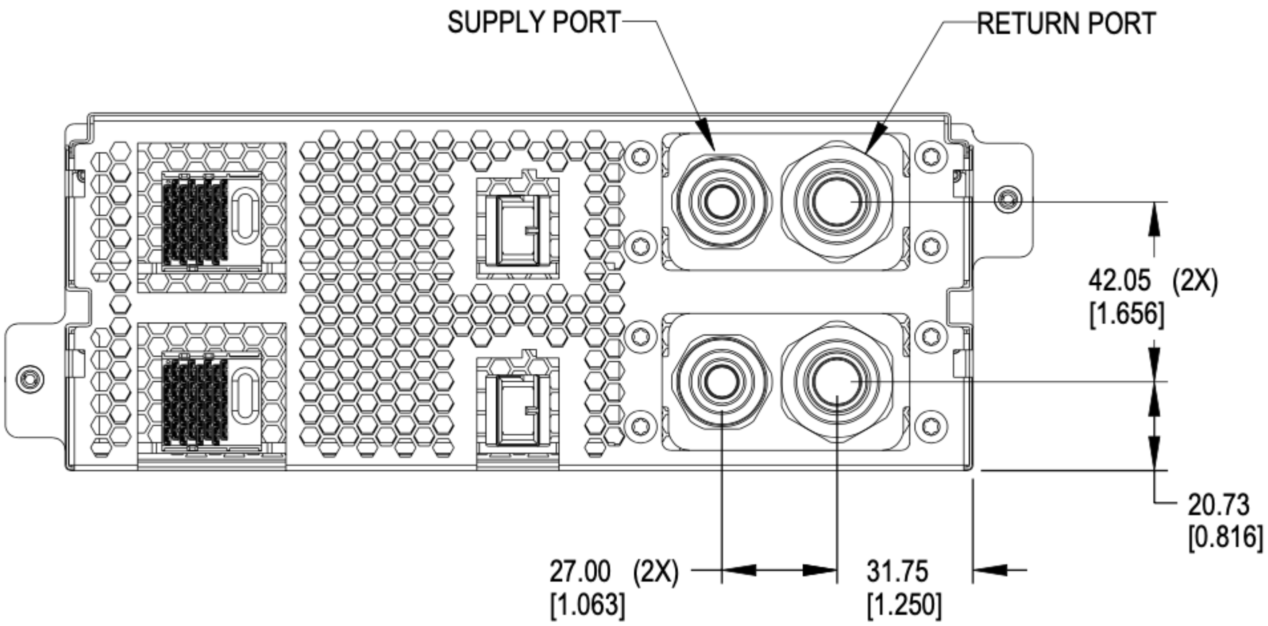


Figure 66. Brick side connectors locations

Two-phase offset depth

The mating surface of the brick side connector sits xx mm behind (outside) the outer edge of the back plane of the brick. Specifying this dimension ensures the server manufacturer and manifold/cage side implementor have a consistent location where the connectors must meet to ensure full mating and thereby full flow.

The reference mating surfaces of each connector are highlighted below along with a drawing showing the specified dimension.

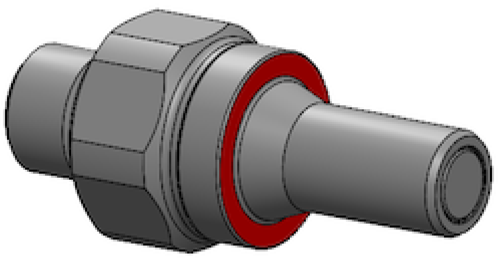


Figure 67. Mating surface BLQ4 insert

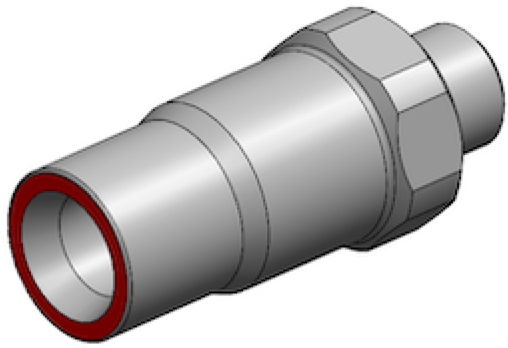


Figure 68. Mating surface BLQ4 body

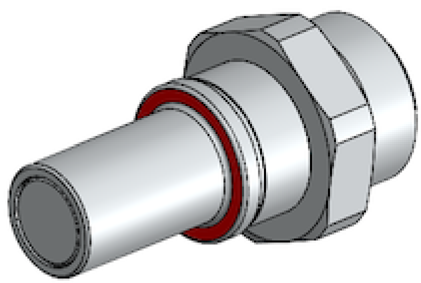


Figure 69. Mating surface BLQ6 insert

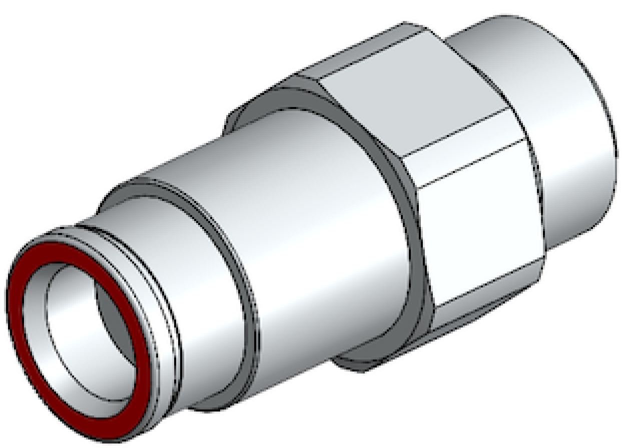


Figure 70. Mating surface BLQ6 body

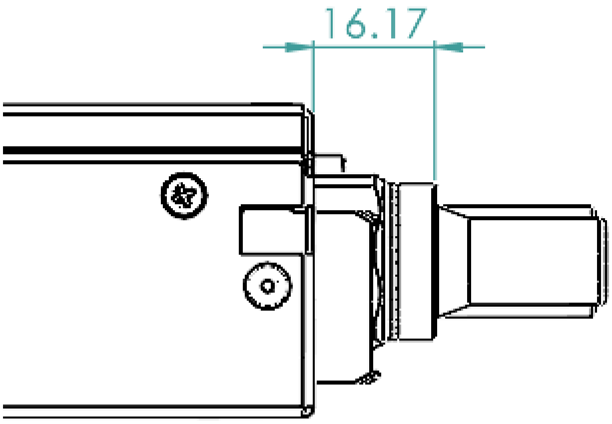


Figure 71. Offset from cage back panel to BLQ4 connector mating interface

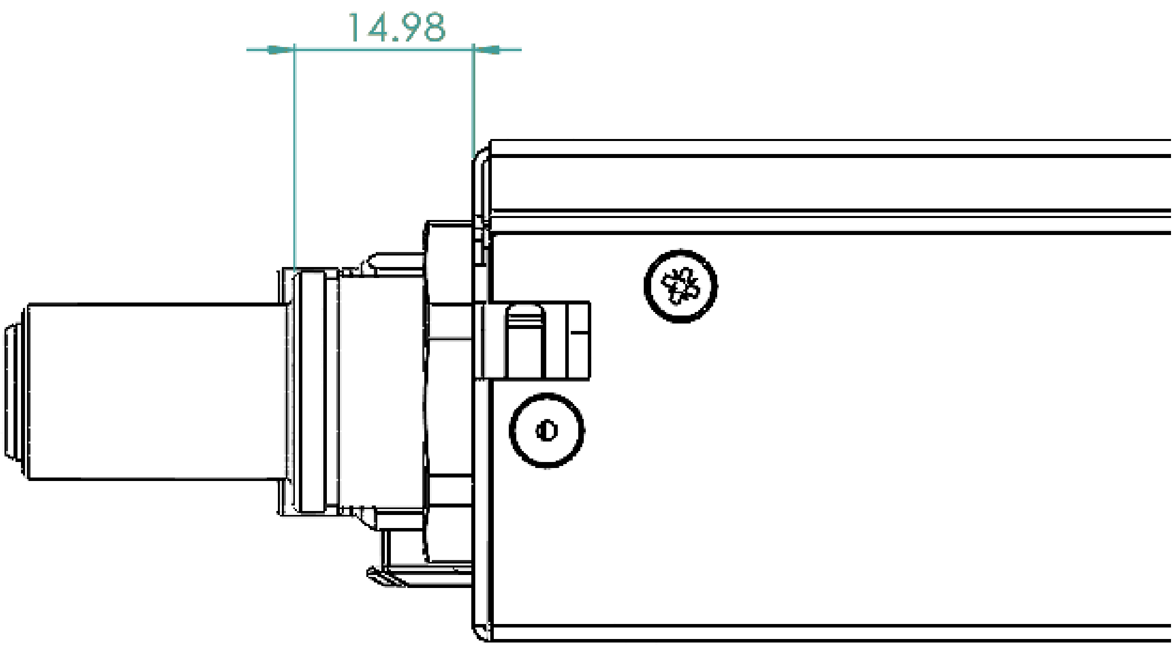


Figure 72. Offset from cage back panel to BLQ6 connector mating interface

The BLQ4 & BLQ6 connectors must be within ±1.5mm of this engaged state to achieve full flow.

4.5.4. Weight

The maximum weight of the brick, when equipped with all drives and/or accessories, is 60lbs (13.6kg). This is an increase of 10lbs from V1.

4.5.5. Design guidance

Design guidance is not a specification requirement. It is simple design guidance that can be implemented to simplify future needs, to meet certification requirements, or satisfy general fit & finish.

4.5.6. Brick faceplate guidance

The faceplate is notched above and below the fastener (See Figure 40, “Brick retention”). Note that the faceplate can accommodate any type of ports or drives as needed and is not required to be fully perforated.

Screws and/or rivets used to secure the top cover are flush to be under-flush. The vertical edges on the rear of the brick chassis base are coined to break sharp edges.

4.5.7. Back panel safety perforation

See below for typical dimensions of the airflow perforation. All dimensions below are critical.

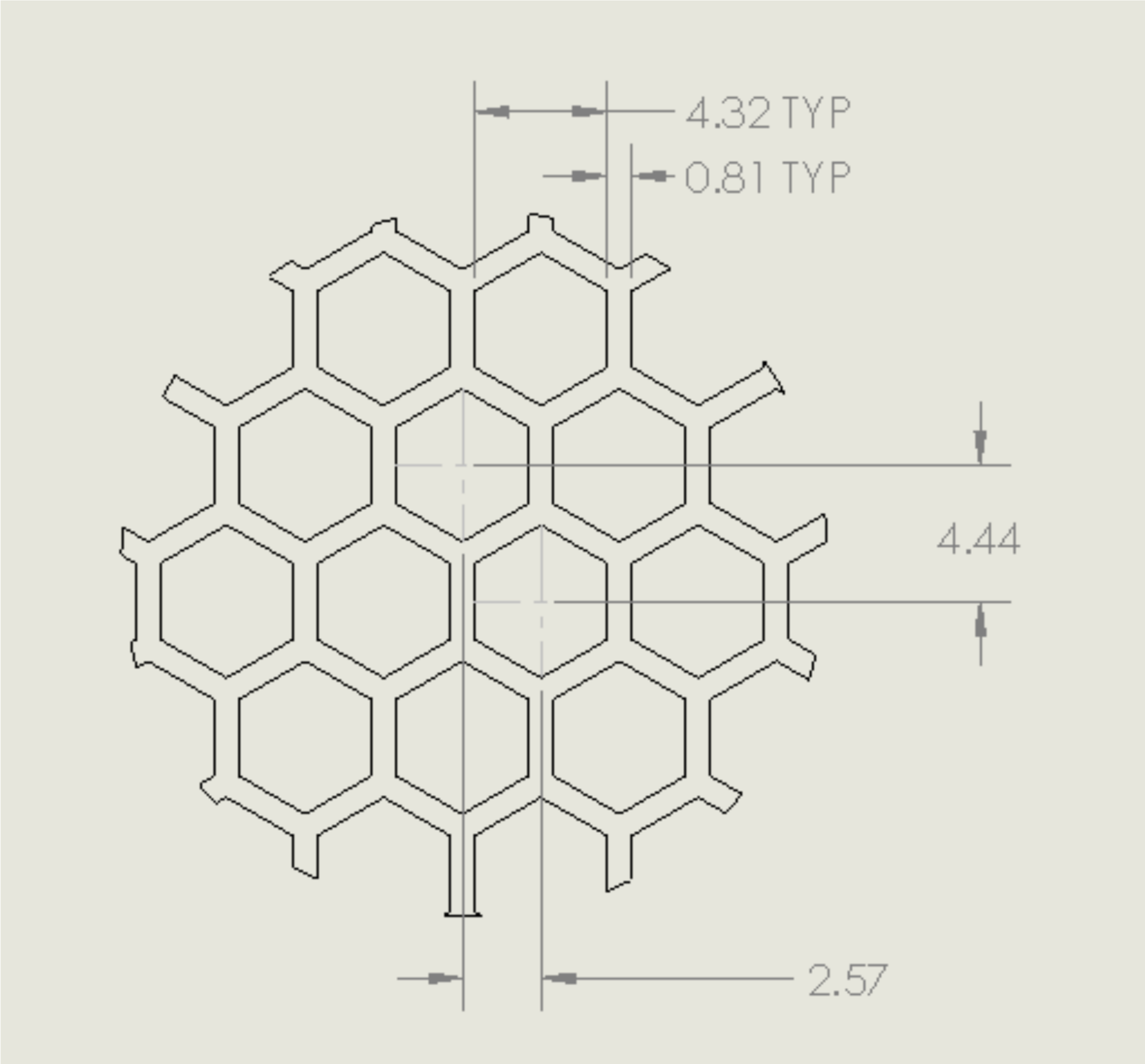


Figure 73. Brick panel perforation

Keep-out zones

Keep-out zones are defined in the brick surrounding the power connector and liquid cooling interfaces. The liquid cooling keep-out spaces "fly" above the PCB, allowing components to be populated below the connections.

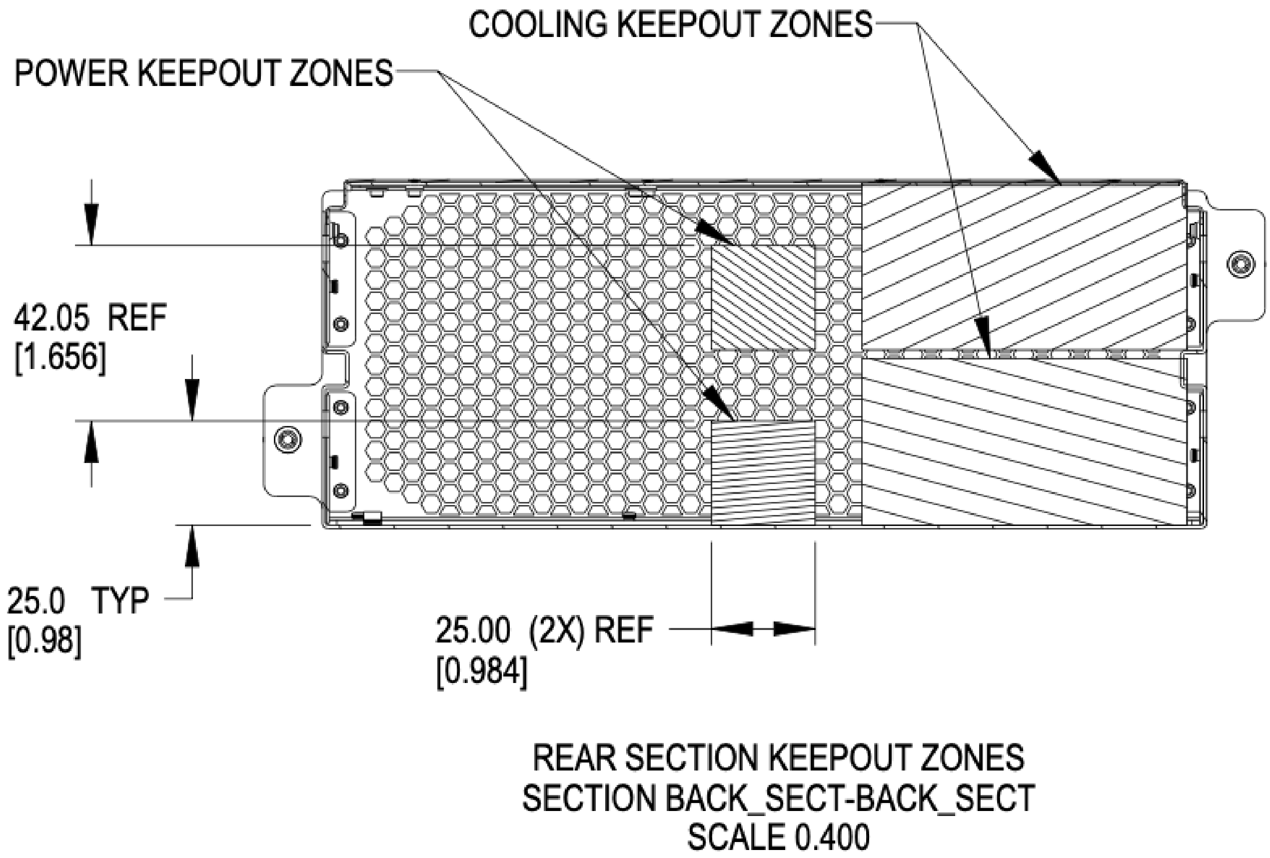


Figure 74. Brick keep-out zone

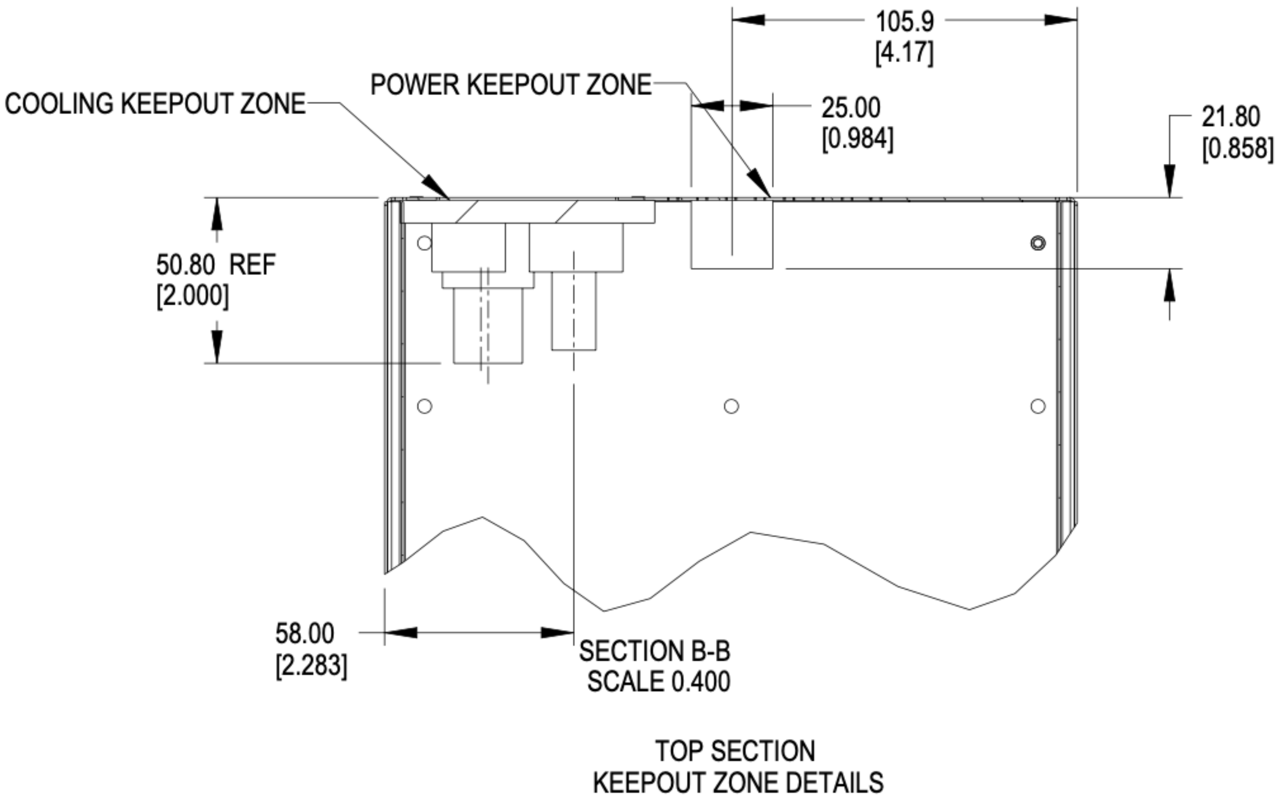


Figure 75. Brick keep-out zone top view

4.6. Brick performance

4.6.1. Brick electrical startup behavior

In order to safely operate the power shelf, we define the following characteristics for brick startup behavior.

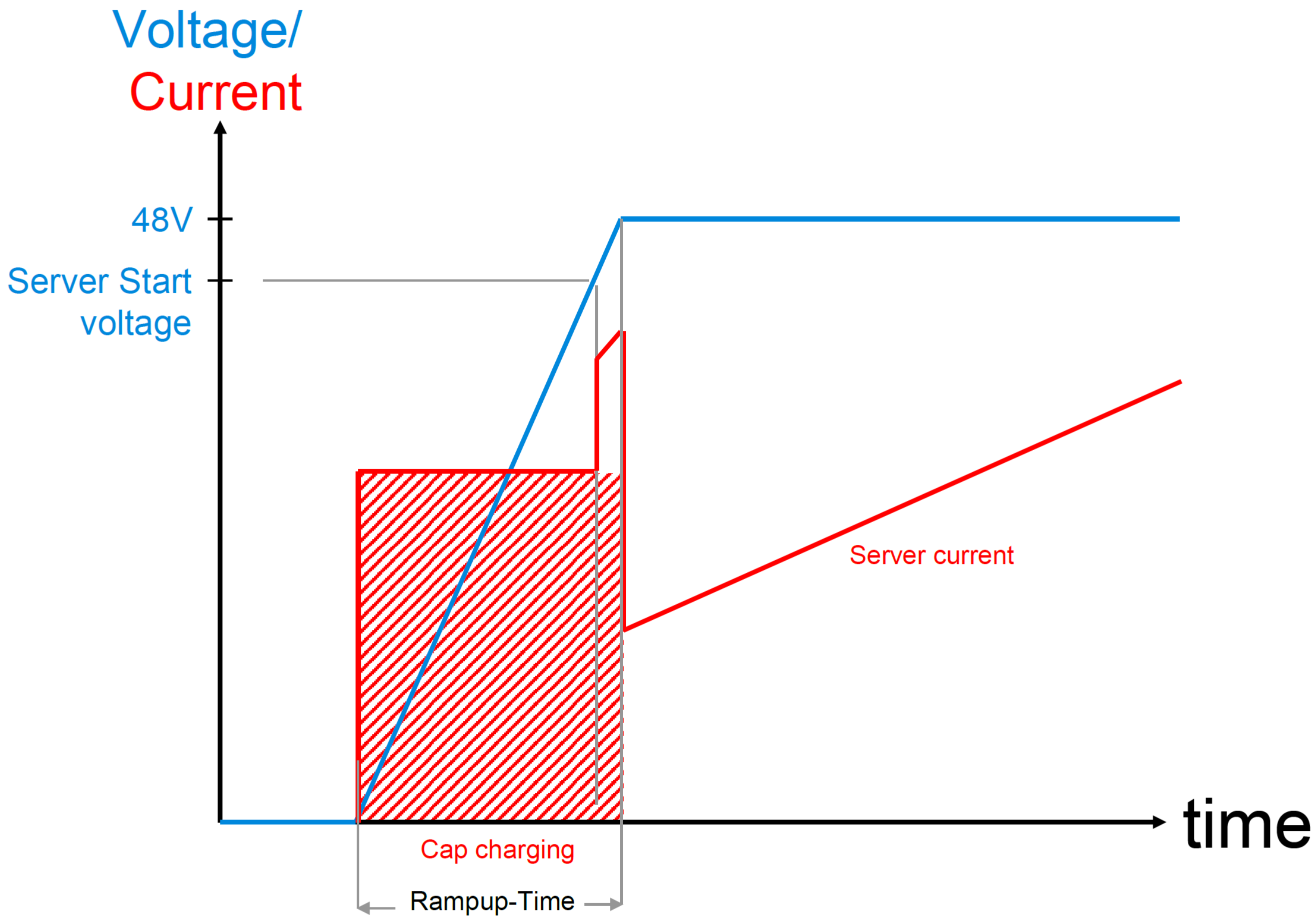


Figure 76. Server startup electrical characteristics

These values are per connector feed from a power shelf.

Two power connectors are available for use.

- Min start-up voltage: +44V DC
- Max input capacitance per brick feed: ⇐1000uF
- Max server current at start-up voltage: 70A
- Ramp time: Follow Intel CRPS spec: ramp time 10-70msec, slew rate 0.5V/msec

https://www.intel.com/content/dam/www/public/us/en/documents/guides/power-supply-design-guide-june.pdf

Chapter 5. Brick Mechanical Specification [SHDW]

5.1. SHDW Brick

The SHDW brick is commonly used as a 1RU server form-factor.

5.2. Brick Dimensions

The brick is designed with the overall dimensions shown.

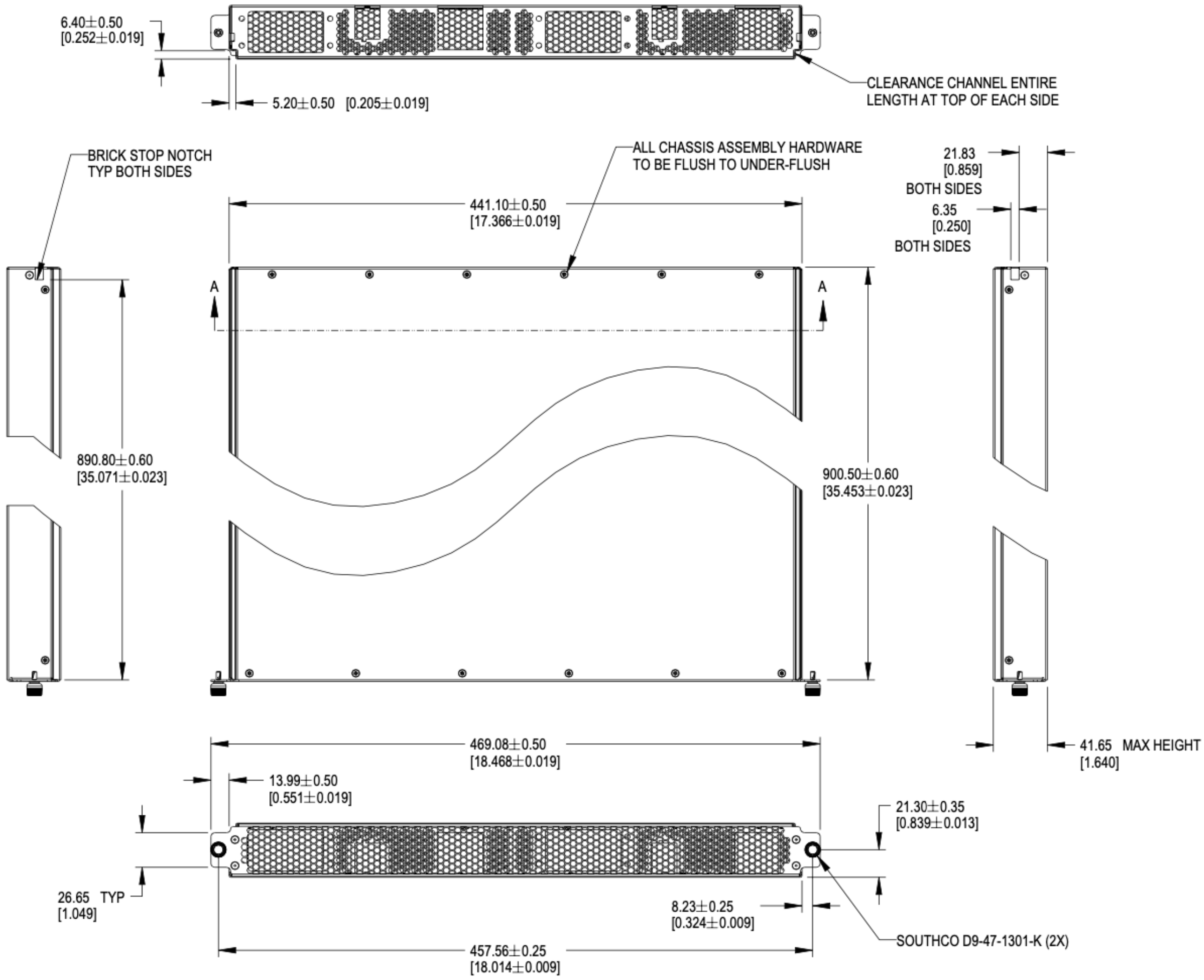


Figure 77. Brick dimensions

Please refer to the drawings, and 3D files for further details.

The V1 specification for bricks defined 901.7mm as the overall length of the brick, including the thickness of the front panel.

The dimension is modified to reference to the inside of the front panel to the rear of the brick. This dimension is 900.50mm +/-0.60 from the rear face of the front panel (mating surface to the cage) to the outside of the rear panel.

5.3. Mechanical features

5.3.1. Brick retention

The brick is secured into the cage with two Southco D9-47-1302-K quarter-turn fastener, located on the left and right side of the faceplate.

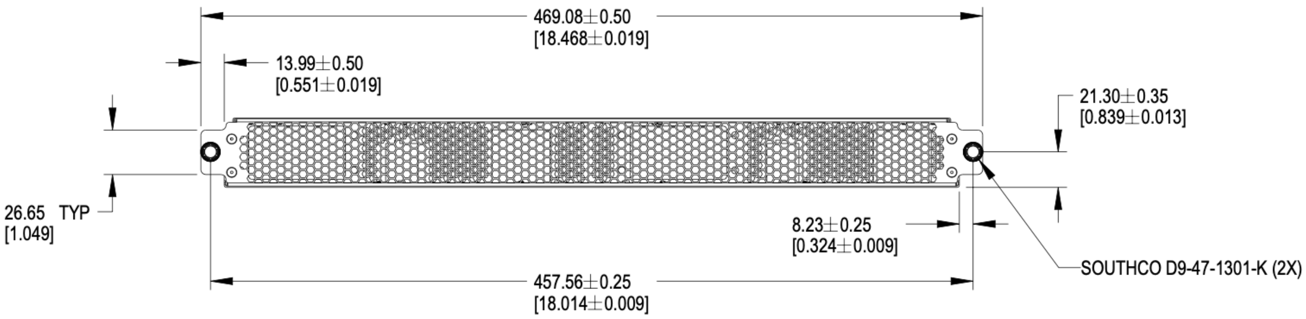


Figure 78. Brick retention

5.3.2. Brick mechanical guide

Clearance channels are required along the top edges of the brick, extending forward from the rear panel to the back of the faceplate.

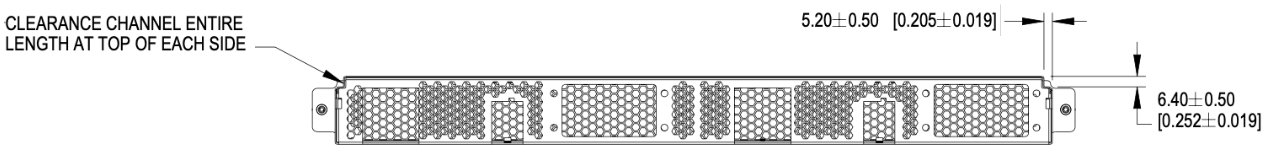


Figure 79. Mechanical guide at top edge of brick

Brick insertion and retention assistance for liquid cooling

For liquid cooling applications, two different mechanical assist for insertion and retention are available in the cage. Bricks can implement features to make use either/both features.

The features are:

- Brick retention standoff, available on both sides in the mechanical guide channel
- Slots, available on both sides of the cage wall

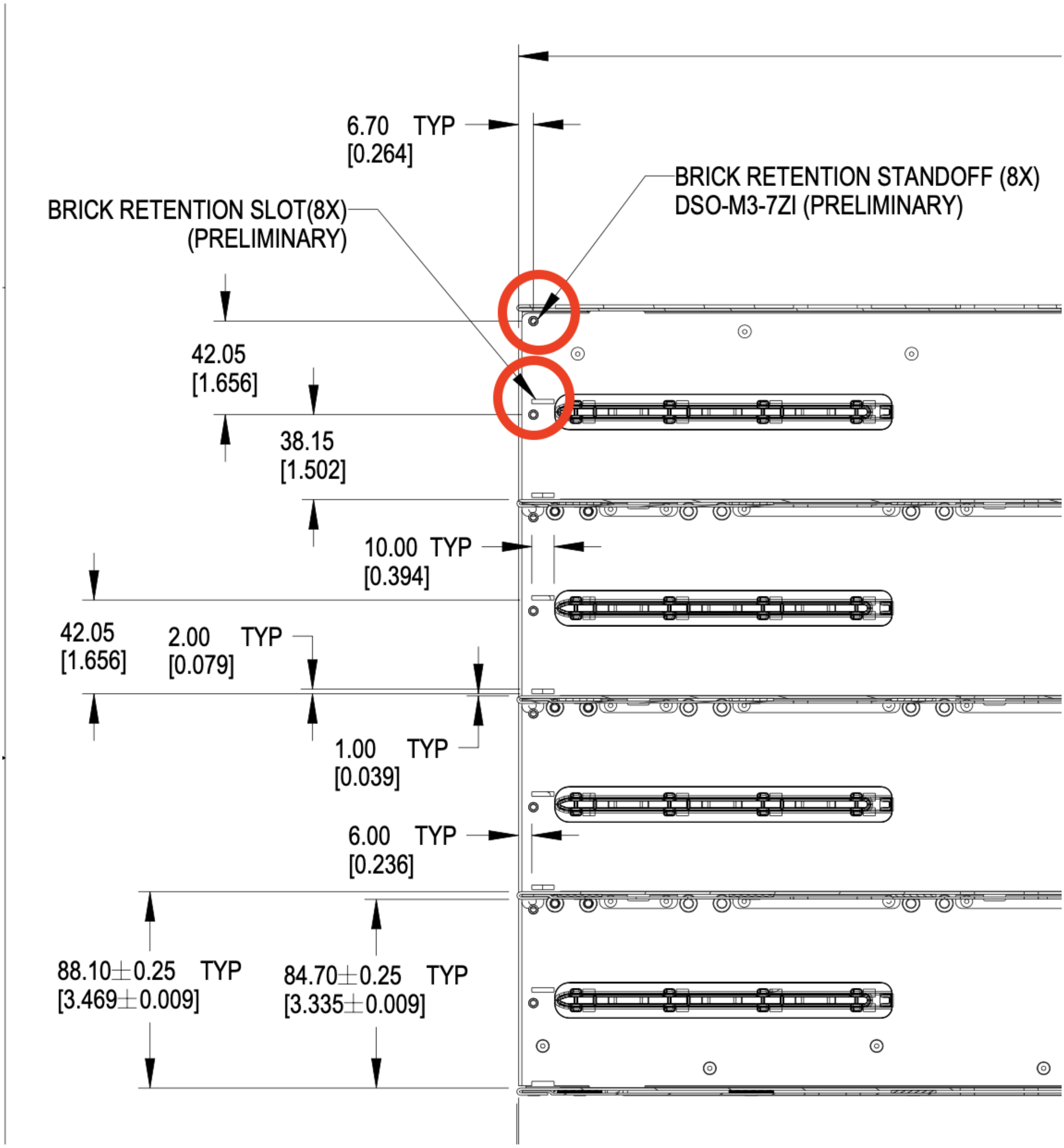


Figure 80. Brick mechanical assistance features in cage

Each feature shall support 100 lbf (445N) force or greater, using the recommended 1.3mm SGCC material.

5.3.3. Brick Mechanical Keying

Two keying notches are positioned in the left and right corners of the sheet metal enclosure as shown below. All dimensions below are critical.

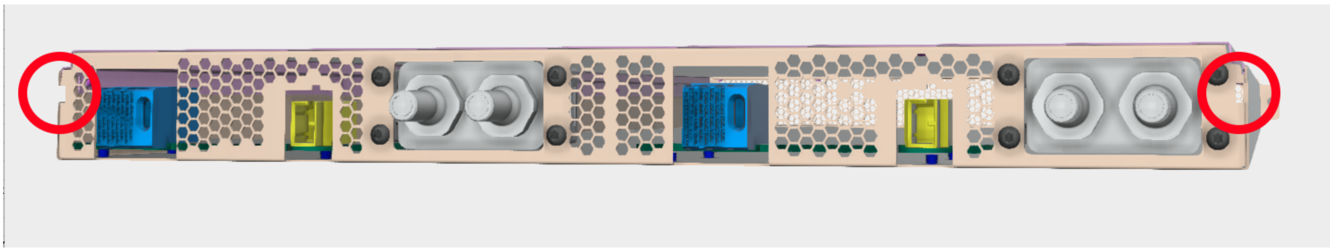


Figure 81. Brick keying

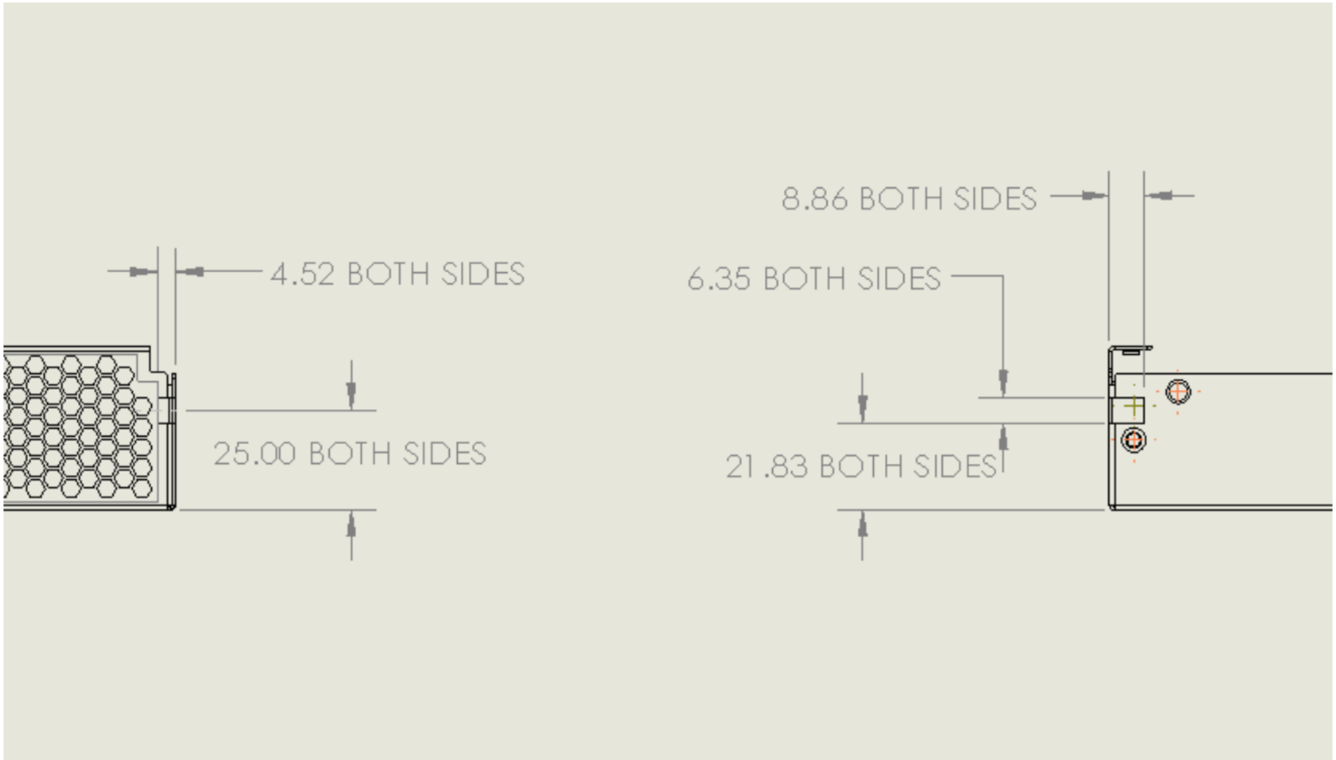


Figure 82. Brick keying dimensions

5.3.4. PCBA

The rear PCBA (Printed Circuit Board Assembly), also referenced as the interposer, provides the +48V DC input power and a data connections. The top of the PCBA is the reference surface, located 6.5mm +/- 0.6mm above the bottom surface of the sheet metal chassis.

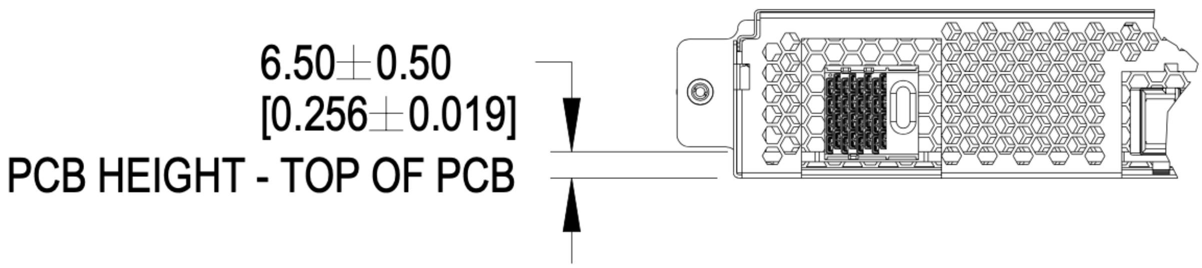


Figure 83. PCBA References

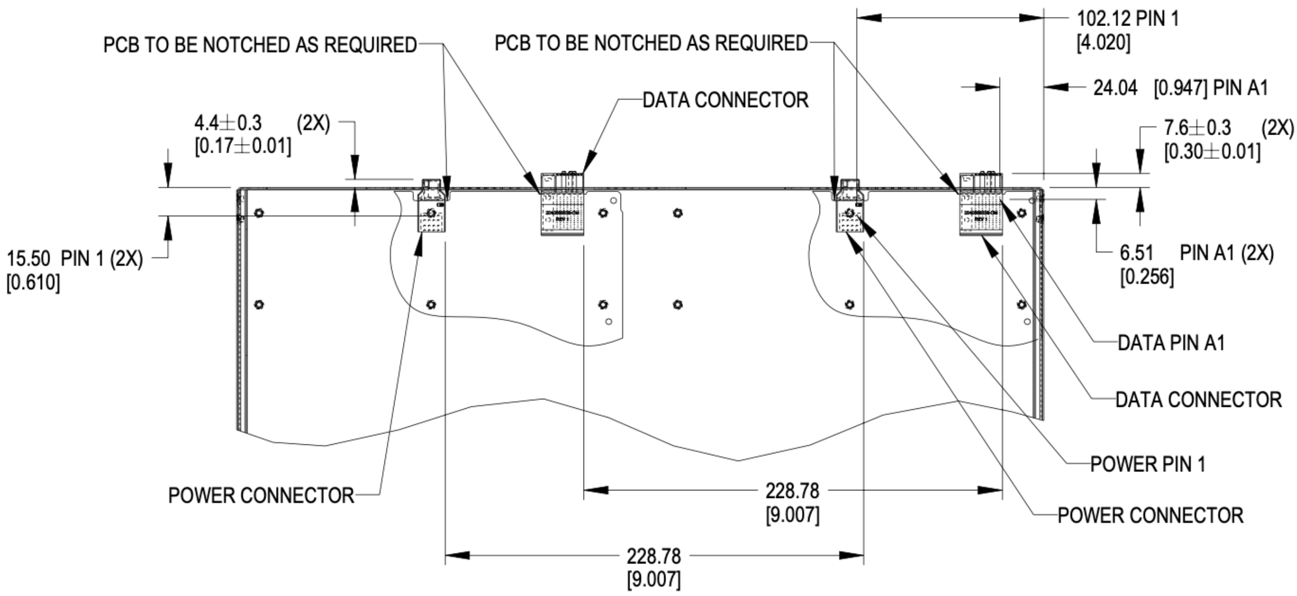


Figure 84. Brick PCB dimensions

5.3.5. Materials

The brick enclosure is constructed of sheet metal with adequate strength and rigidity. Suggested material is 0.8mm SGCC for the chassis base and 0.6mm SGCC for the top cover.

5.3.6. Blind mate interfaces

A brick is equipped with at least one set of blind mate connectors for receiving +48V DC power and transferring data with an external device, typically a network switch. Introduced in this version is are blind mate connections to facilitate liquid cooling. The connections mate with cable and manifold projections from the rear wall of the cage.

- Power connector: Molex Ten60 PN: 220370-0001 (35A) 22037-0000 (70A), refer to Molex document PSD-220370-0000
- Data connector: Molex 4x8 Impel with guide receptacle PN: 204066-9038 (PCB thickness 2.2-4.1mm), 204066-9138 (PCB thickness 1.0-1.9mm), refer to Molex document PSD-204066-9001

The power connector has two ratings:

- Power: Not-To-Exceed (NTE) 70A at +48V DC
- Power: Not-To-Exceed (NTE) 35A at +48V DC

The ratings shall be coordinated with the power shelf. 35A bricks shall be enabled in 70A capable infrastructure. 70A bricks are keyed to prevent unsafe operation in 35A infrastructure.

The brick may populate a second set of connectors, as required.

5.3.7. Weight

The maximum weight of the brick, when equipped with all drives and/or accessories, is 60lbs (13.6kg). This is an increase of 10lbs from V1.

5.4. Connector Pinout

5.4.1. Brick power connector

The brick blind-mate power connector for receiving 48V DC (51V nominal) power from the power shelf. The connector mates with the power cable connector projected from the brick cage wall.

Two variations of the brick connector exist, with different Not-to-exceed (NTE) ratings:

- 35A Power Connector: Molex 2203700001 (NTE 35 AMPS) No Key
- 70A Power Connector: Molex 2203700000 (NTE 70 AMPS) Keyed

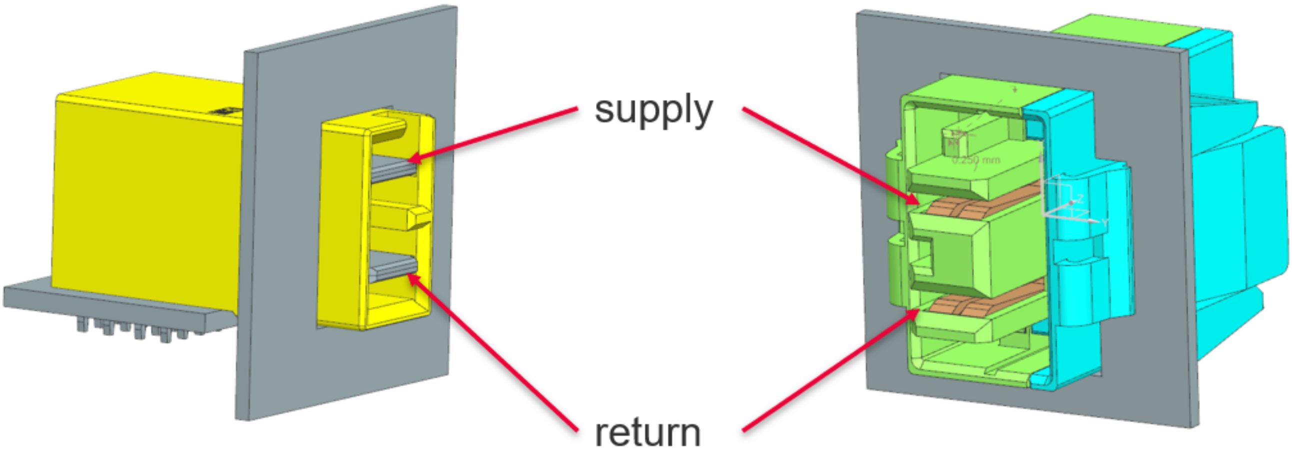


Figure 85. Power connector supply/return definition

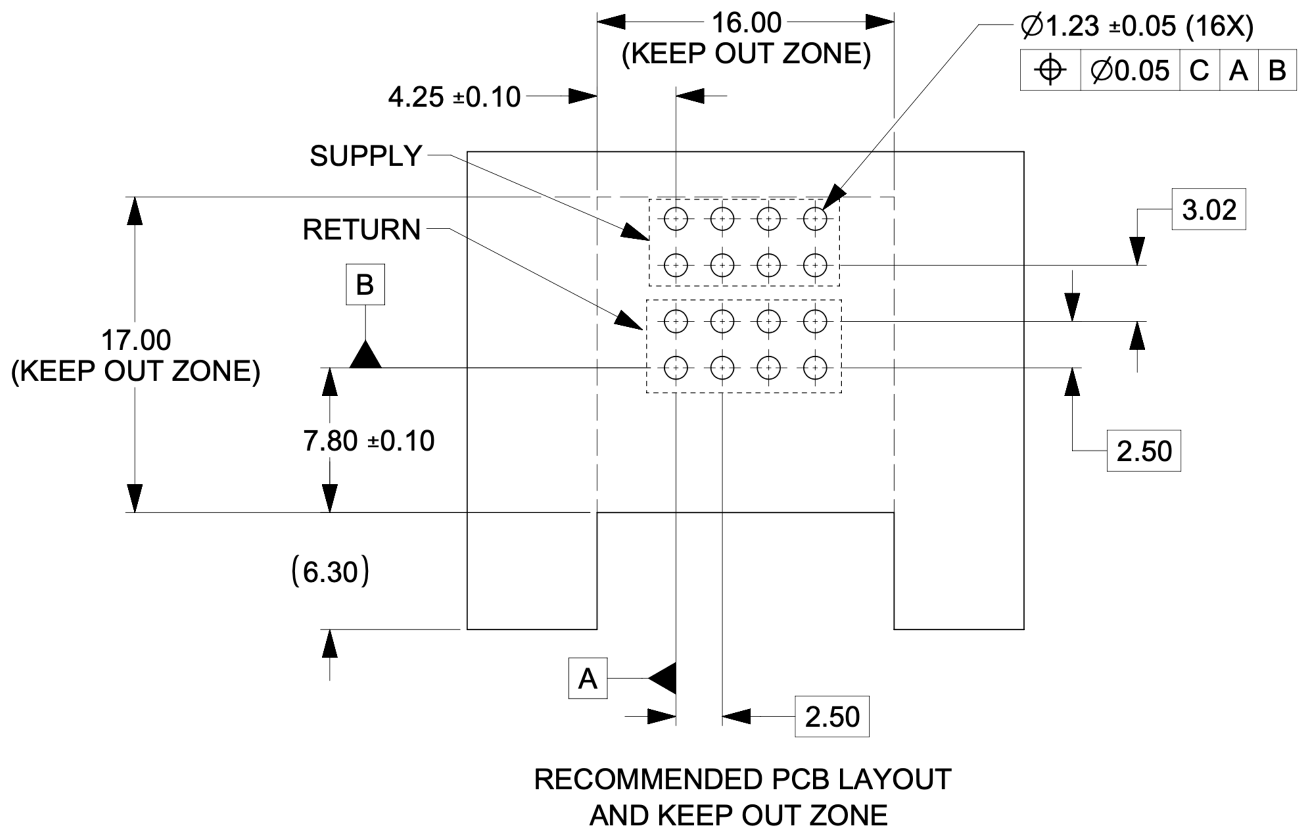


Figure 86. Power connector footprint definition

Connector drawings are available: Molex Brick power connector

5.4.2. Brick critical dimensions

The brick power connector has a critical dimension for the projection from the brick panel. Spacing between panels is 6.23mm. The power connector shall be proud of the panel 4.40mm.

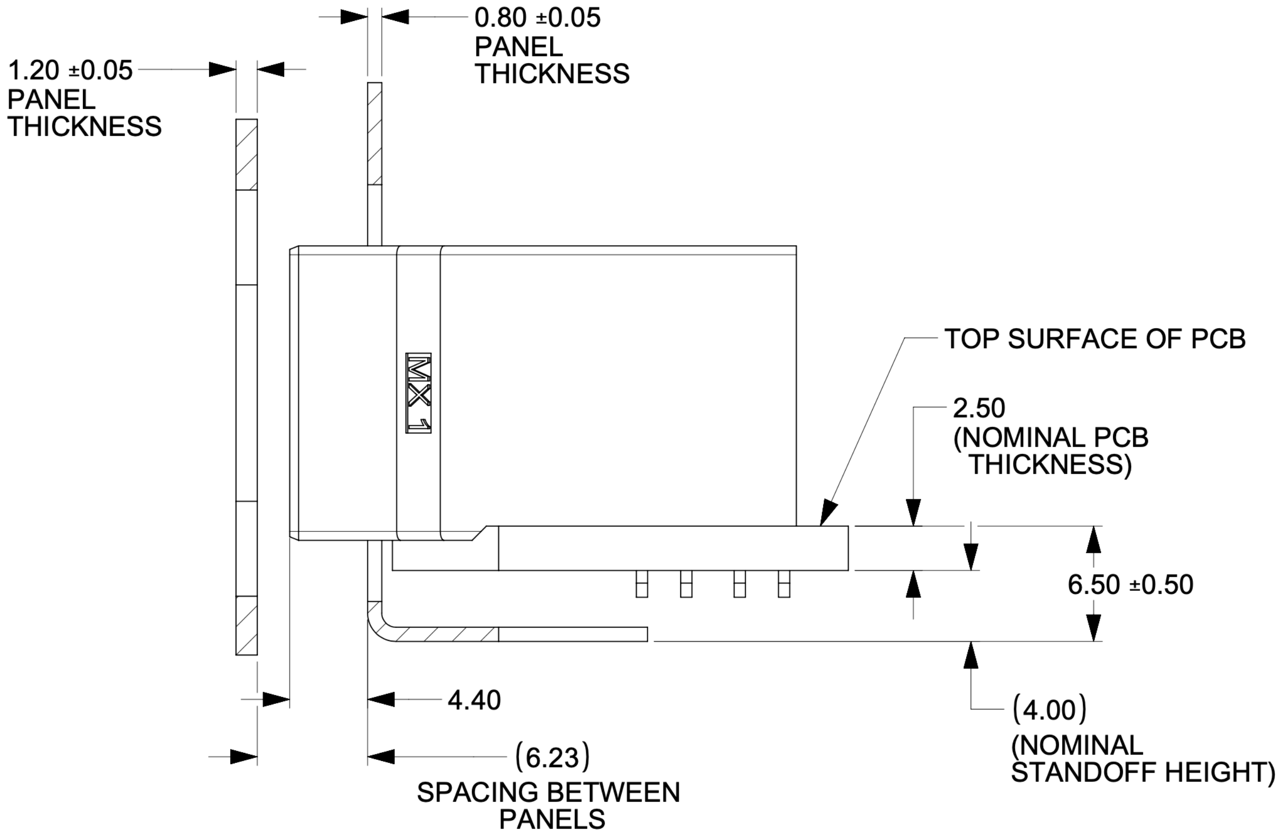


Figure 87. Power connector critical dimensions

5.4.3. Brick voltage

Open19 V2 power shares Open Compute ORV3 power characteristics. Open19 bricks shall safely operate in the DC range 48-52V and shall remain functional when exposed to 0-54V. The power rail shall not exceed 52.5V for more than 200mS. DC voltage shall never exceed 54V.

- Operating range: 48-52V
- NTE voltage: 54V
- Max voltage: ⇐52.5V for <200mS

5.4.4. Data Connector

The data connector is rated for 56G PAM4 with an air dielectric. Variations of cables deliver different lane counts, data rates, and physical interfaces. In the reference Ethernet implementation, the data connector pinout explicitly supports a dedicated 10/100/1000Base-T management port, four 25G NRZ/50G PAM4 lanes. Optionally, a single QSFP-DD or two QSFP interfaces are supported.

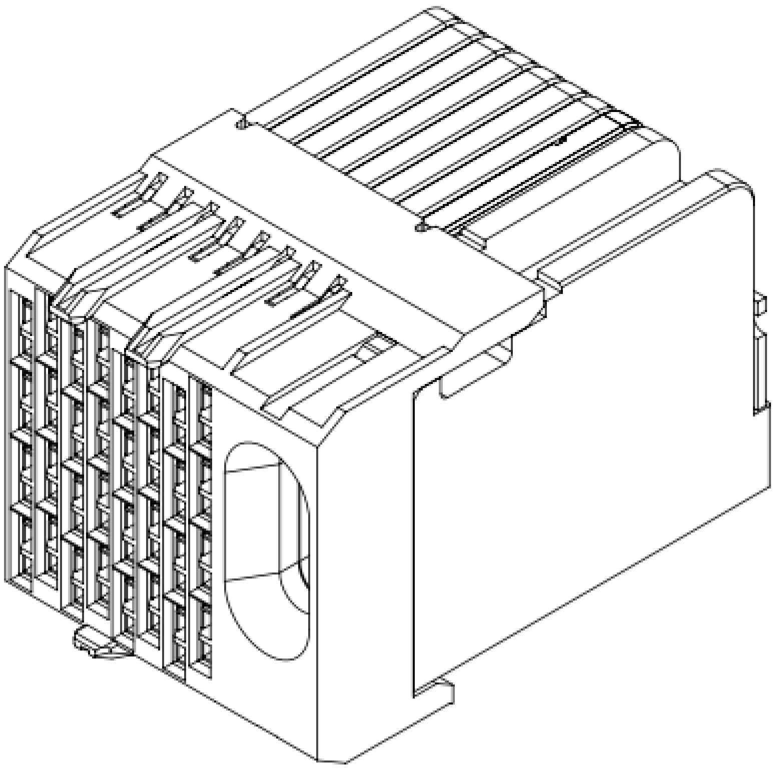


Figure 88. Impel Connector with guide feature

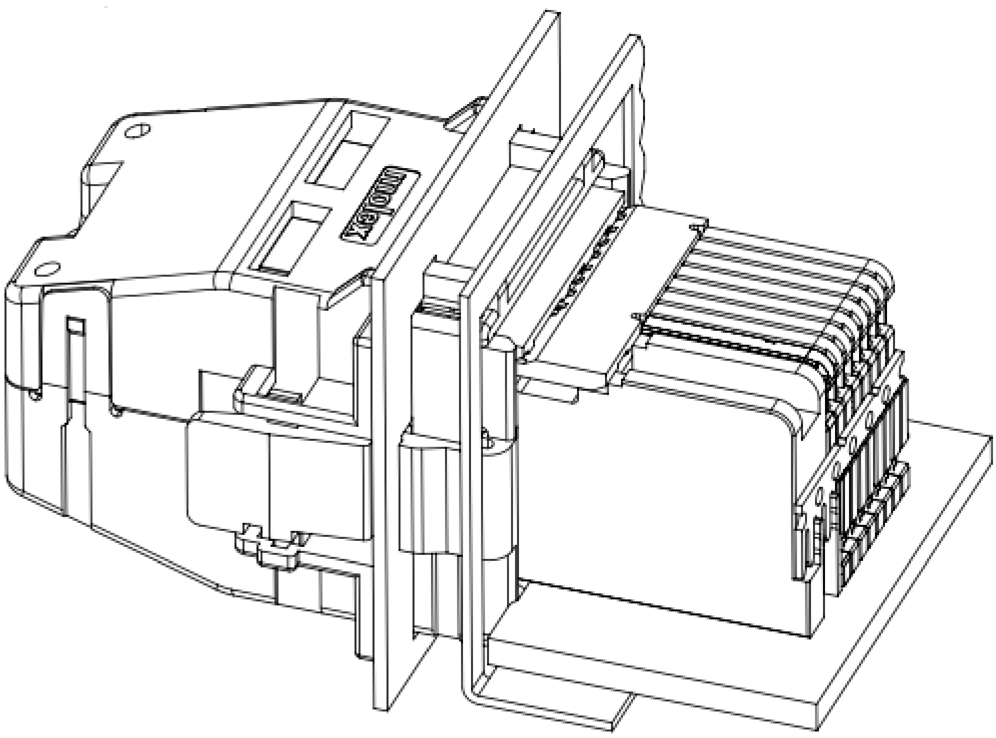


Figure 89. Impel Mated connector with cable backshell

The data connector pinout is defined in the [Appendix]

There is no attempt to support 50G (2x25G NRZ) nor 50G (1x50G NRZ).

5.4.5. Ethernet interface inter-operability

The inter-operability of bricks at deployment with Ethernet interfaces is critical. To support 25G NRZ, 50G PAM4, 100G (4x25G NRZ), 200G (4x50G PAM4), and 400G (8x50G PAM4), the following table:

<table>
<thead>
<tr><th>Port Count</th><th>Data Rate</th><th>Implementation Interface and order</th></tr>
</thead>
<tbody>
<tr><td>1</td><td rowspan="6">25G NRZ

50G PAM4</td><td>SFP1</td></tr>
<tr><td>2</td><td>SFP1, SFP2</td></tr>
<tr><td>3</td><td>SFP1, SFP2, SFP3</td></tr>
<tr><td>4</td><td>SFP1, SPF2, SFP3, SPF4</td></tr>
<tr><td>5</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1</td></tr>
<tr><td>6</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1, QSFP2 LN 1</td></tr>
<tr><td>1</td><td rowspan="4">25G NRZ

50G PAM4</td><td>SFP1</td></tr>
<tr><td>2</td><td>SFP1, SFP2</td></tr>
<tr><td>3</td><td>SFP1, SFP2, SFP3</td></tr>
<tr><td>4</td><td>SFP1, SPF2, SFP3, SPF4</td></tr>
<tr><td>5</td><td rowspan="2">100G (4x25G)

200G (4x50G PAM4)</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1</td></tr>
<tr><td>6</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1, QSFP2</td></tr>
<tr><td>1</td><td rowspan="4">25G NRZ

50G PAM4</td><td>SFP1</td></tr>
<tr><td>2</td><td>SFP1, SFP2</td></tr>
<tr><td>3</td><td>SFP1, SFP2, SFP3</td></tr>
<tr><td>4</td><td>SFP1, SPF2, SFP3, SPF4</td></tr>
<tr><td>5</td><td>400G (8x50G PAM4)</td><td>QSFP-DD (QSFP1+QSFP2)

200G (4x50G PAM4)</td></tr>
<tr><td>QSFP (SFP1+SFP2+SFP3+SFP4)</td><td>2</td><td>QSFP, QSFP1</td></tr>
</tbody>
</table>

5.5. Liquid cooling

5.5.1. Brick liquid cooling

Brick liquid cooling is divided into two broad categories, single-phase and two phase. They are mutually incompatible and the designs are mechanically incompatible by design. All bricks inside a single cage are expected to be the same liquid cooling technology.

Unified cutout

Open19 V2 liquid cooling implements a unified liquid cooling cutout. This allows for a consistent brick design to implement both liquid cooling solutions. Implementing the unified cutout is not required, provided all the requirements of a single solution are met.

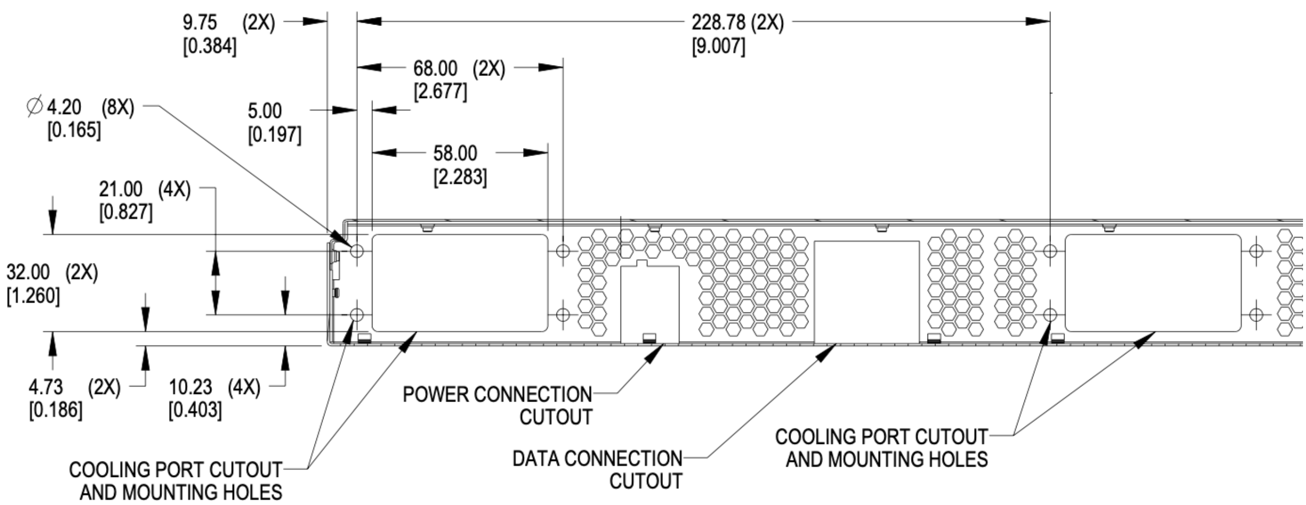


Figure 90. Brick unified cooling cutout

5.5.2. Single phase

Single-phase Connector selection

Single-phase cooling solutions shall use the CPC BLQ4 family of interfaces.

Figure 91. BLQ4 body for manifold

Figure 92. BLQ4 insert for bricks

BLQ4_SpecSheet.pdf

The BLQ4 connectors are blind mate. Both sides of the connector have G 3/8" threads which thread into manifold (body) and brick (insert), respectively.

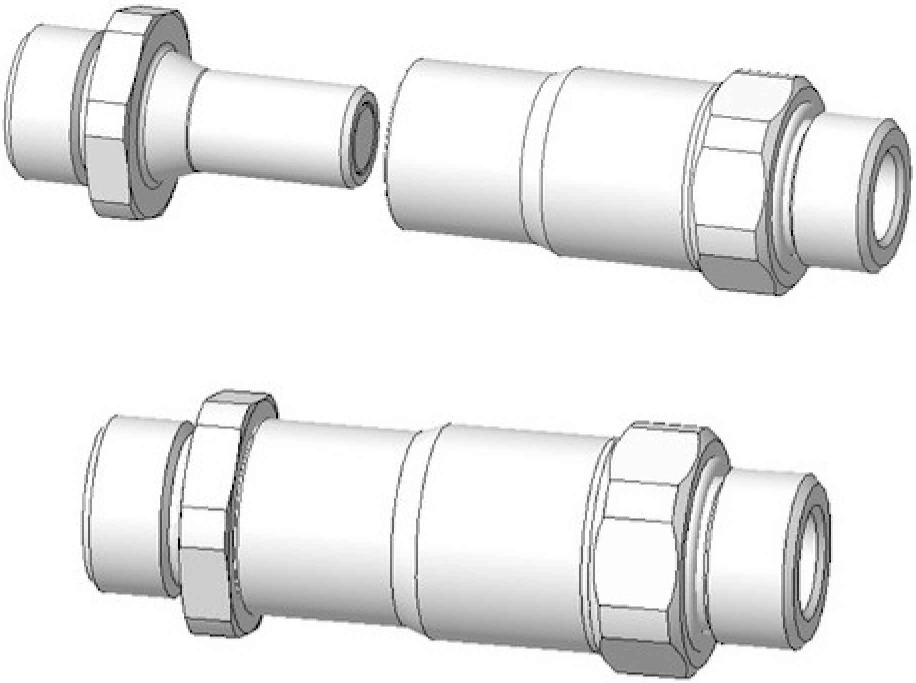


Figure 93. BLQ4 pair

Single-phase connector mounting and polarization

Bricks shall have male blind-mate connectors (inserts), BLQ4D47006 or compatible part.

When viewed from the rear of the brick, in reference to the server: Liquid return shall be located to the left. Liquid supply shall be located in the to the right.

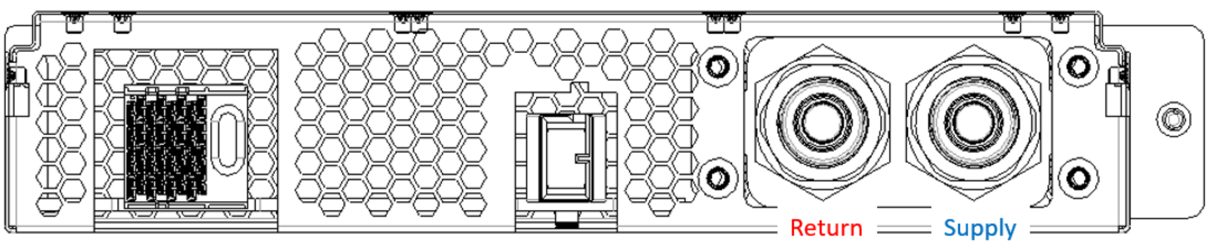


Figure 94. Brick side supply and return

Single-phase brick side connector locations

The supply connector should be 29.5mm from the outer side of the brick (opposite the thumbscrew) in the x-direction and 20.7mm from the bottom of face of the brick in the y-direction. THe two

connectors should be lined up in the horizontal plane and spaced 28.5mm from each other.

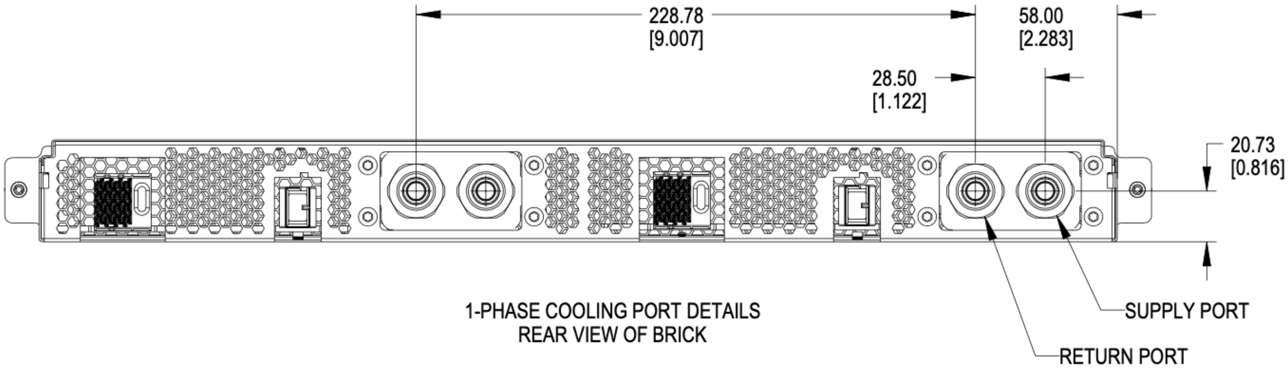


Figure 95. Brick side liquid connector locations for single-phase

Single-phase offset depth

The mating surface of the brick side connector sits xx mm behind (outside) the outer edge of the back plane of the brick. Specifying this dimension ensures the server manufacturer and manifold/cage side implementor have a consistent location where the connectors must meet to ensure full mating and thereby full flow.

The reference mating surfaces of each connector are highlighted below along with a drawing showing the specified dimension.

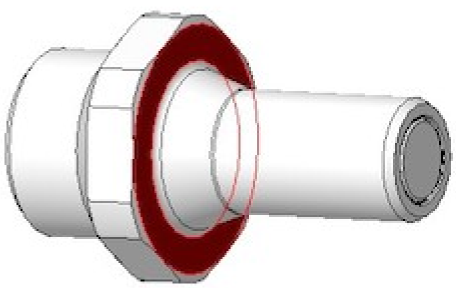


Figure 96. Mating surface BLQ4 insert

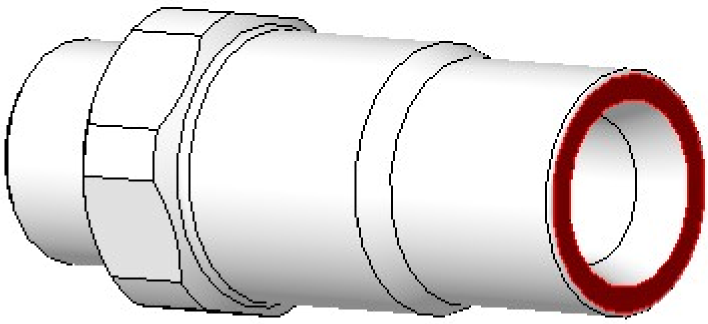


Figure 97. Mating surface BLQ4 body

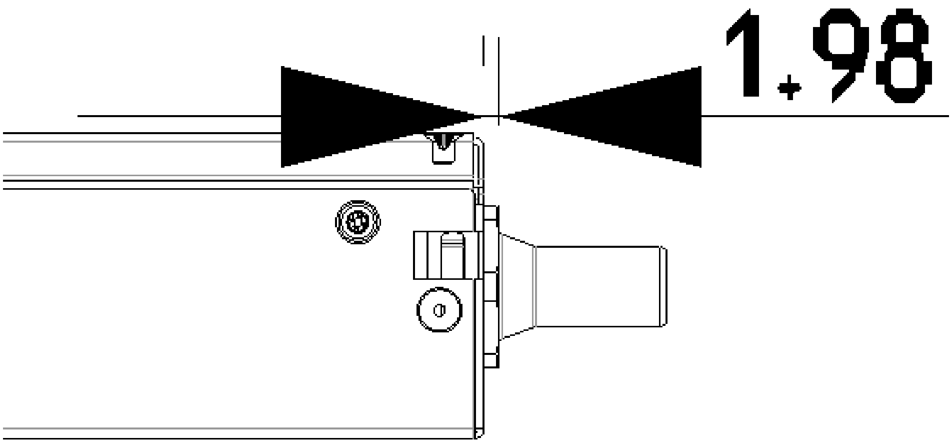


Figure 98. Offset from cage back panel to connector mating interface

The BLQ4 connectors must be within 3mm of this fully engaged state to achieve full flow.

5.5.3. Two phase

Two-phase connector selection

The two-phase cooling solution shall use the CPC BLQ family of interfaces.

On brick side

Figure 99. Liquid - CPC BLQ4 series insert P/N BLQ4D47004PROTO (chrome plated brass) or P/N 4649800PROTO (Aluminum)

Figure 100. Vapor - CPC BLQ6 series insert P/N 4650000PROTO (chrome plated brass)

On the Manifold side

Figure 101. Liquid - CPC BLQ4 series body P/N BLQ4D31004PROTO (chrome plated brass) or P/N 4649900PROTO (Aluminum)

Figure 102. Vapor - CPC BLQ6 series body P/N 4650200PROTO (chrome plated brass)

Two-phase connector mounting polarization

Cage manifolds shall have body (female) blind mate connectors. Bricks shall have insert (male) blind-mate connectors. When viewed from rear of the brick, vapor connection shall be located at the right position. Liquid connection shall be located at the left position.

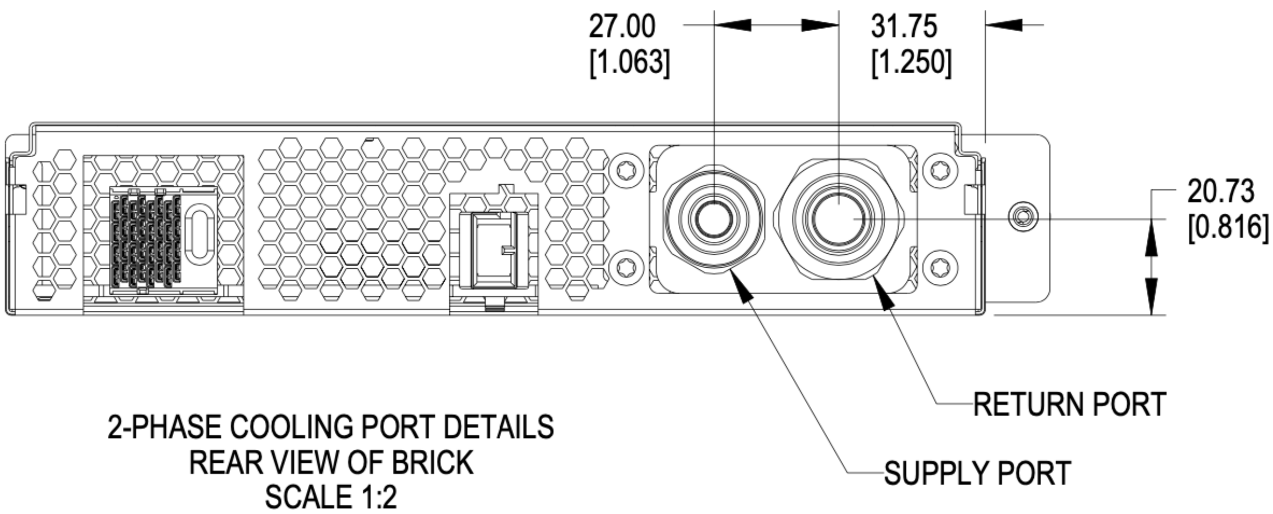


Figure 103. Brick liquid supply and Vapor return

Two-phase brick side connector locations

Using the right-side panel of the brick as a reference, the Vapor return connector on the brick should be 31.85mm in the x direction and 20.73 upwards in the y direction.

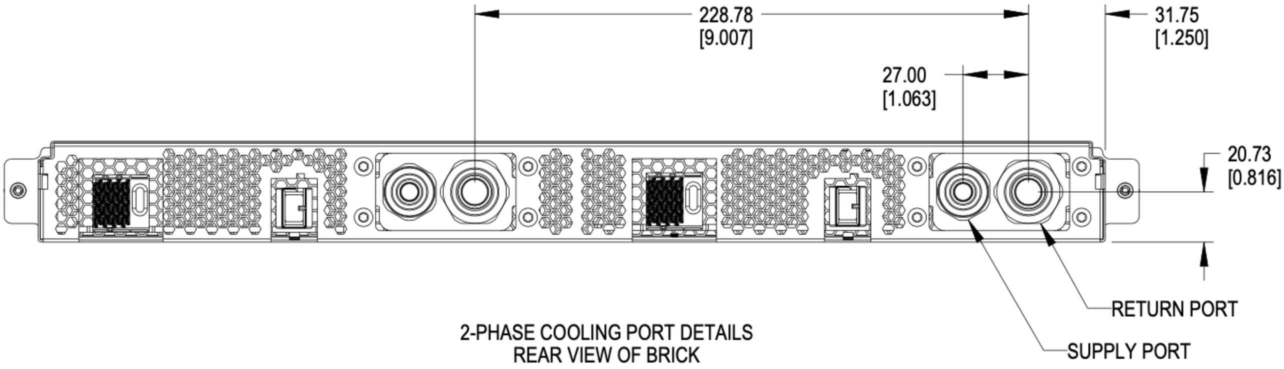


Figure 104. Brick side connectors locations

Two-phase offset depth

The mating surface of the brick side connector sits xx mm behind (outside) the outer edge of the back plane of the brick. Specifying this dimension ensures the server manufacturer and manifold/cage side implementor have a consistent location where the connectors must meet to ensure full mating and thereby full flow.

The reference mating surfaces of each connector are highlighted below along with a drawing showing the specified dimension.

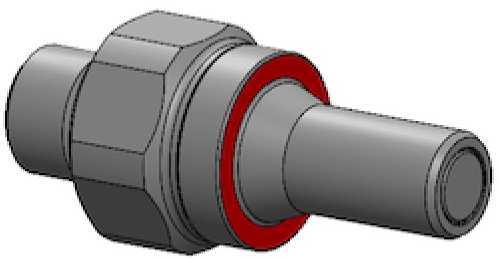


Figure 105. Mating surface BLQ4 insert

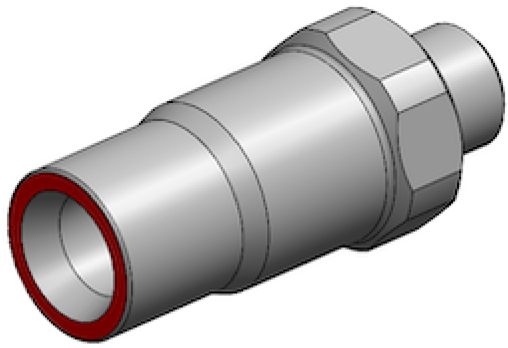


Figure 106. Mating surface BLQ4 body

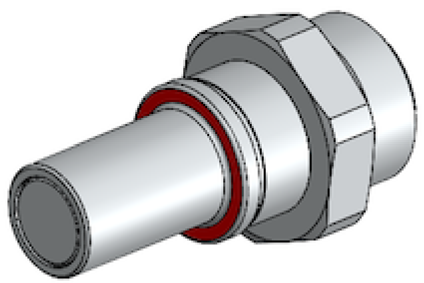


Figure 107. Mating surface BLQ6 insert

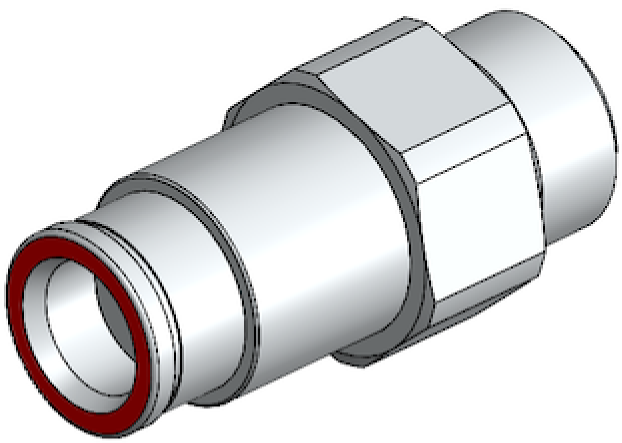


Figure 108. Mating surface BLQ6 body

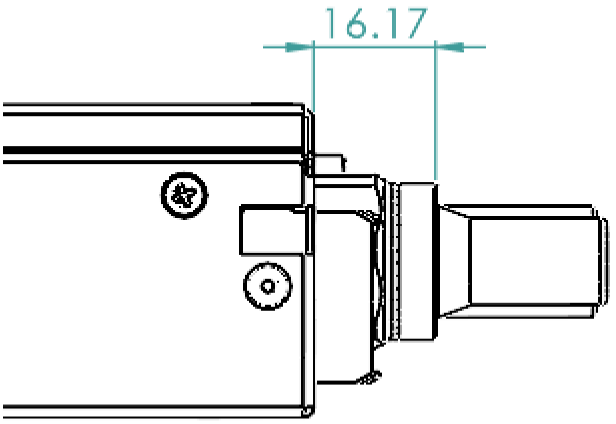


Figure 109. Offset from cage back panel to BLQ4 connector mating interface

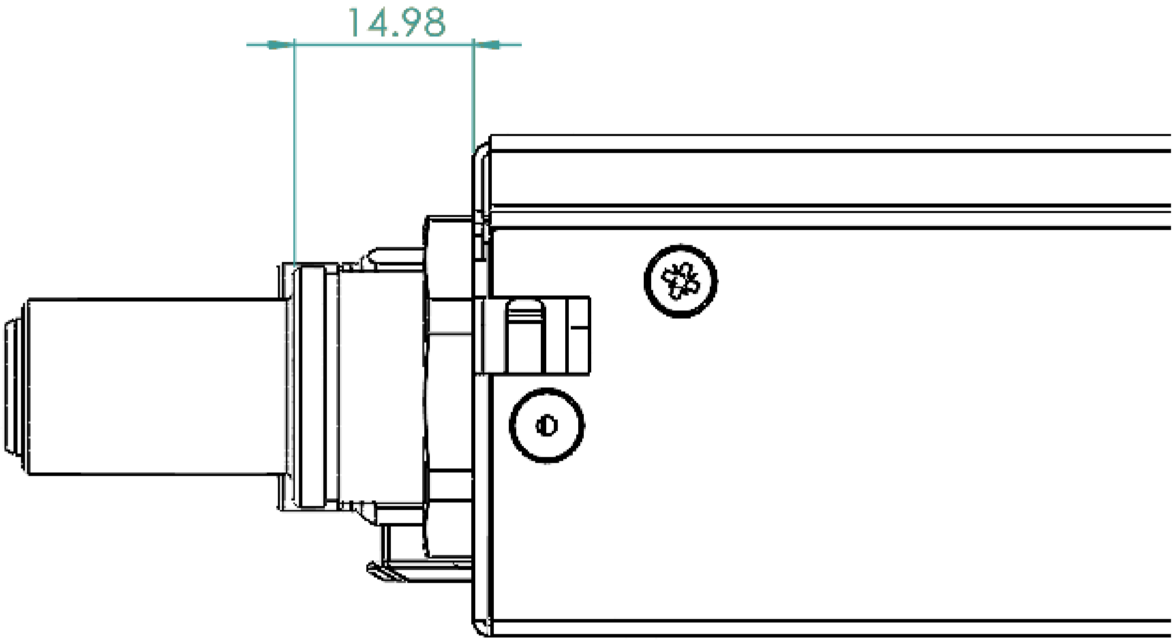


Figure 110. Offset from cage back panel to BLQ6 connector mating interface

The BLQ4 & BLQ6 connectors must be within ±1.5mm of this engaged state to achieve full flow.

5.6. Design guidance

Design guidance is not a specification requirement. It is simple design guidance that can be implemented to simplify future needs, to meet certification requirements, or satisfy general fit & finish.

5.6.1. Brick faceplate guidance

The faceplate is notched above and below the fastener (See Figure 78, “Brick retention”). Note that the faceplate can accommodate any type of ports or drives as needed and is not required to be fully perforated.

Screws and/or rivets used to secure the top cover are flush to be under-flush. The vertical edges on the rear of the brick chassis base are coined to break sharp edges.

5.6.2. Back panel safety perforation

See below for typical dimensions of the airflow perforation. All dimensions below are critical.

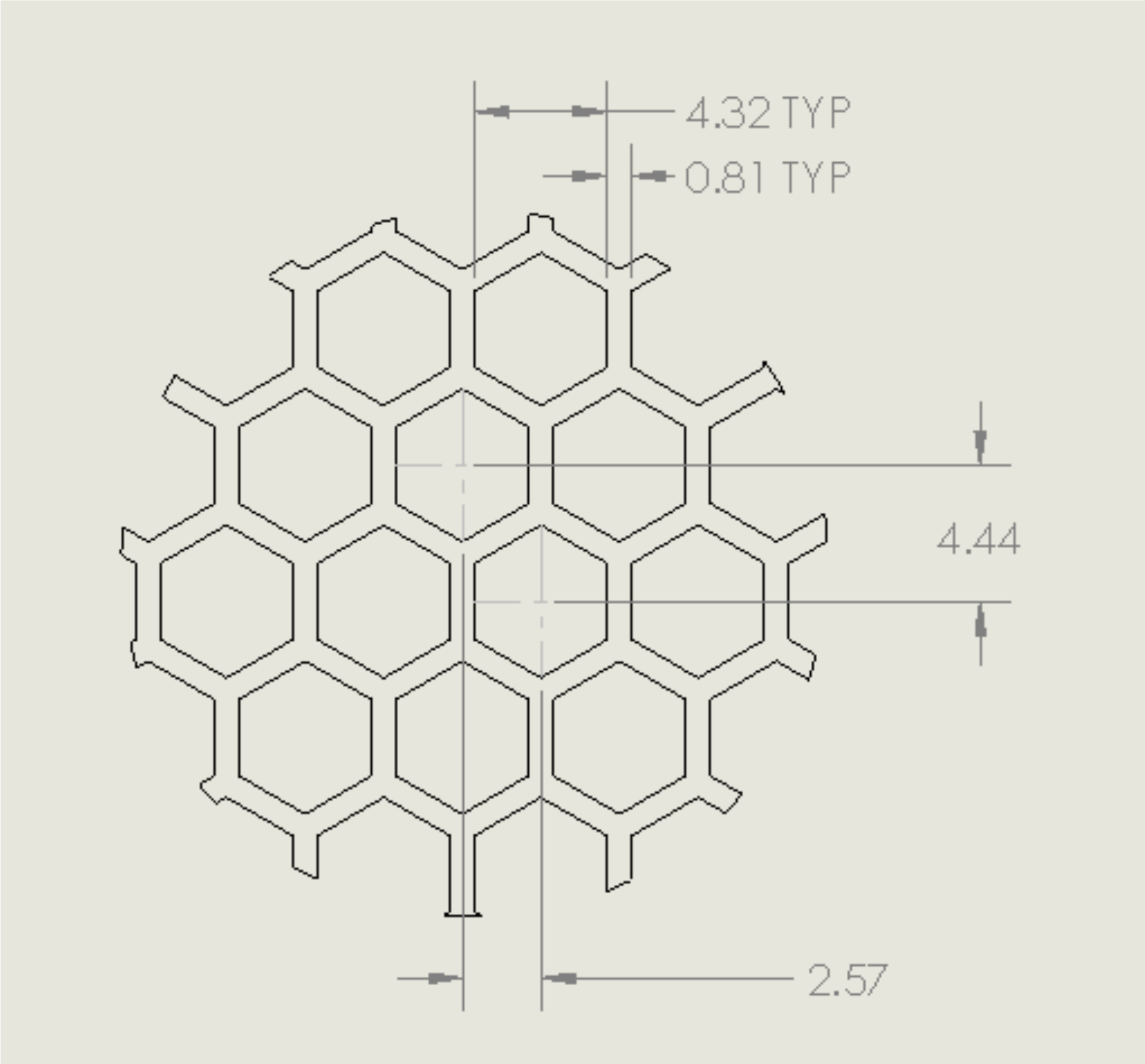


Figure 111. Brick panel perforation

Keep-out zones

Keep-out zones are defined in the brick surrounding the power connector and liquid cooling interfaces. The liquid cooling keep-out spaces "fly" above the PCB, allowing components to be populated below the connections.

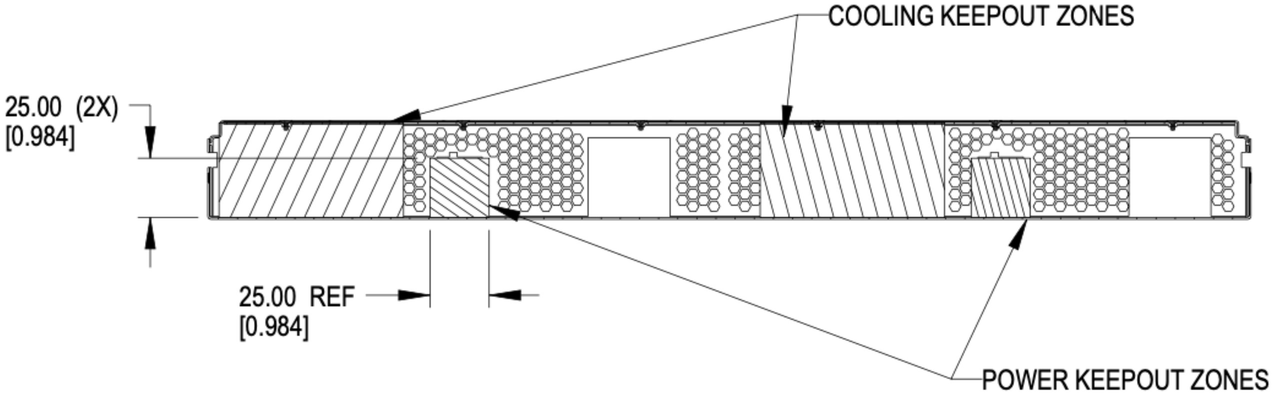


Figure 112. Brick keep-out zone

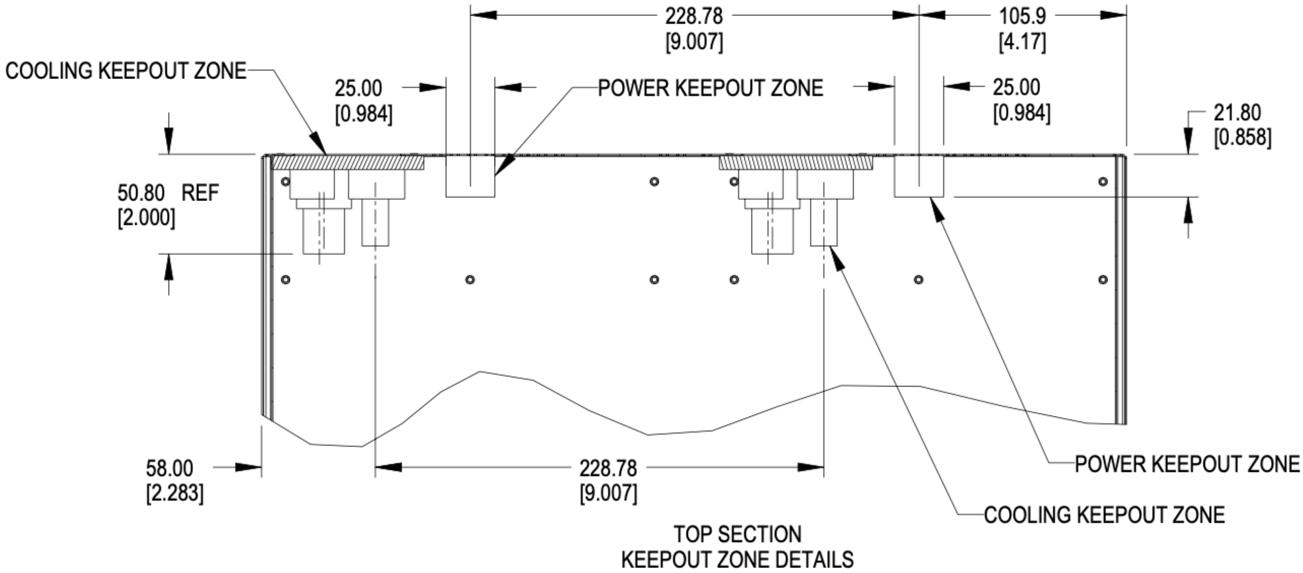


Figure 113. Brick keep-out zone top view

5.7. Brick performance

5.7.1. Brick electrical startup behavior

In order to safely operate the power shelf, we define the following characteristics for brick startup behavior.

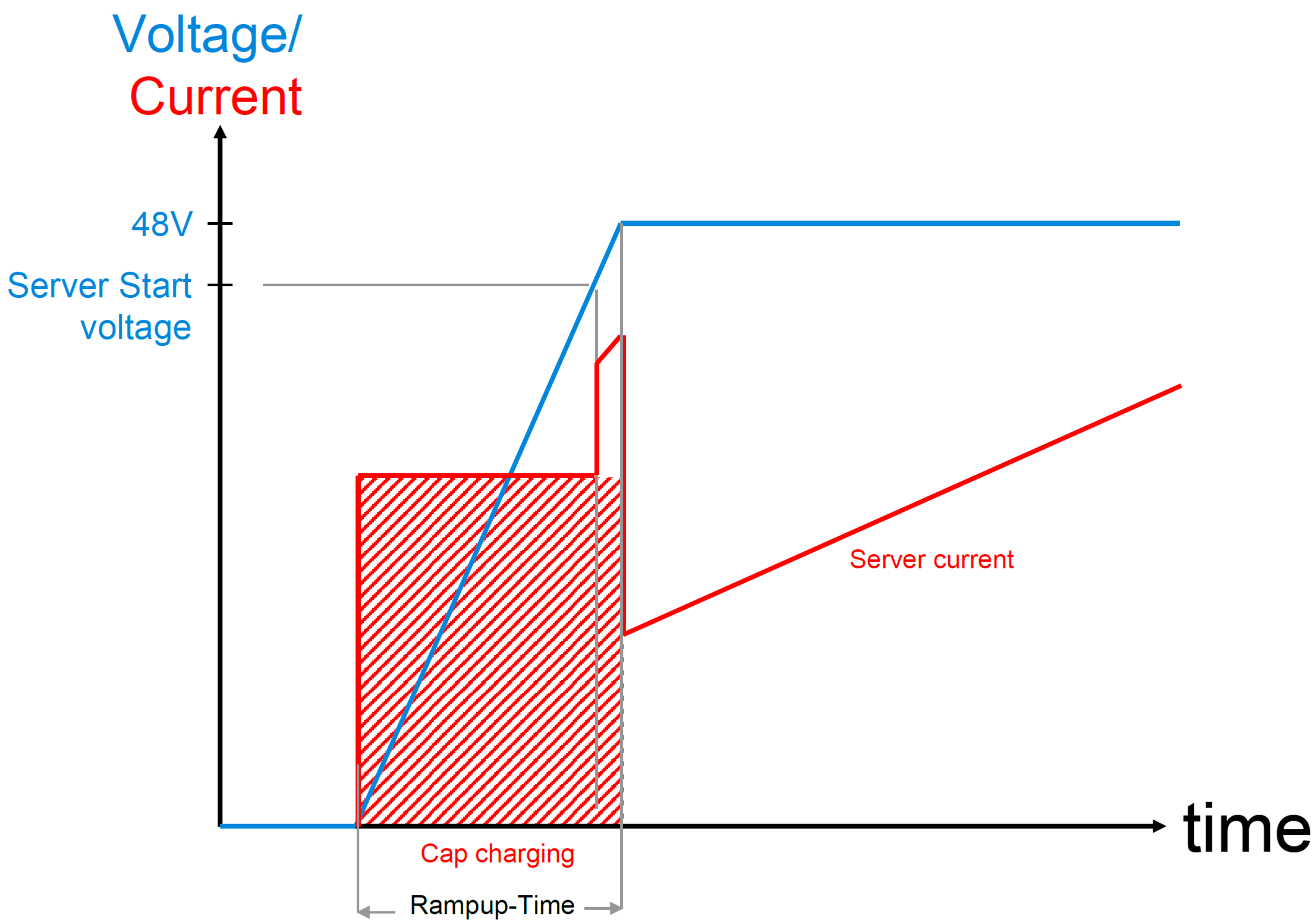


Figure 114. Server startup electrical characteristics

These values are per connector feed from a power shelf.

Two power connectors are available for use.

- Min start-up voltage: +44V DC
- Max input capacitance per brick feed: ⇐1000uF
- Max server current at start-up voltage: 70A
- Ramp time: Follow Intel CRPS spec: ramp time 10-70msec, slew rate 0.5V/msec

https://www.intel.com/content/dam/www/public/us/en/documents/guides/power-supply-design-guide-june.pdf

Chapter 6. Brick Mechanical Specification [DHDW]

6.1. DHDW Brick

The DHDW brick is commonly used as a 2U server form-factor.

6.2. Brick Dimensions

The brick is designed with the overall dimensions shown.

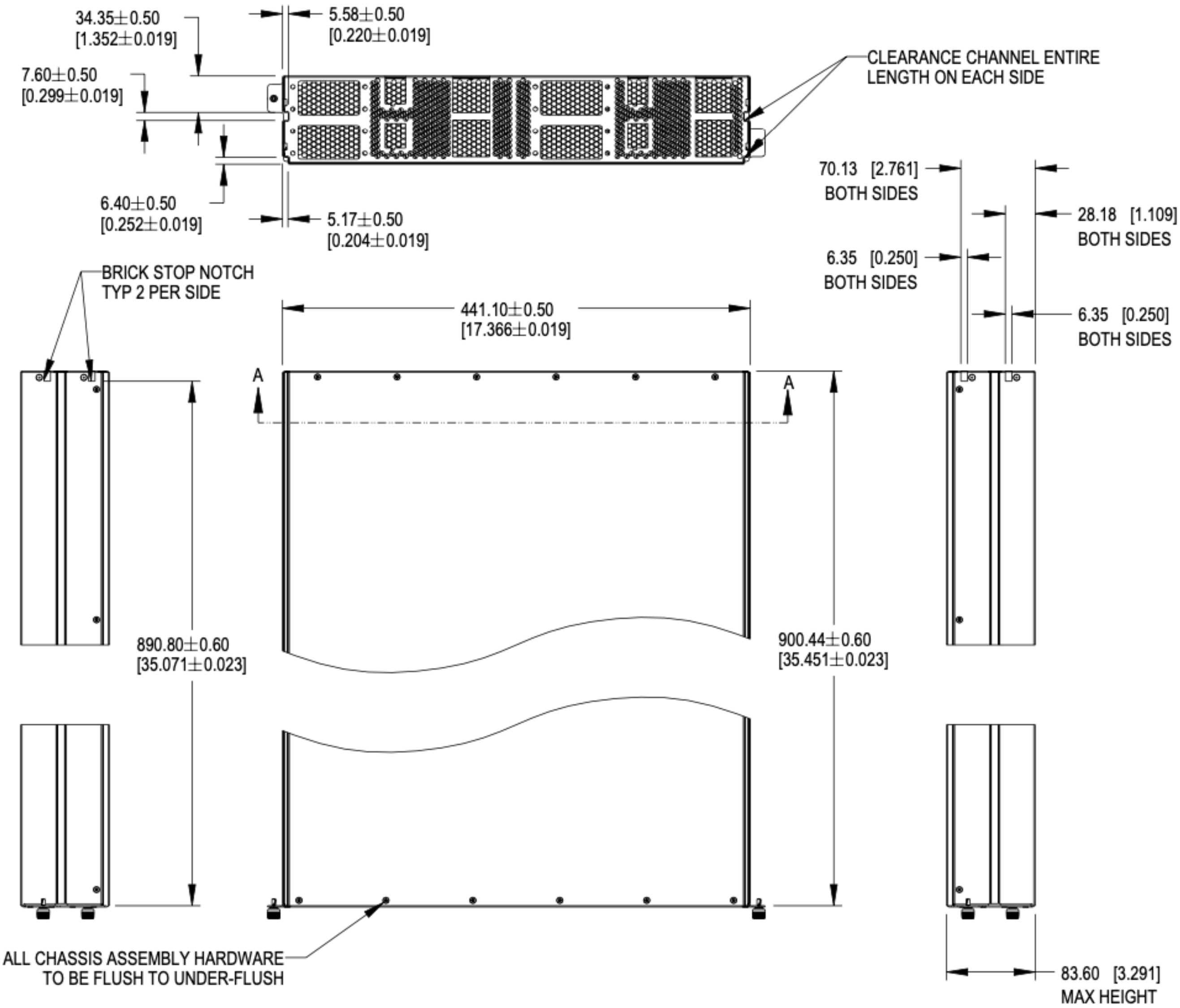


Figure 115. Brick dimensions

Please refer to the drawings, and 3D files for further details.

The V1 specification for bricks defined 901.7mm as the overall length of the brick, including the thickness of the front panel.

The dimension is modified to reference to the inside of the front panel to the rear of the brick. This dimension is 900.50mm +/-0.60 from the rear face of the front panel (mating surface to the cage) to the outside of the rear panel.

6.3. Mechanical features

6.3.1. Brick retention

The brick is secured into the cage with up to four Southco D9-47-1302-K quarter-turn fasteners, located on the left and right side of the faceplate.

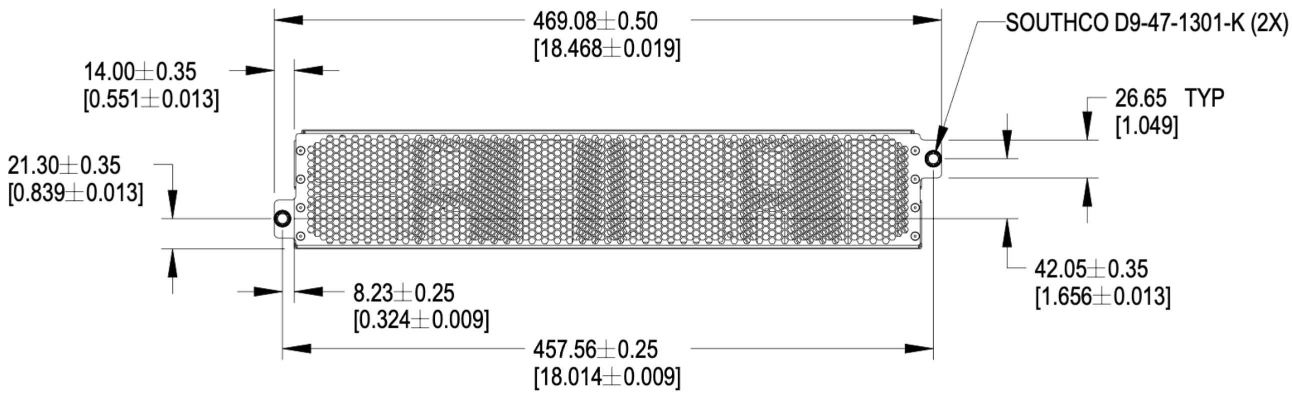


Figure 116. Brick retention

6.3.2. Brick mechanical guide

Clearance channels are required along the top edges of the brick and the sidewalls, extending forward from the rear panel to the back of the faceplate.

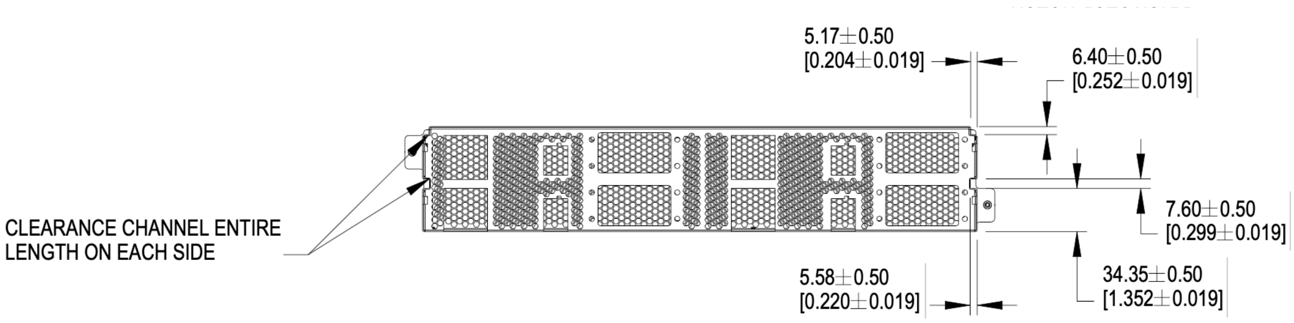


Figure 117. Mechanical guide at top edge and side walls of brick

Brick insertion and retention assistance for liquid cooling

For liquid cooling applications, two different mechanical assist for insertion and retention are available in the cage. Bricks can implement features to make use either/both features.

The features are:

- Brick retention standoff, available on both sides in the mechanical guide channel
- Slots, available on both sides of the cage wall

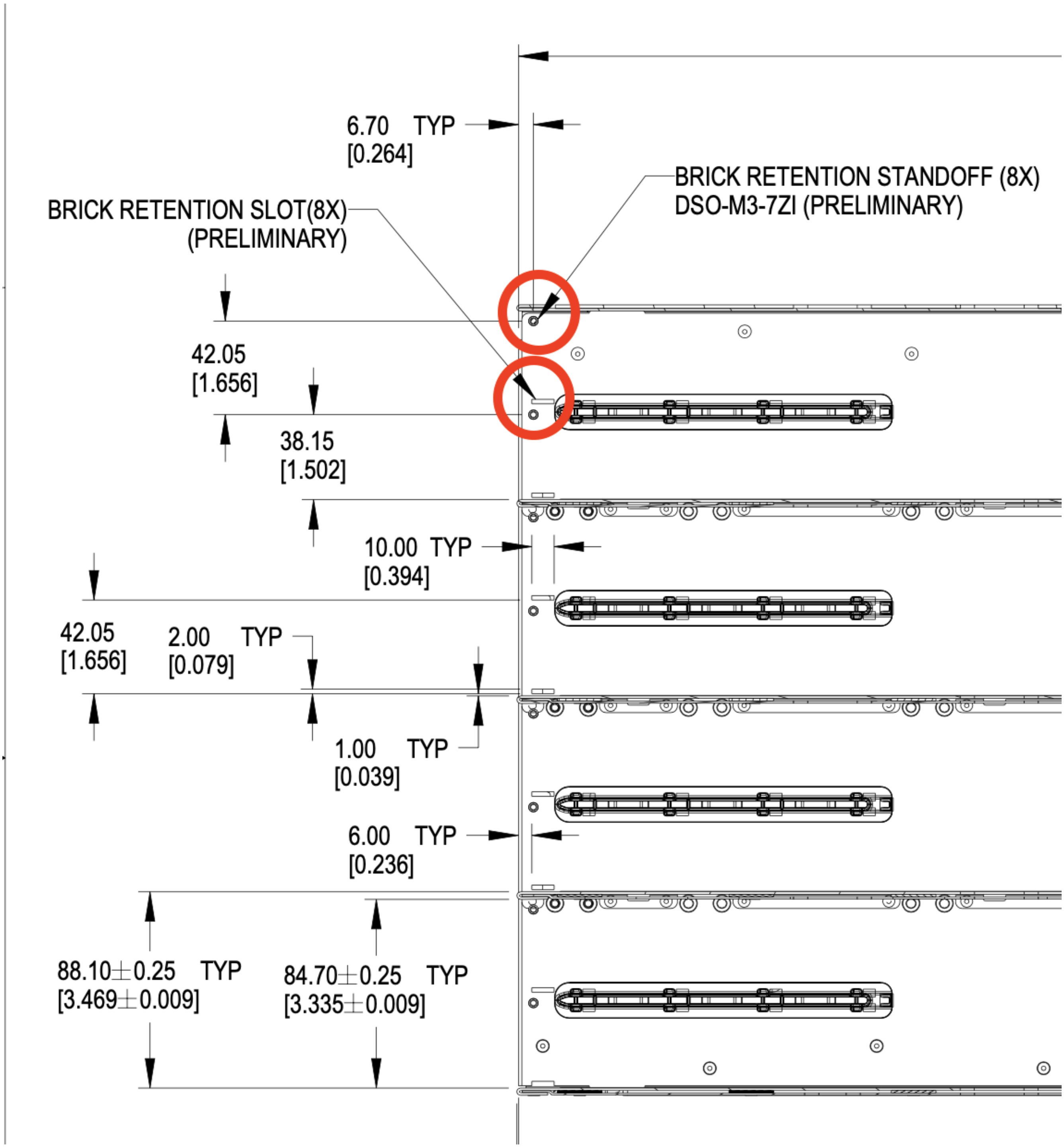


Figure 118. Brick mechanical assistance features in cage

Each feature shall support 100 lbf (445N) force or greater, using the recommended 1.3mm SGCC material.

6.3.3. Brick Mechanical Keying

Two keying notches are positioned in the left and right corners of the sheet metal enclosure as shown below. All dimensions below are critical.

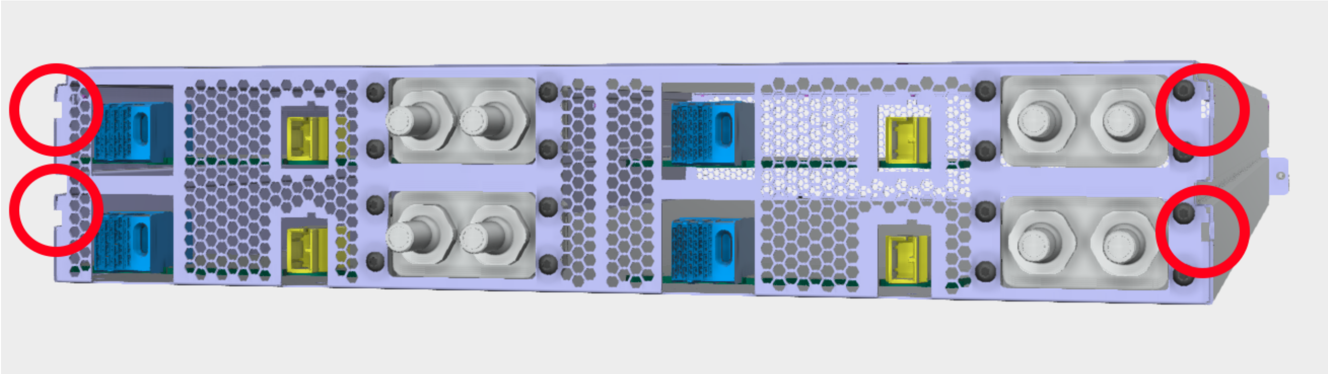


Figure 119. Brick keying

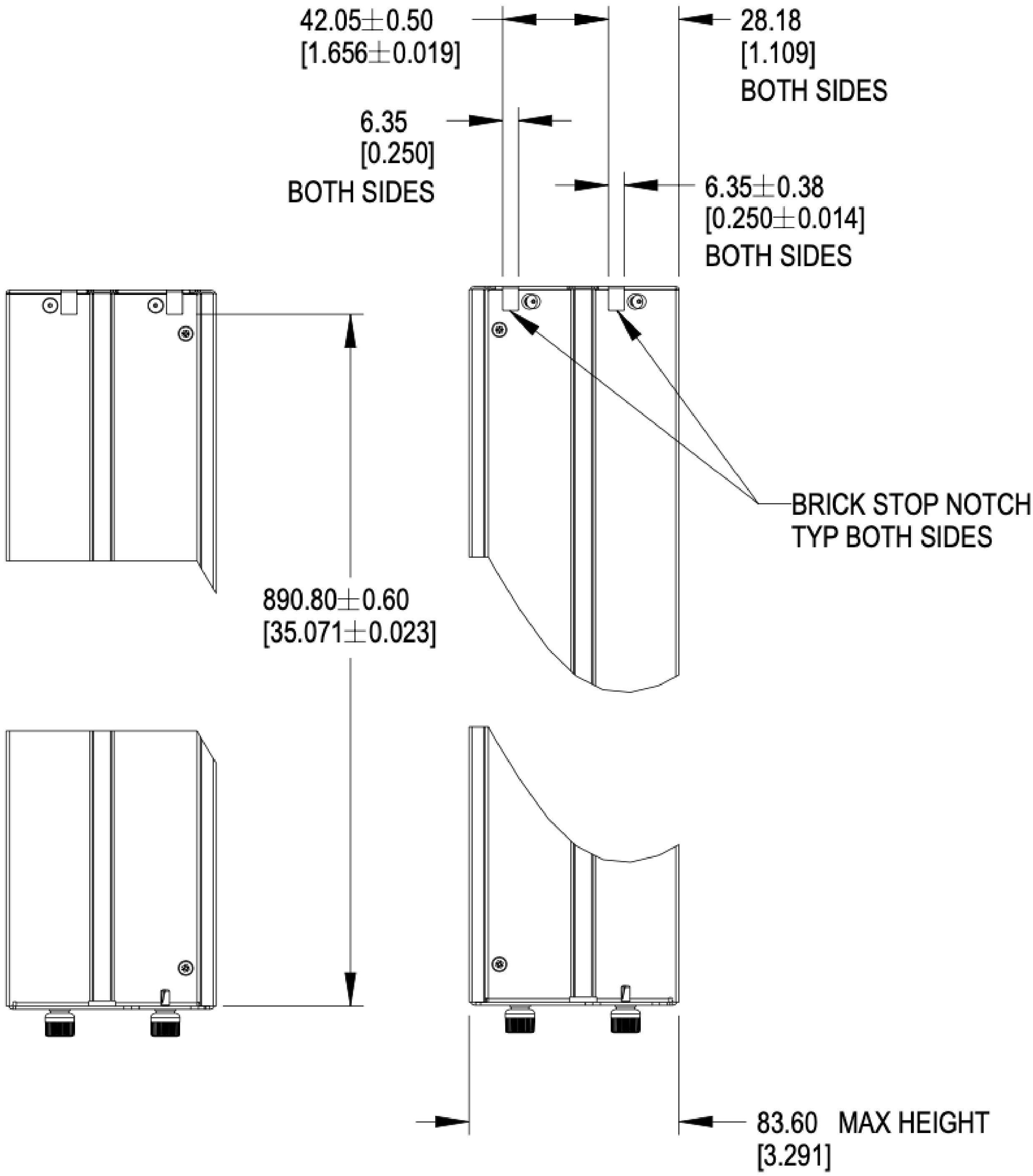


Figure 120. Brick keying dimensions

6.3.4. PCBA

The rear PCBA (Printed Circuit Board Assembly), also referenced as the interposer, provides the +48V DC input power and a data connections. The top of the PCBA is the reference surface, located 6.5mm +/- 0.6mm above the bottom surface of the sheet metal chassis.

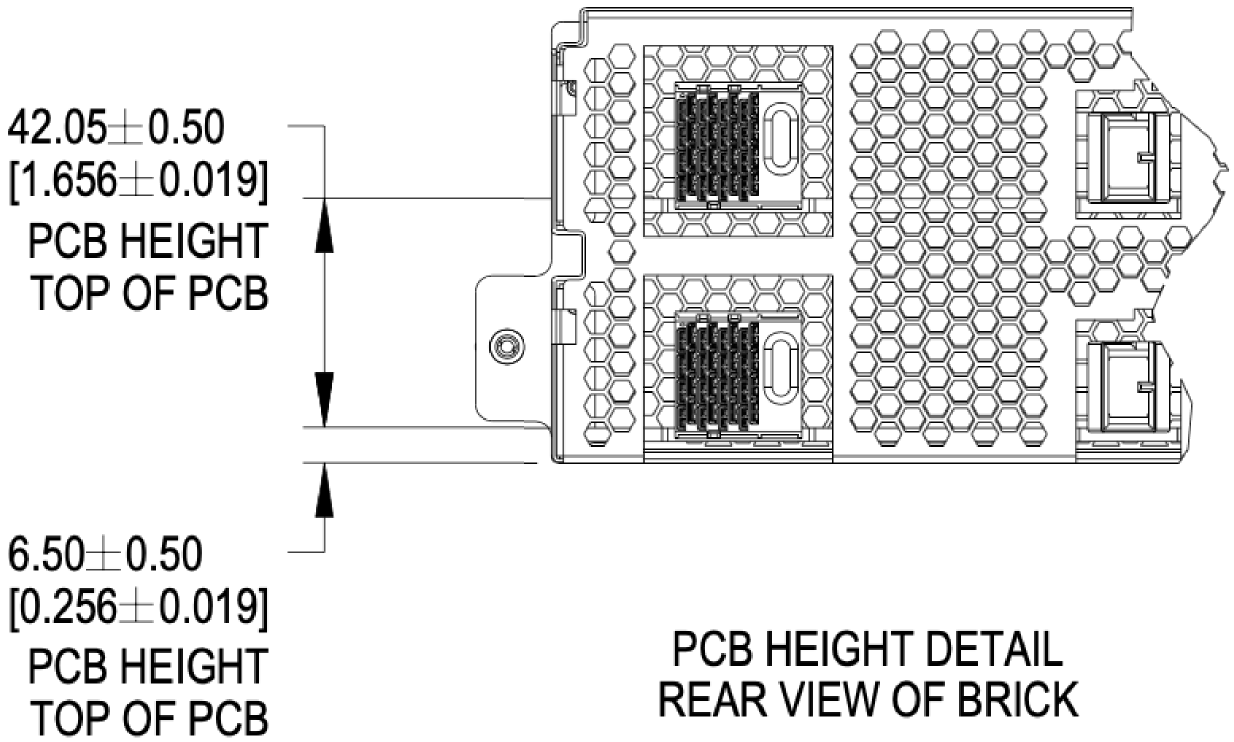


Figure 121. PCBA References

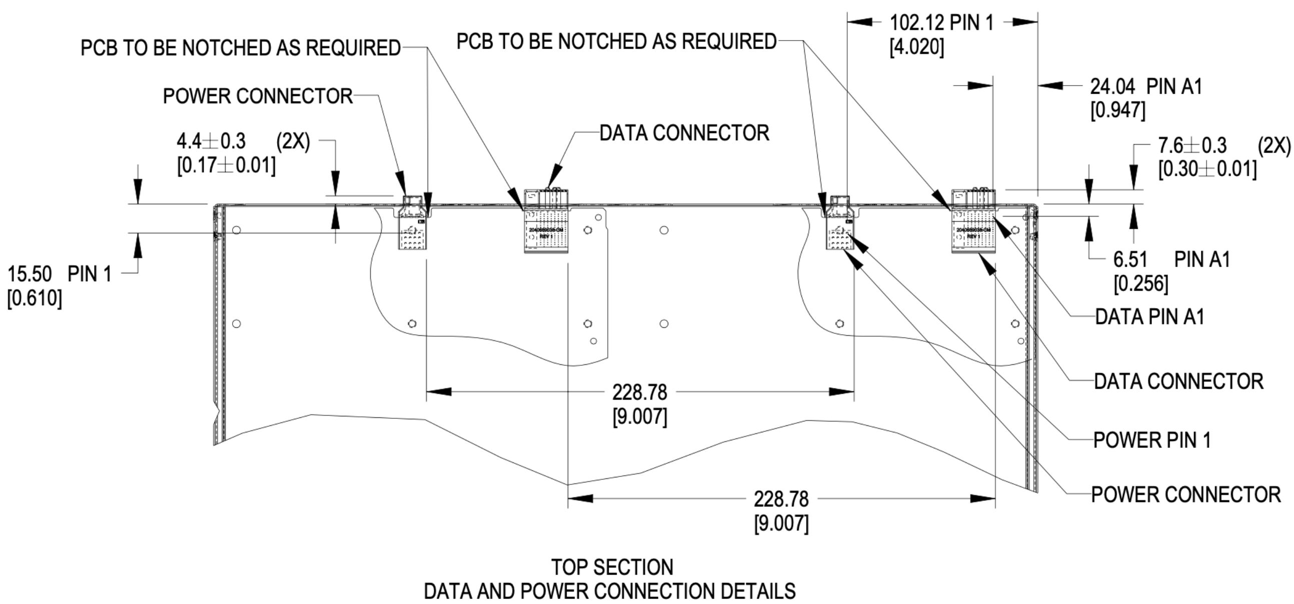


Figure 122. Brick PCB dimensions

6.3.5. Materials

The brick enclosure is constructed of sheet metal with adequate strength and rigidity. Suggested material is 0.8mm SGCC for the chassis base and 0.6mm SGCC for the top cover.

6.3.6. Blind mate interfaces

A brick is equipped with at least one set of blind mate connectors for receiving +48V DC power and transferring data with an external device, typically a network switch. Introduced in this version is are blind mate connections to facilitate liquid cooling. The connections mate with cable and manifold projections from the rear wall of the cage.

- Power connector: Molex Ten60 PN: 220370-0001 (35A) 22037-0000 (70A), refer to Molex document PSD-220370-0000
- Data connector: Molex 4x8 Impel with guide receptacle PN: 204066-9038 (PCB thickness 2.2-4.1mm), 204066-9138 (PCB thickness 1.0-1.9mm), refer to Molex document PSD-204066-9001

The power connector has two ratings:

- Power: Not-To-Exceed (NTE) 70A at +48V DC
- Power: Not-To-Exceed (NTE) 35A at +48V DC

The ratings shall be coordinated with the power shelf. 35A bricks shall be enabled in 70A capable infrastructure. 70A bricks are keyed to prevent unsafe operation in 35A infrastructure.

The brick may populate up to four sets of connectors, as required.

6.3.7. Weight

The maximum weight of the brick, when equipped with all drives and/or accessories, is 120lbs (54.43kg). This is an increase of 20lbs from V1.

6.4. Connector Pinout

6.4.1. Brick power connector

The brick blind-mate power connector for receiving 48V DC (51V nominal) power from the power shelf. The connector mates with the power cable connector projected from the brick cage wall.

Two variations of the brick connector exist, with different Not-to-exceed (NTE) ratings:

- 35A Power Connector: Molex 2203700001 (NTE 35 AMPS) No Key
- 70A Power Connector: Molex 2203700000 (NTE 70 AMPS) Keyed

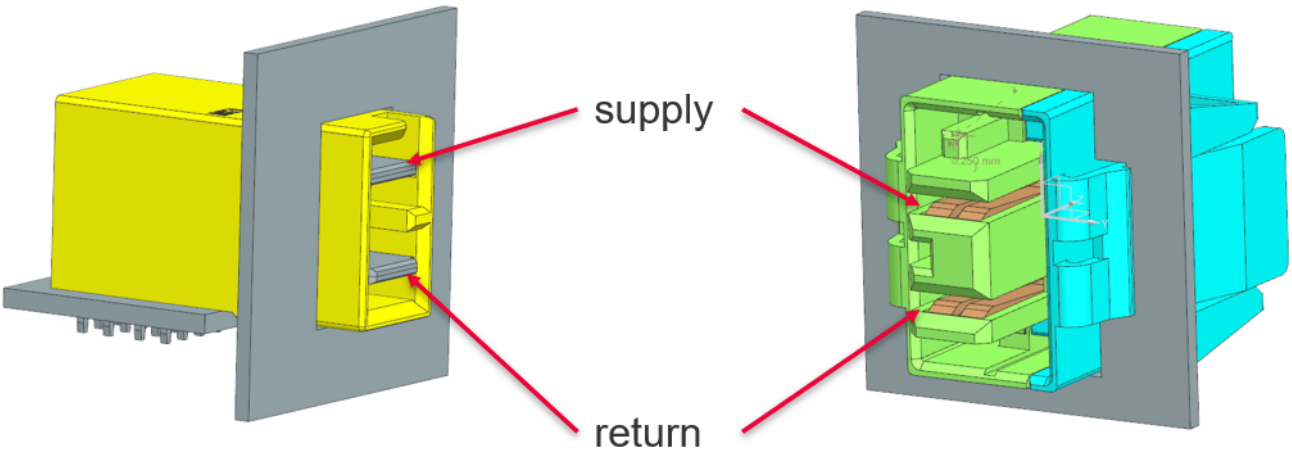


Figure 123. Power connector supply/return definition

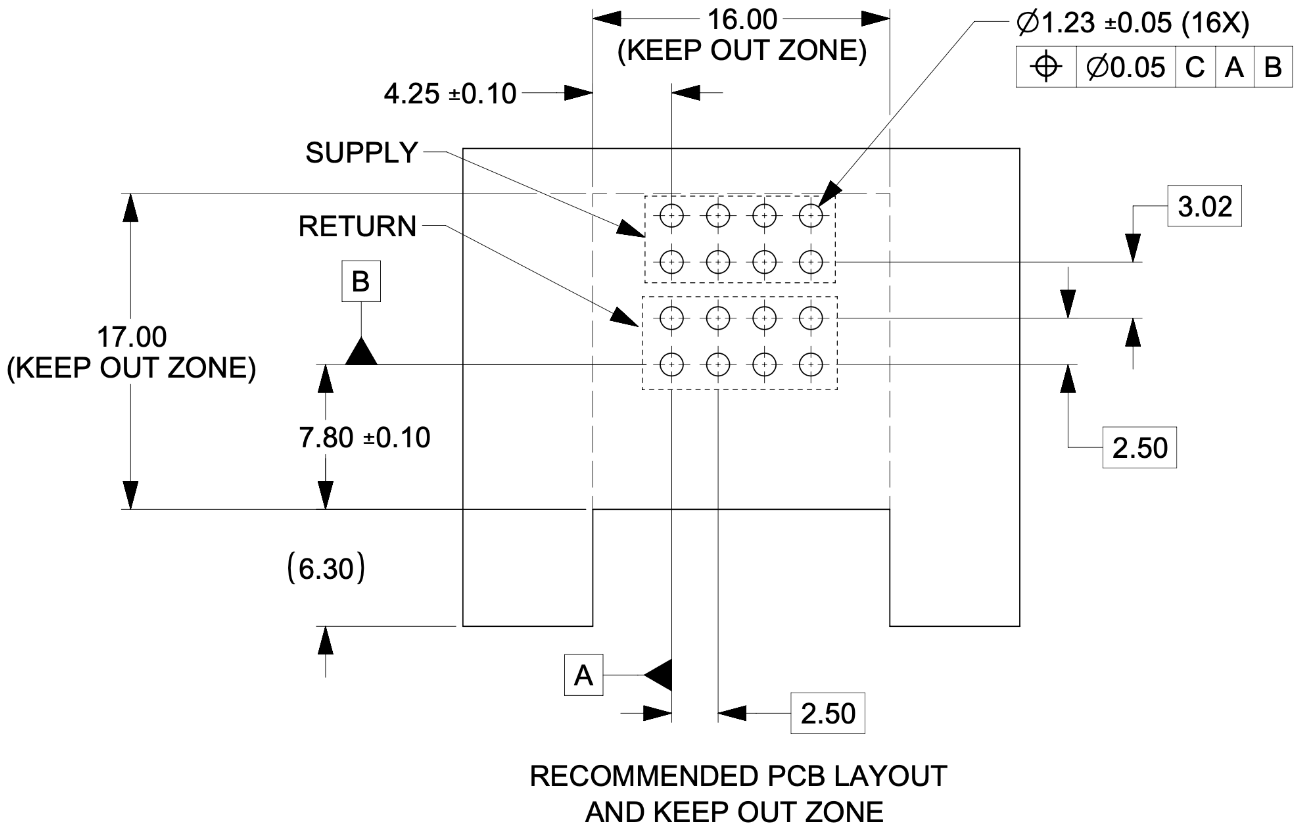


Figure 124. Power connector footprint definition

Connector drawings are available: Molex Brick power connector

6.4.2. Brick critical dimensions

The brick power connector has a critical dimension for the projection from the brick panel. Spacing between panels is 6.23mm. The power connector shall be proud of the panel 4.40mm.

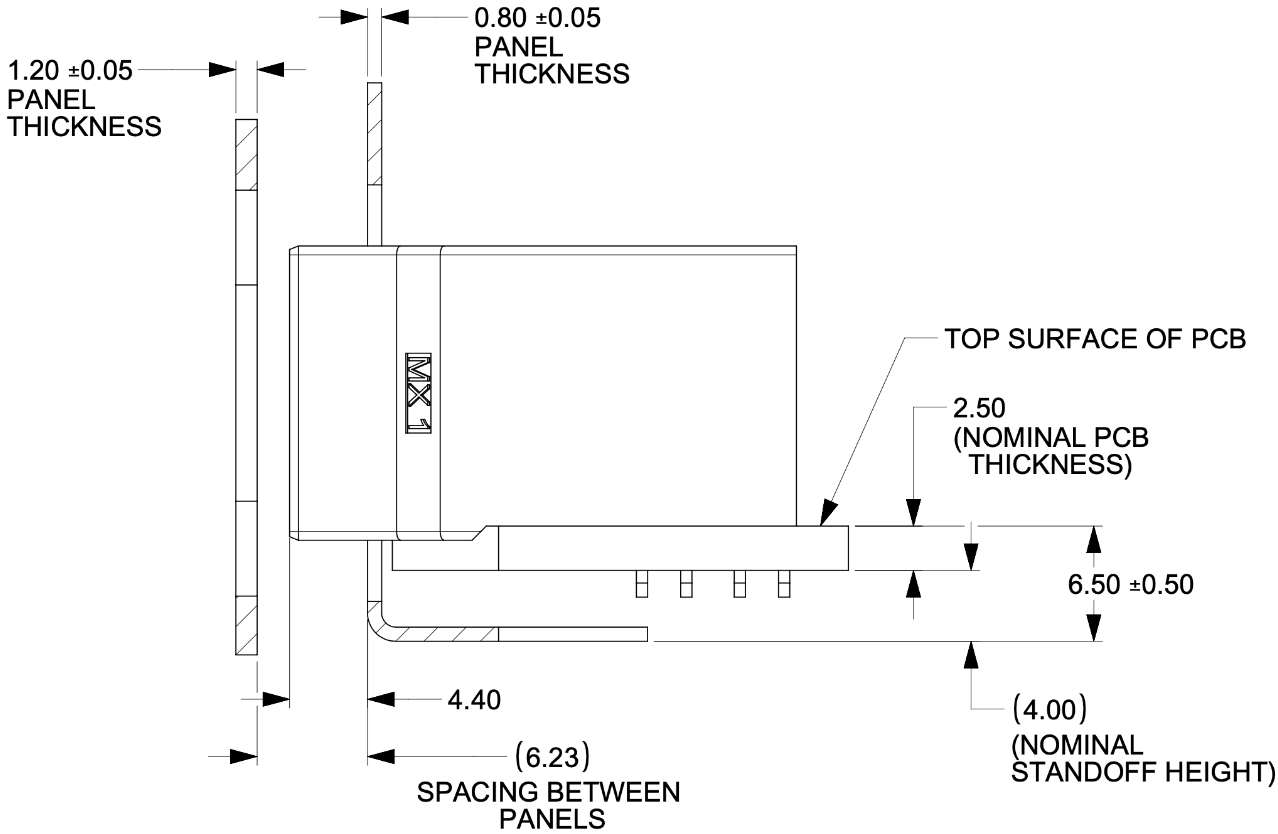


Figure 125. Power connector critical dimensions

6.4.3. Brick voltage

Open19 V2 power shares Open Compute ORV3 power characteristics. Open19 bricks shall safely operate in the DC range 48-52V and shall remain functional when exposed to 0-54V. The power rail shall not exceed 52.5V for more than 200mS. DC voltage shall never exceed 54V.

- Operating range: 48-52V
- NTE voltage: 54V
- Max voltage: ⇐52.5V for <200mS

6.4.4. Data Connector

The data connector is rated for 56G PAM4 with an air dielectric. Variations of cables deliver different lane counts, data rates, and physical interfaces. In the reference Ethernet implementation, the data connector pinout explicitly supports a dedicated 10/100/1000Base-T management port, four 25G NRZ/50G PAM4 lanes. Optionally, a single QSFP-DD or two QSFP interfaces are supported.

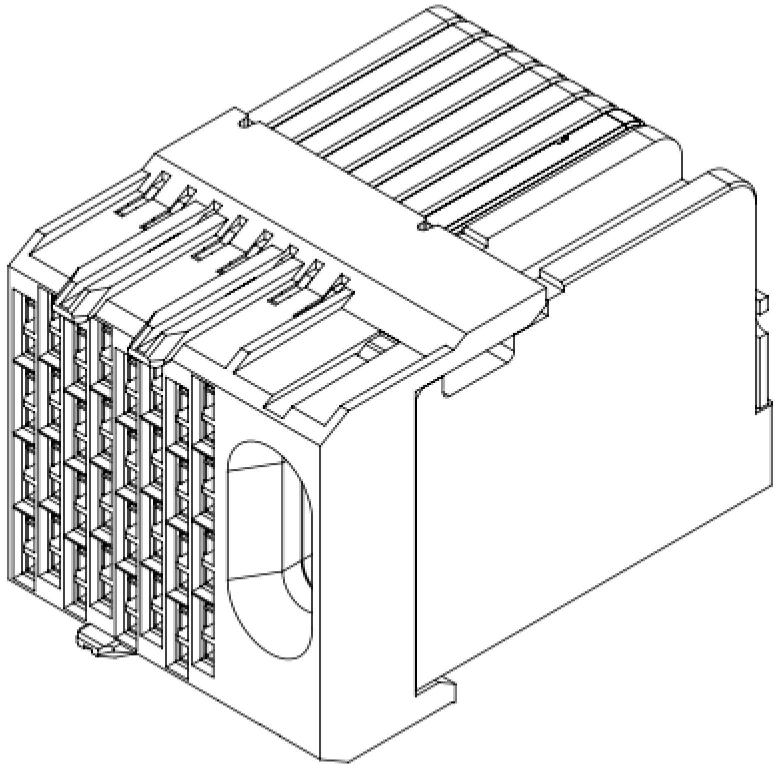


Figure 126. Impel Connector with guide feature

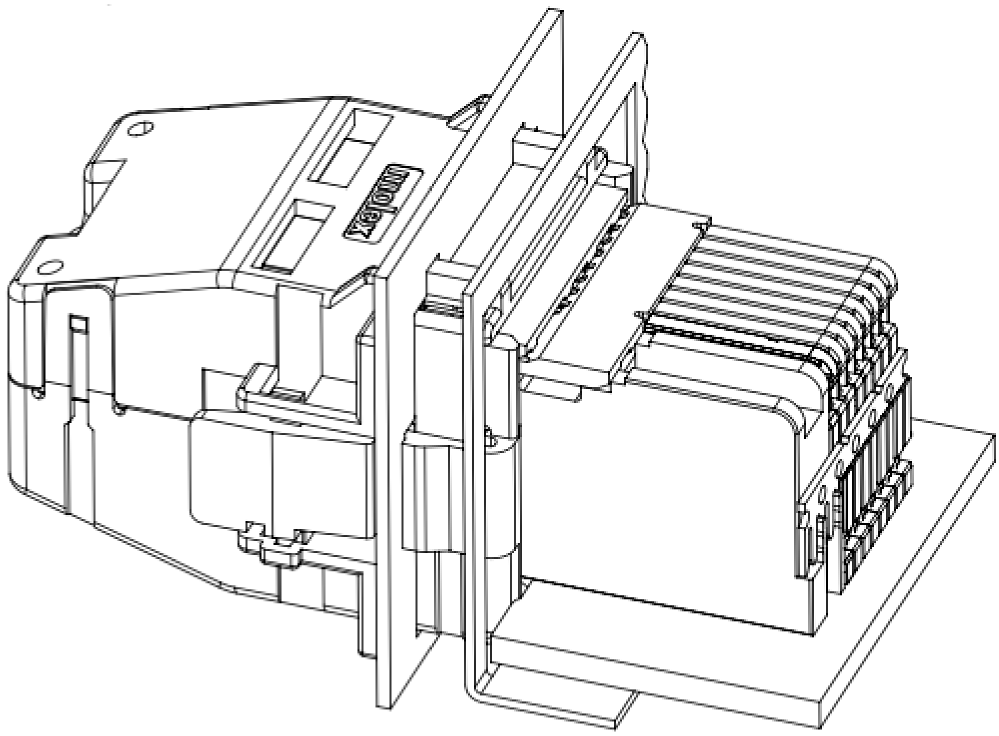


Figure 127. Impel Mated connector with cable backshell

The data connector pinout is defined in the [Appendix]

There is no attempt to support 50G (2x25G NRZ) nor 50G (1x50G NRZ).

6.4.5. Ethernet interface inter-operability

The inter-operability of bricks at deployment with Ethernet interfaces is critical. To support 25G NRZ, 50G PAM4, 100G (4x25G NRZ), 200G (4x50G PAM4), and 400G (8x50G PAM4), the following table:

<table>
<tr><th>Port Count</th><th>Data Rate</th><th>Implementation Interface and order</th></tr>
<tr><td>1</td><td rowspan="6">25G NRZ

50G PAM4</td><td>SFP1</td></tr>
<tr><td>2</td><td>SFP1, SFP2</td></tr>
<tr><td>3</td><td>SFP1, SFP2, SFP3</td></tr>
<tr><td>4</td><td>SFP1, SPF2, SFP3, SPF4</td></tr>
<tr><td>5</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1</td></tr>
<tr><td>6</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1 LN 1, QSFP2 LN 1</td></tr>
<tr><td>1</td><td rowspan="4">25G NRZ

50G PAM4</td><td>SFP1</td></tr>
<tr><td>2</td><td>SFP1, SFP2</td></tr>
<tr><td>3</td><td>SFP1, SFP2, SFP3</td></tr>
<tr><td>4</td><td>SFP1, SPF2, SFP3, SPF4</td></tr>
<tr><td>5</td><td rowspan="2">100G (4x25G)

200G (4x50G PAM4)</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1</td></tr>
<tr><td>6</td><td>SFP1, SPF2, SFP3, SPF4, QSFP1, QSFP2</td></tr>
<tr><td>1</td><td rowspan="4">25G NRZ

50G PAM4</td><td>SFP1</td></tr>
<tr><td>2</td><td>SFP1, SFP2</td></tr>
<tr><td>3</td><td>SFP1, SFP2, SFP3</td></tr>
<tr><td>4</td><td>SFP1, SPF2, SFP3, SPF4</td></tr>
<tr><td>5</td><td>400G (8x50G PAM4)</td><td>QSFP-DD (QSFP1+QSFP2)

200G (4x50G PAM4)</td></tr>
<tr><td>QSFP (SFP1+SFP2+SFP3+ SFP4)</td><td>2</td><td>QSFP, QSFP1</td></tr>
</table>

6.5. Liquid cooling

6.5.1. Brick liquid cooling

Brick liquid cooling is divided into two broad categories, single-phase and two phase. They are mutually incompatible and the designs are mechanically incompatible by design. All bricks inside a single cage are expected to be the same liquid cooling technology.

Unified cutout

Open19 V2 liquid cooling implements a unified liquid cooling cutout. This allows for a consistent brick design to implement both liquid cooling solutions. Implementing the unified cutout is not required, provided all the requirements of a single solution are met.

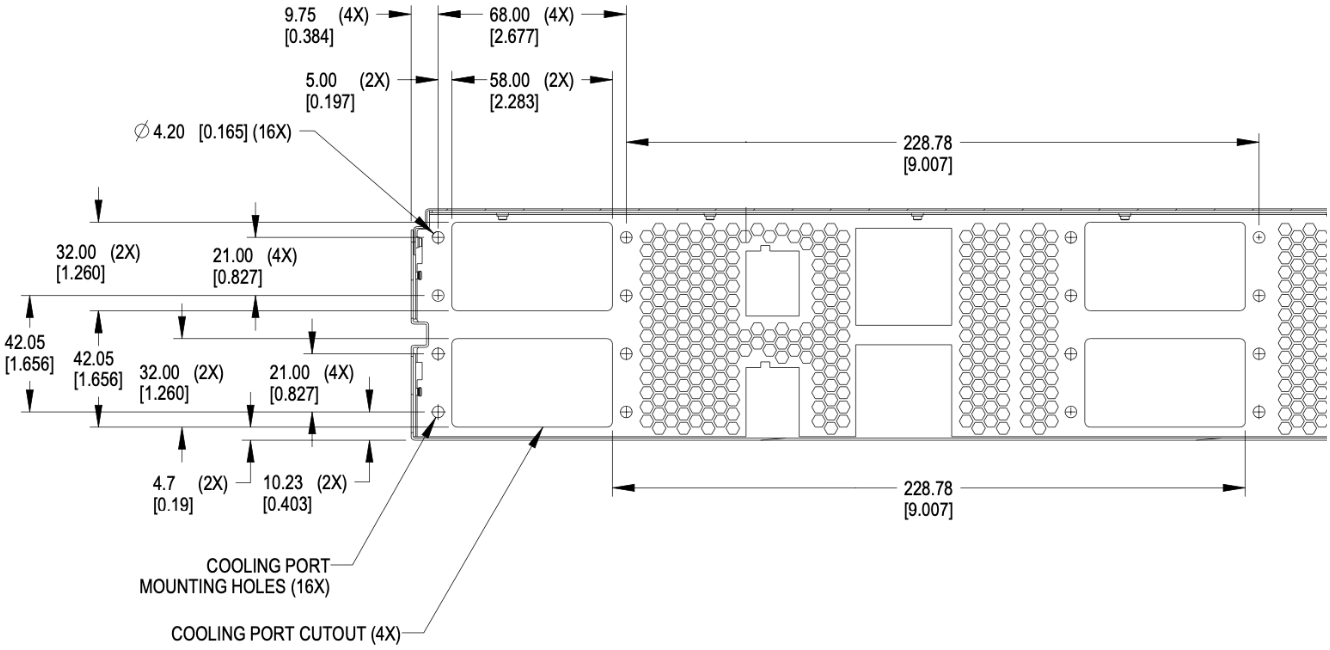


Figure 128. Brick unified cooling cutout

6.5.2. Single phase

Single-phase Connector selection

Single-phase cooling solutions shall use the CPC BLQ4 family of interfaces.

Figure 129. BLQ4 body for manifold

Figure 130. BLQ4 insert for bricks

BLQ4_SpecSheet.pdf

The BLQ4 connectors are blind mate. Both sides of the connector have G 3/8" threads which thread into manifold (body) and brick (insert), respectively.

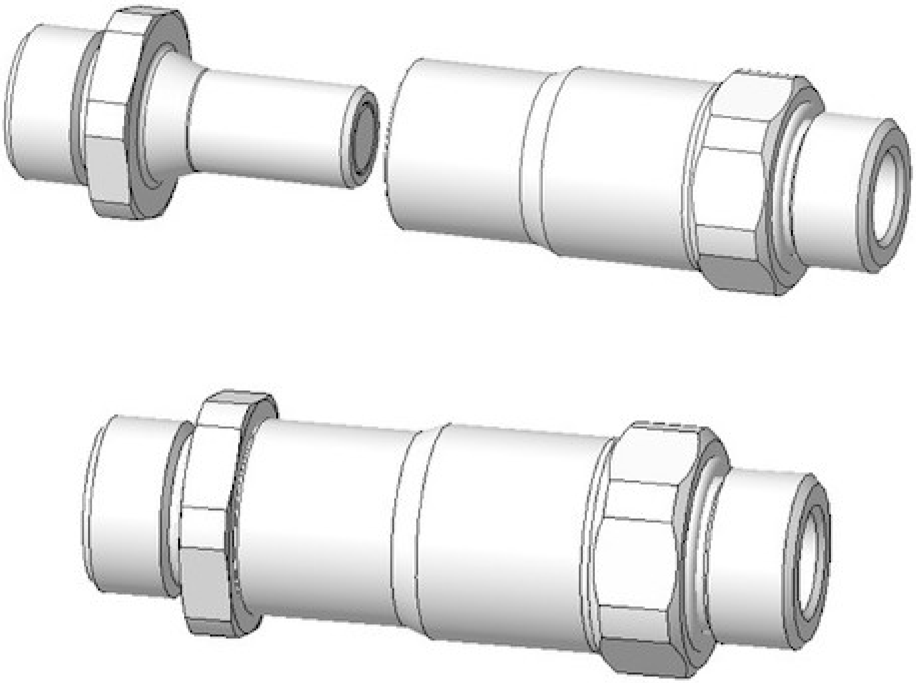


Figure 131. BLQ4 pair

Single-phase connector mounting and polarization

Bricks shall have male blind-mate connectors (inserts), BLQ4D47006 or compatible part.

When viewed from the rear of the brick, in reference to the server: Liquid return shall be located to the left. Liquid supply shall be located in the to the right.

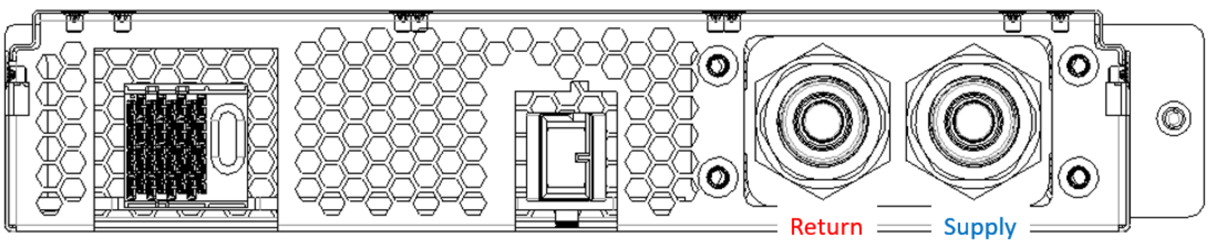


Figure 132. Brick side supply and return

Single-phase brick side connector locations

The supply connector should be 29.5mm from the outer side of the brick (opposite the thumbscrew) in the x-direction and 20.73mm from the bottom of face of the brick in the y-direction. THe two

connectors should be lined up in the horizontal plane and spaced 28.5mm from each other.

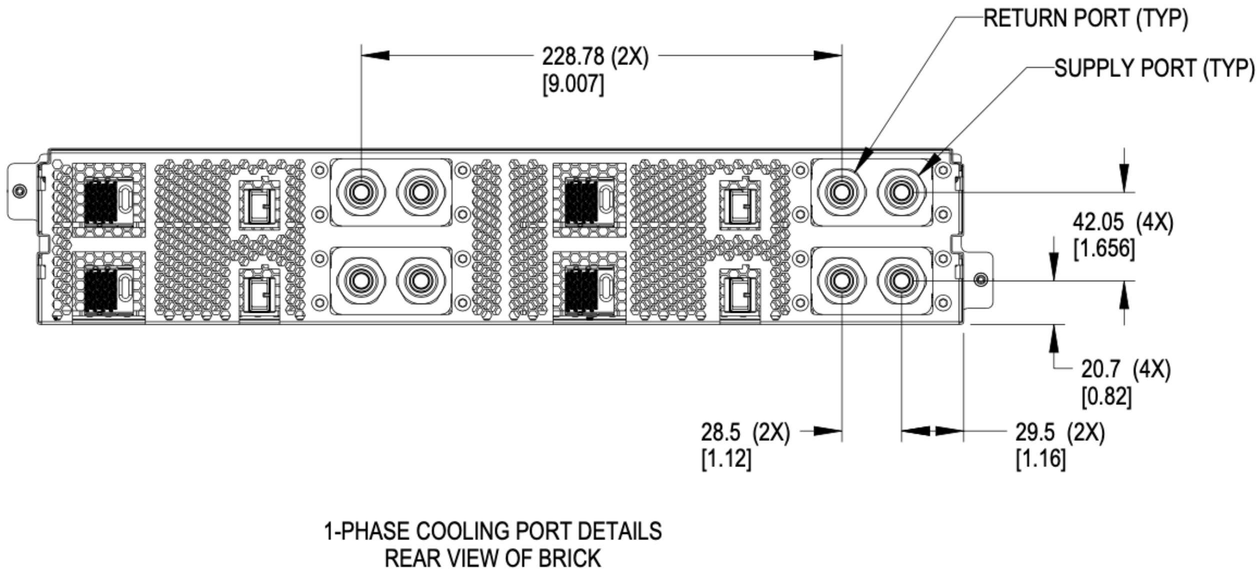


Figure 133. Brick side liquid connector locations for single-phase

Single-phase offset depth

The mating surface of the brick side connector sits xx mm behind (outside) the outer edge of the back plane of the brick. Specifying this dimension ensures the server manufacturer and manifold/cage side implementor have a consistent location where the connectors must meet to ensure full mating and thereby full flow.

The reference mating surfaces of each connector are highlighted below along with a drawing showing the specified dimension.

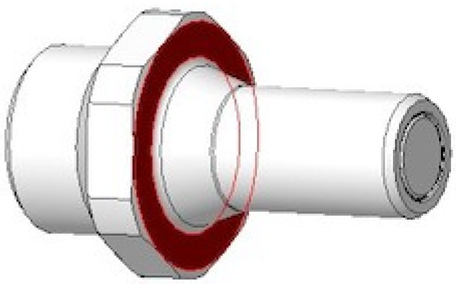


Figure 134. Mating surface BLQ4 insert

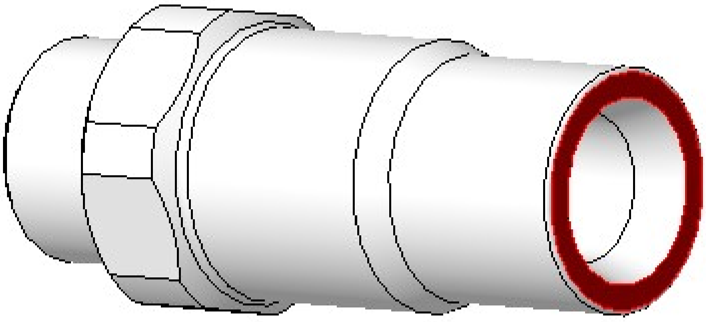


Figure 135. Mating surface BLQ4 body

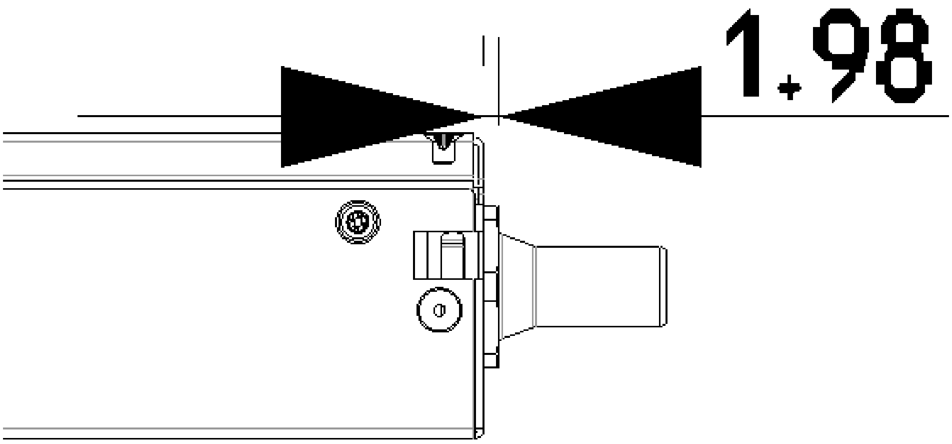


Figure 136. Offset from cage back panel to connector mating interface

The BLQ4 connectors must be within 3mm of this fully engaged state to achieve full flow.

6.5.3. Two phase

Two-phase connector selection

The two-phase cooling solution shall use the CPC BLQ family of interfaces.

On brick side

Figure 137. Liquid - CPC BLQ4 series insert P/N BLQ4D47004PROTO (chrome plated brass) or P/N 4649800PROTO (Aluminum)

Figure 138. Vapor - CPC BLQ6 series insert P/N 4650000PROTO (chrome plated brass)

On the Manifold side

Figure 139. Liquid - CPC BLQ4 series body P/N BLQ4D31004PROTO (chrome plated brass) or P/N 4649900PROTO (Aluminum)

Figure 140. Vapor - CPC BLQ6 series body P/N 4650200PROTO (chrome plated brass)

Two-phase connector mounting polarization

Cage manifolds shall have body (female) blind mate connectors. Bricks shall have insert (male) blind-mate connectors. When viewed from rear of the brick, vapor connection shall be located at the right position. Liquid connection shall be located at the left position.

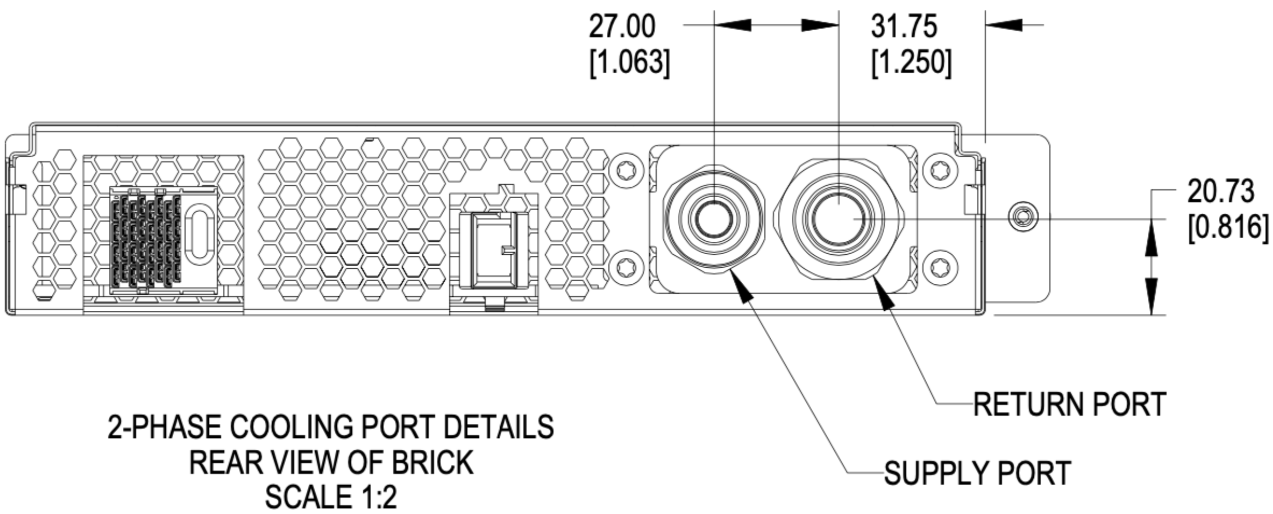


Figure 141. Brick liquid supply and Vapor return

Two-phase brick side connector locations

Using the right-side panel of the brick as a reference, the Vapor return connector on the brick should be 31.85mm in the x direction and 20.73 upwards in the y direction.

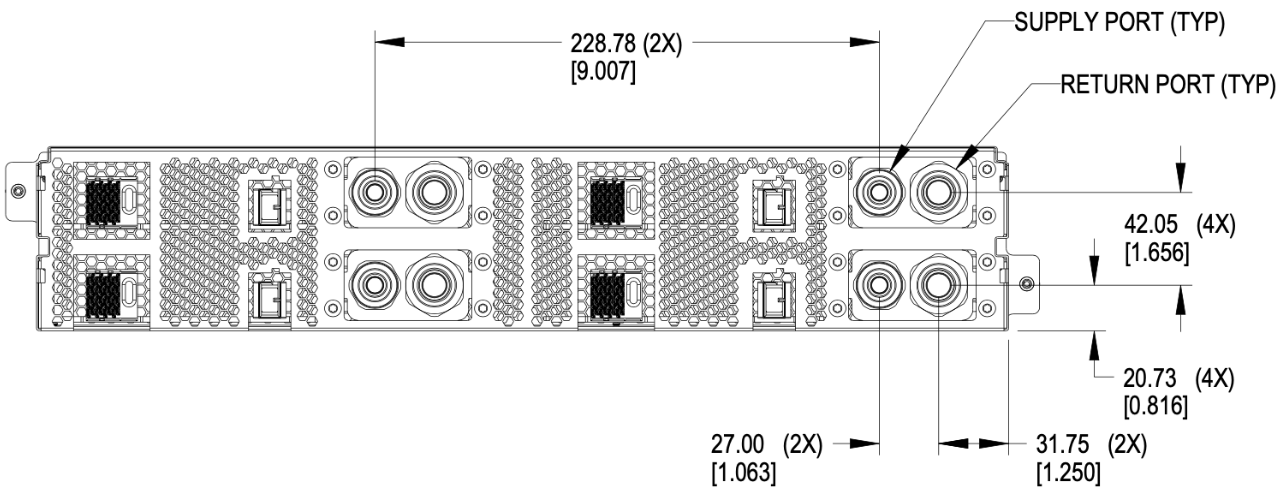


Figure 142. Brick side connectors locations

Two-phase offset depth

The mating surface of the brick side connector sits xx mm behind (outside) the outer edge of the back plane of the brick. Specifying this dimension ensures the server manufacturer and manifold/cage side implementor have a consistent location where the connectors must meet to ensure full mating and thereby full flow.

The reference mating surfaces of each connector are highlighted below along with a drawing showing the specified dimension.

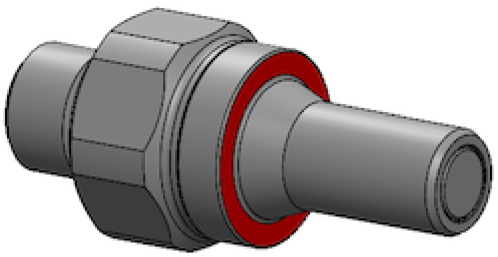


Figure 143. Mating surface BLQ4 insert

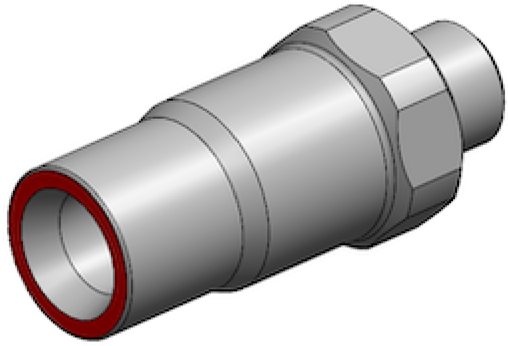


Figure 144. Mating surface BLQ4 body

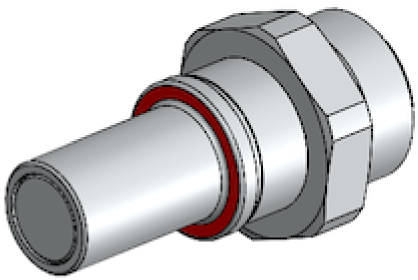


Figure 145. Mating surface BLQ6 insert

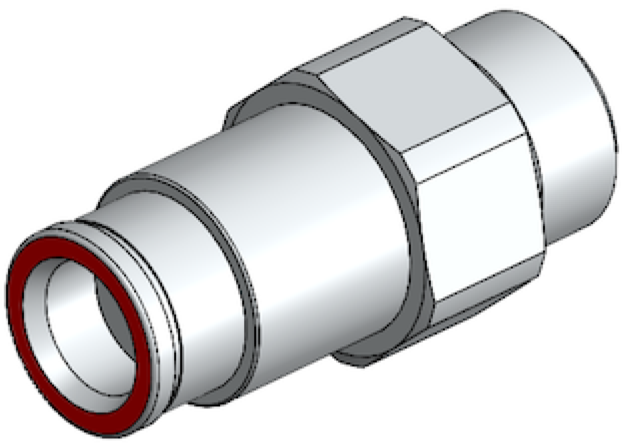


Figure 146. Mating surface BLQ6 body

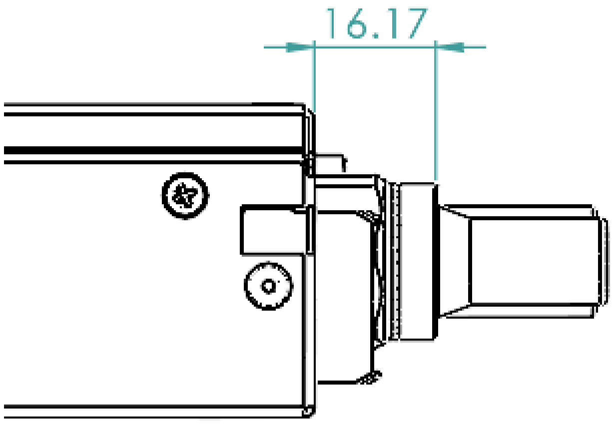


Figure 147. Offset from cage back panel to BLQ4 connector mating interface

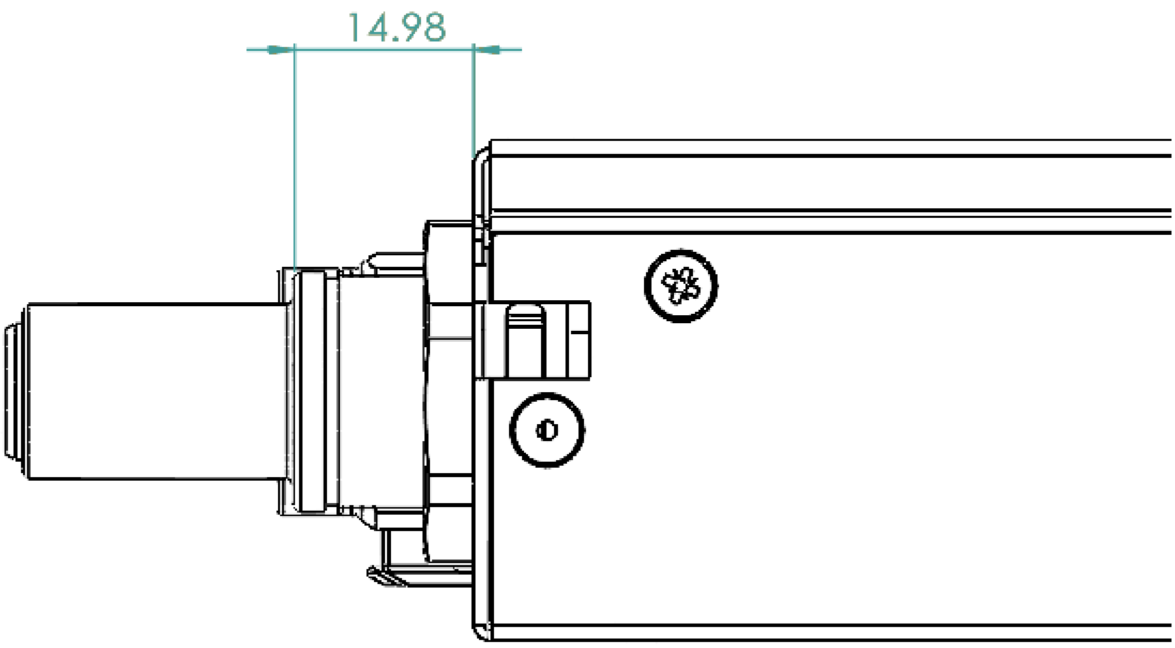


Figure 148. Offset from cage back panel to BLQ6 connector mating interface

The BLQ4 & BLQ6 connectors must be within ±1.5mm of this engaged state to achieve full flow.

6.6. Design guidance

Design guidance is not a specification requirement. It is simple design guidance that can be implemented to simplify future needs, to meet certification requirements, or satisfy general fit & finish.

6.6.1. Brick faceplate guidance

The faceplate is notched above and below the fastener (See Figure 116, “Brick retention”). Note that the faceplate can accommodate any type of ports or drives as needed and is not required to be fully perforated.

Screws and/or rivets used to secure the top cover are flush to be under-flush. The vertical edges on the rear of the brick chassis base are coined to break sharp edges.

6.6.2. Back panel safety perforation

See below for typical dimensions of the airflow perforation. All dimensions below are critical.

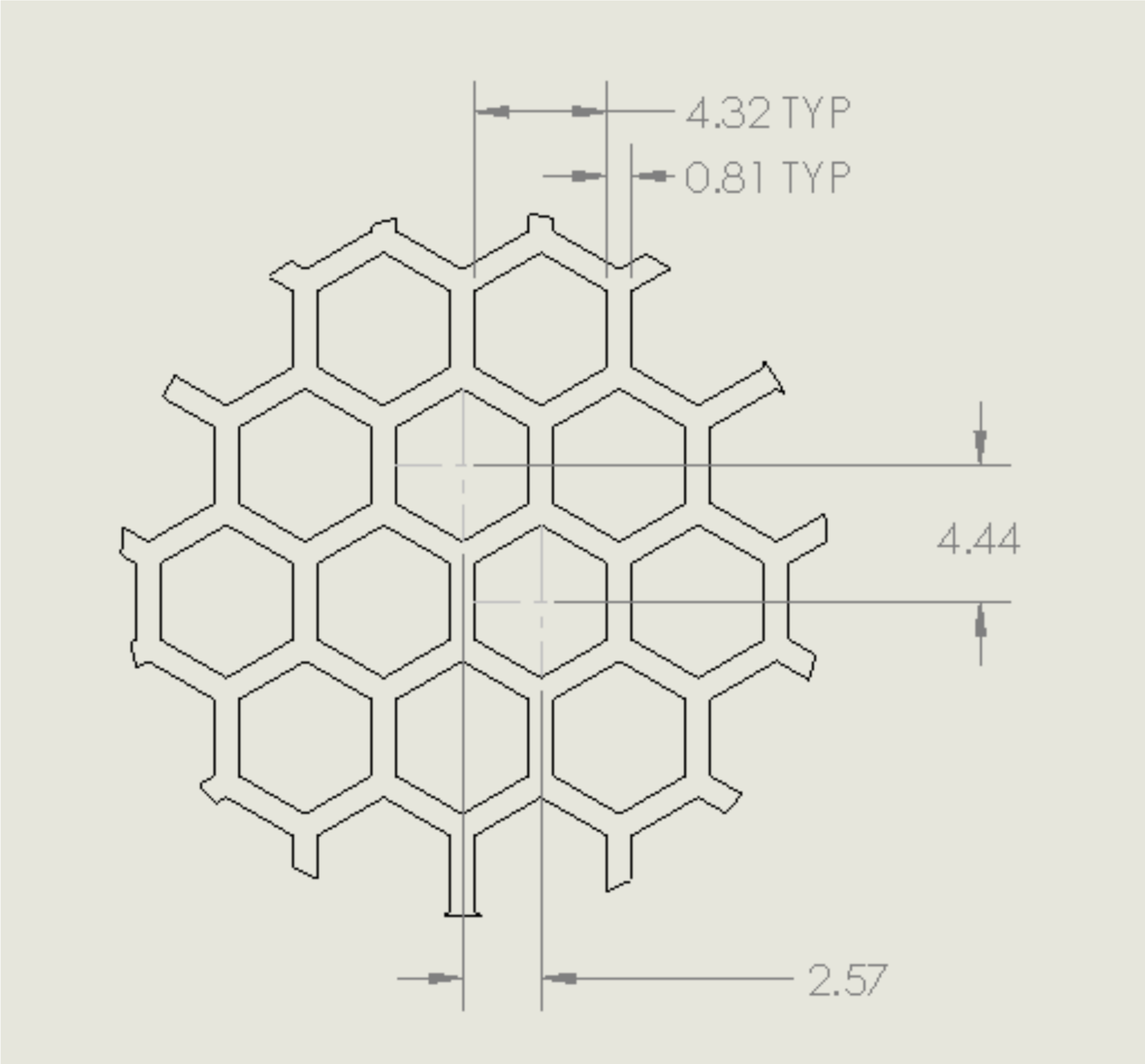


Figure 149. Brick panel perforation

Keep-out zones

Keep-out zones are defined in the brick surrounding the power connector and liquid cooling interfaces. The liquid cooling keep-out spaces "fly" above the PCB, allowing components to be populated below the connections.

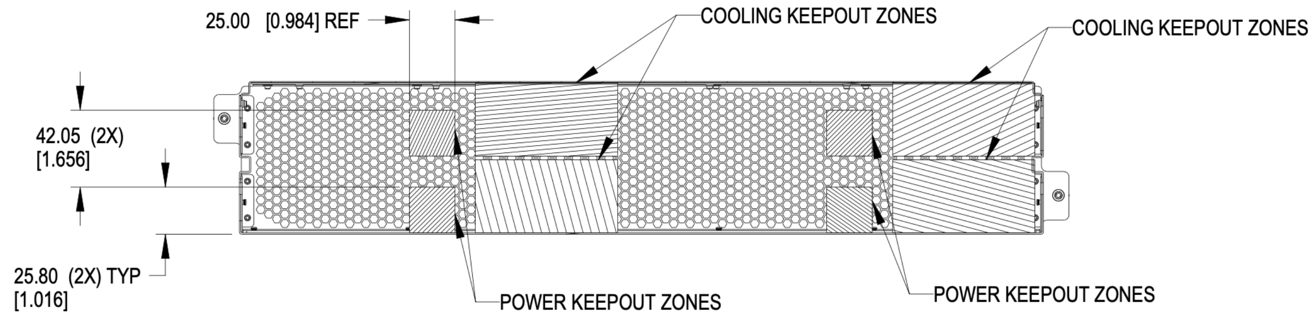


Figure 150. Brick keep-out zone

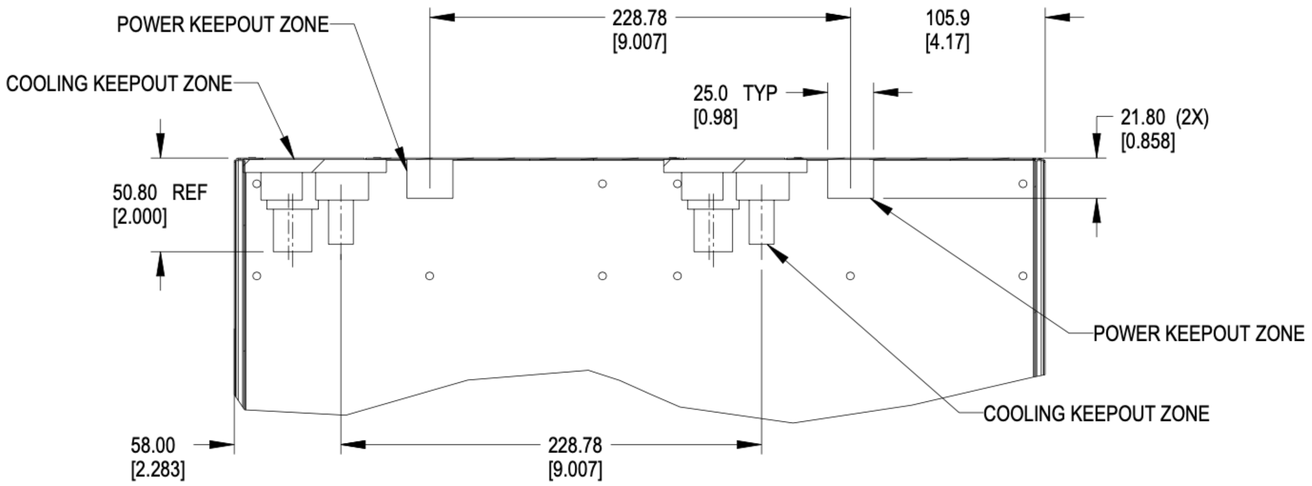


Figure 151. Brick keep-out zone top view

6.7. Brick performance

6.7.1. Brick electrical startup behavior

In order to safely operate the power shelf, we define the following characteristics for brick startup behavior.

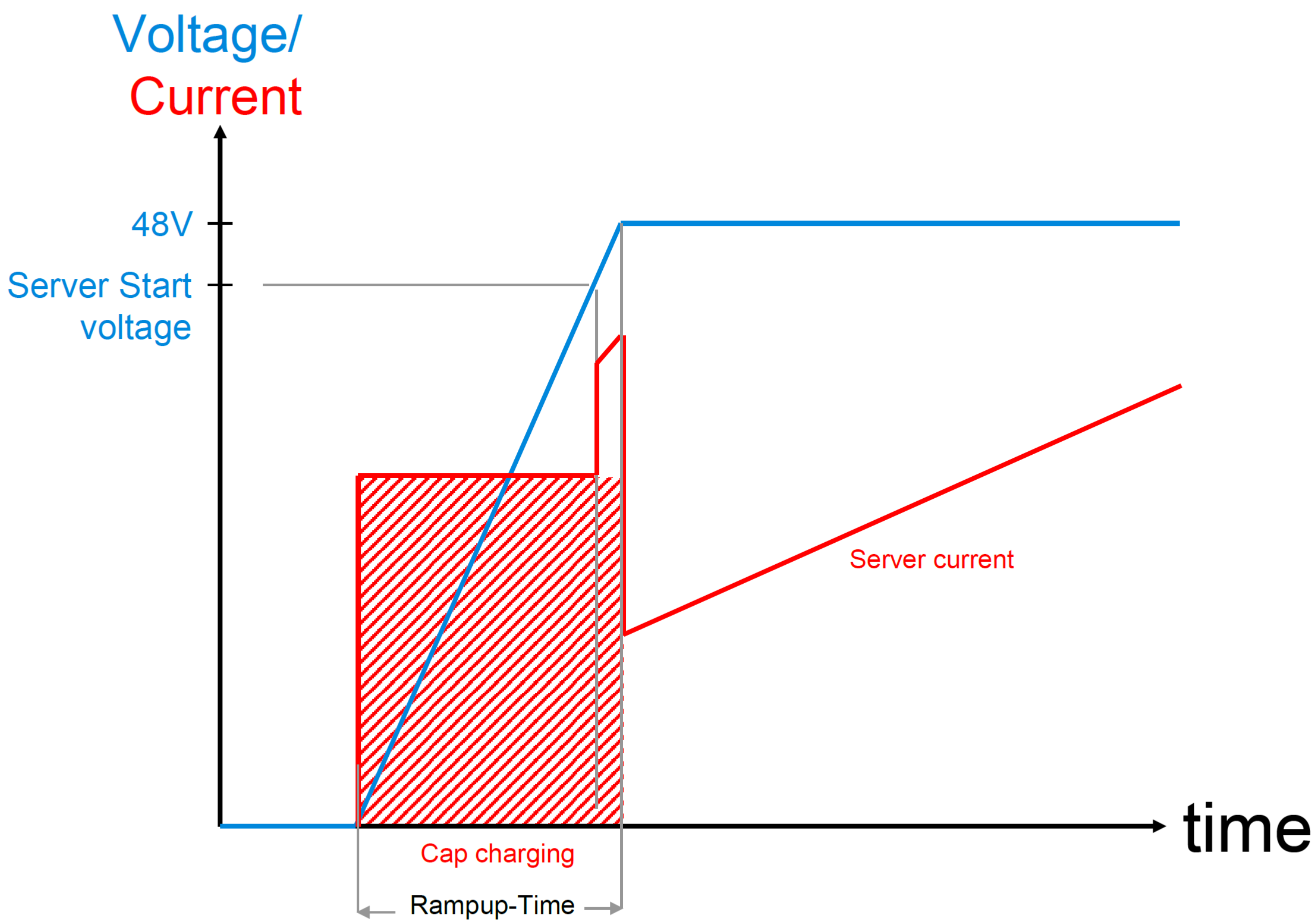


Figure 152. Server startup electrical characteristics

These values are per connector feed from a power shelf.

Four power connectors are available for use.

- Min start-up voltage: +44V DC
- Max input capacitance per brick feed: ⇐1000uF
- Max server current at start-up voltage: 70A
- Ramp time: Follow Intel CRPS spec: ramp time 10-70msec, slew rate 0.5V/msec

https://www.intel.com/content/dam/www/public/us/en/documents/guides/power-supply-design-guide-june.pdf

Data connector pin assignments

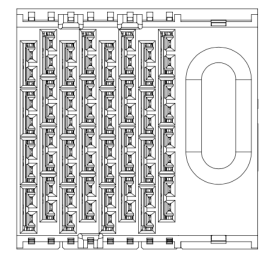
WHEN VIEWED FROM PCB DAUGHTER CARD CONNECTOR REAR BRICK PANEL
DIRECTLY FACING CONNECTOR. PIN AC1 is BOTTOM LEFT CORNER

2022 December 4

	C1	C2	C3	C4	C5	C6	C7	C8	
H	DP4-	DP8-	DP12-	DP16-	DP20-	DP24-	DP28-	DP32-	Key
G	DP4+	DP8+	DP12+	DP16+	DP20+	DP24+	DP28+	DP32+	
F	DP3-	DP7-	DP11-	DP15-	DP19-	DP23-	DP27-	DP31-	
E	DP3+	DP7+	DP11+	DP15+	DP19+	DP23+	DP27+	DP31+	
D	DP2-	DP6-	DP10-	DP14-	DP18-	DP22-	DP26-	DP30-	
C	DP2+	DP6+	DP10+	DP14+	DP18+	DP22+	DP26+	DP30+	
B	DP1-	DP5-	DP9-	DP13-	DP17-	DP21-	DP25-	DP29-	
A	DP1+	DP5+	DP9+	DP13+	DP17+	DP21+	DP25+	DP29+	

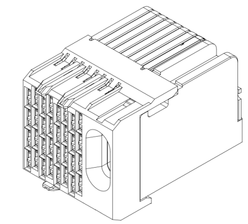
GROUND SHIELD TO GROUND

TOP



KEY

TOP



KEY

	C1	C2	C3	C4	C5	C6	C7	C8	
H									Key
G			SFP	SFP					
F									
E	RJ45	RJ45			QSFP		QSFP		
D									
C			SFP	SFP					
B									
A									

	C1	C2	C3	C4	C5	C6	C7	C8	
H									Key
G			SFP2 RX	SFP4 RX					
F									
E	RJ45_1	RJ45_2	SFP1 RX	SFP3 RX	QSFP1		QSFP2		
D									
C			SFP2 TX	SFP4 TX					
B									
A			SFP1 TX	SFP3 TX					

	C1	C2	C3	C4	C5	C6	C7	C8	
H	RJ45 PIN8	RJ45 PIN8	SFP2	SFP4	QSFP1	QSFP1	QSFP2	QSFP2	Key
G	RJ45 PIN7	RJ45 PIN7	LN1 RX	LN1 RX	LN2 RX	LN1 RX	LN2 RX	LN1 RX	
F	RJ45 PIN5	RJ45 PIN5	SFP1	SFP3	QSFP1	QSFP1	QSFP2	QSFP2	
E	RJ45 PIN4	RJ45 PIN4	LN1 RX	LN1 RX	LN4 RX	LN3 RX	LN4 RX	LN3 RX	
D	RJ45 PIN6	RJ45 PIN6	SFP2	SFP4	QSFP1	QSFP1	QSFP2	QSFP2	
C	RJ45 PIN3	RJ45 PIN3	LN1 TX	LN1 TX	LN3 TX	LN4 TX	LN3 TX	LN4 TX	
B	RJ45 PIN2	RJ45 PIN2	SFP1	SFP3	QSFP1	QSFP1	QSFP2	QSFP2	
A	RJ45 PIN1	RJ45 PIN1	LN1 TX	LN1 TX	LN1 TX	LN2 TX	LN1 TX	LN2 TX	

GROUND SHIELD TO GROUND

	C1	C2	C3	C4	C5	C6	C7	C8	
H	RJ45_1_BI_DD-	RJ45_2_BI_DD-	SFP2_LN1_RX_N	SFP4_LN1_RX_N	QSFP1_LN2_RX_N	QSFP1_LN1_RX_N	QSFP2_LN2_RX_N	QSFP2_LN1_RX_N	Key
G	RJ45_1_BI_DD+	RJ45_2_BI_DD+	SFP2_LN1_RX_P	SFP4_LN1_RX_P	QSFP1_LN2_RX_P	QSFP1_LN1_RX_P	QSFP2_LN2_RX_P	QSFP2_LN1_RX_P	
F	RJ45_1_BI_DC-	RJ45_2_BI_DC-	SFP1_LN1_RX_N	SFP3_LN1_RX_N	QSFP1_LN4_RX_N	QSFP1_LN3_RX_N	QSFP2_LN4_RX_N	QSFP2_LN3_RX_N	
E	RJ45_1_BI_DC+	RJ45_2_BI_DC+	SFP1_LN1_RX_P	SFP3_LN1_RX_P	QSFP1_LN4_RX_P	QSFP1_LN3_RX_P	QSFP2_LN4_RX_P	QSFP2_LN3_RX_P	
D	RJ45_1_BI_DB-	RJ45_2_BI_DB-	SFP2_LN1_TX_N	SFP4_LN1_TX_N	QSFP1_LN3_TX_N	QSFP1_LN4_TX_N	QSFP2_LN3_TX_N	QSFP2_LN4_TX_N	
C	RJ45_1_BI_DB+	RJ45_2_BI_DB+	SFP2_LN1_TX_P	SFP4_LN1_TX_P	QSFP1_LN3_TX_P	QSFP1_LN4_TX_P	QSFP2_LN3_TX_P	QSFP2_LN4_TX_P	
B	RJ45_1_BI_DA-	RJ45_2_BI_DA-	SFP1_LN1_TX_N	SFP3_LN1_TX_N	QSFP1_LN1_TX_N	QSFP1_LN2_TX_N	QSFP2_LN1_TX_N	QSFP2_LN2_TX_N	
A	RJ45_1_BI_DA+	RJ45_2_BI_DA+	SFP1_LN1_TX_P	SFP3_LN1_TX_P	QSFP1_LN1_TX_P	QSFP1_LN2_TX_P	QSFP2_LN1_TX_P	QSFP2_LN2_TX_P	

GROUND SHIELD TO GROUND

AB = CD = Tx FOR SFP & QSFP

EF = GH = Rx FOR SFP & QSFP

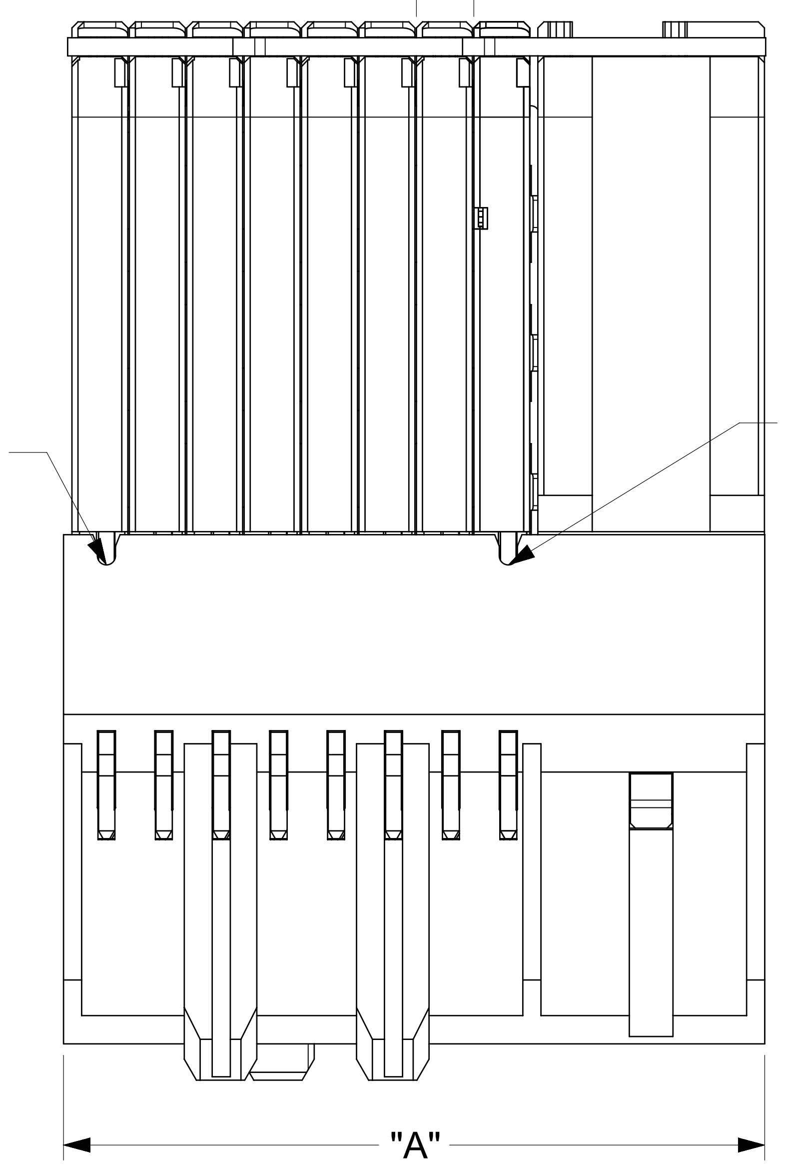
AB = CD = Tx FOR SFP & QSFP

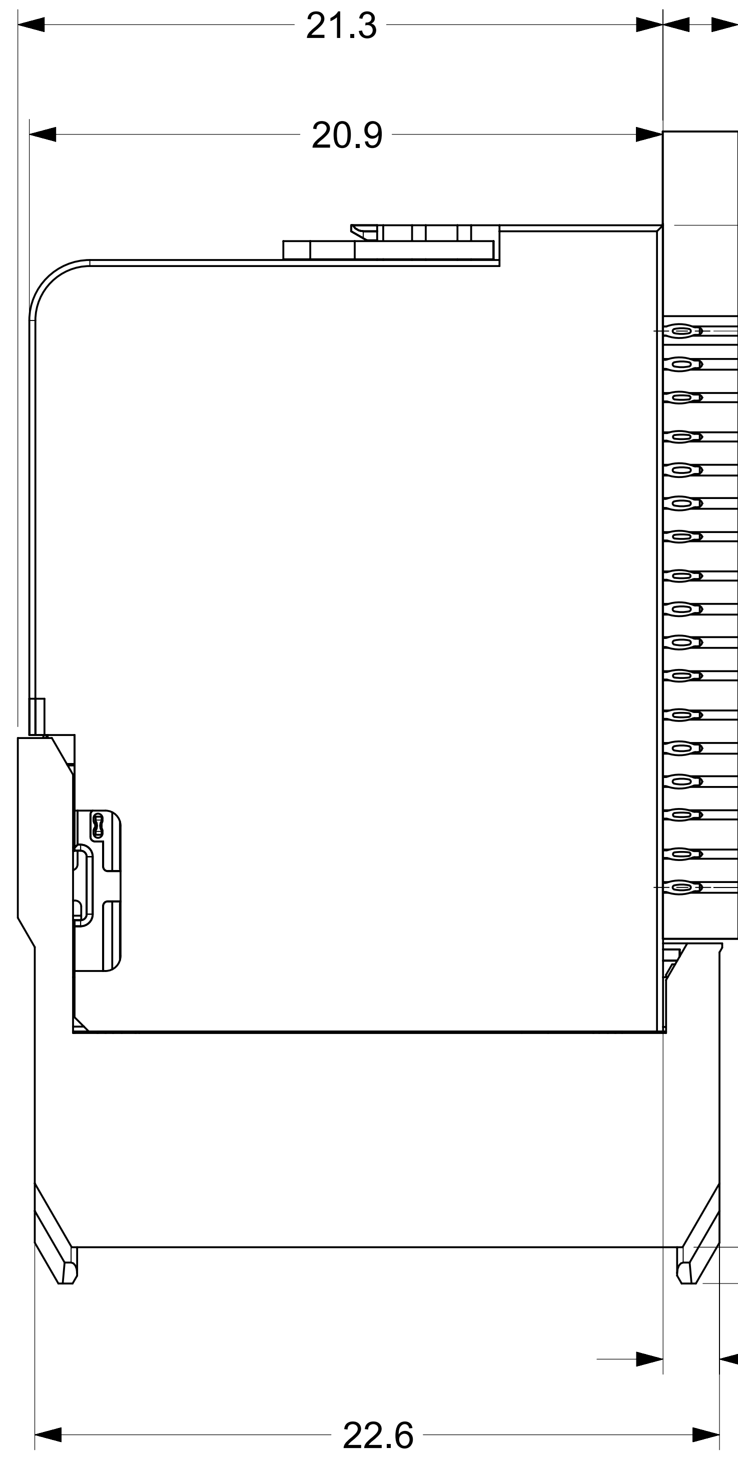
EF = GH = Rx FOR SFP & QSFP

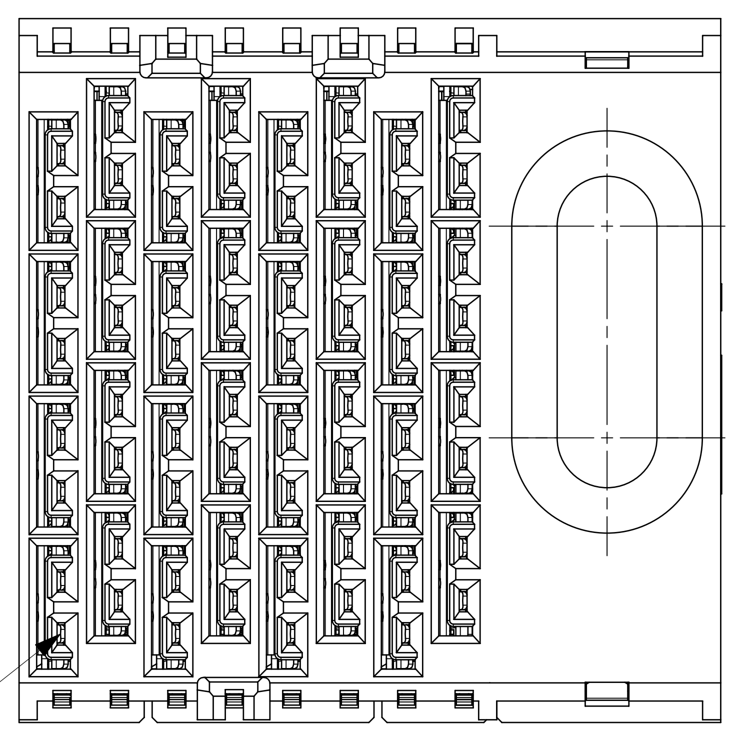
Chapter 7. Private Appendix

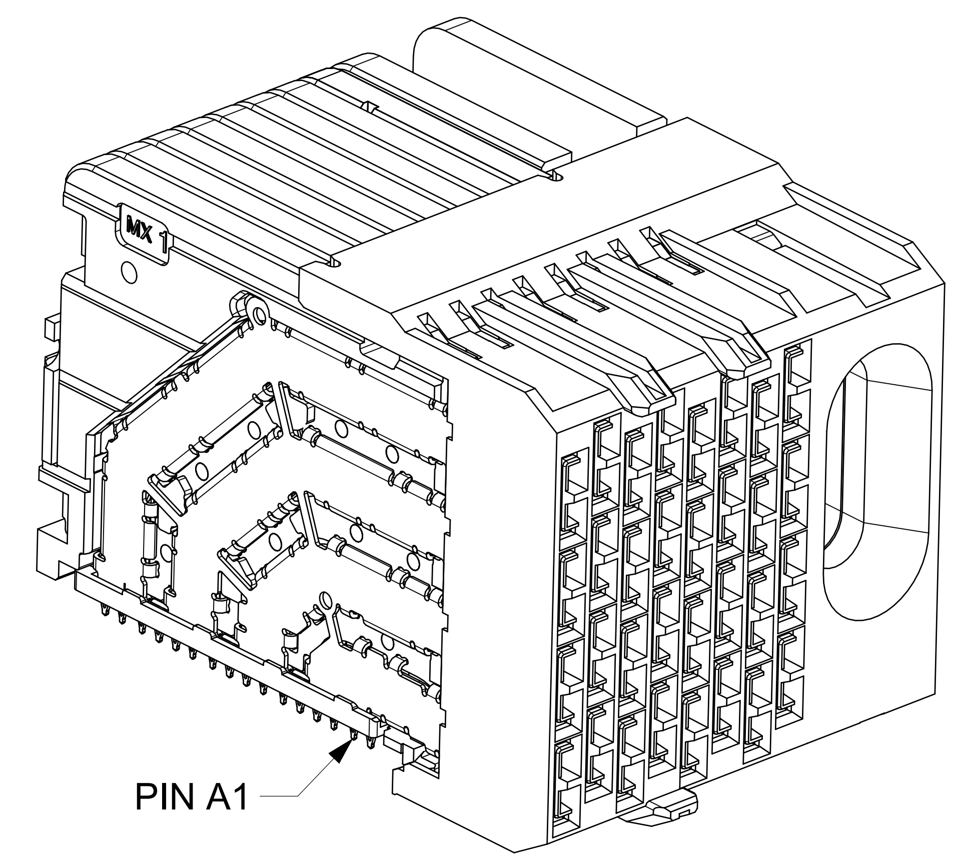
The following material is included for the convenience of Open19 members only. No redistribution to the public is permitted.

7.1. Data connector sales drawing









NOTES:

1. MATERIALS: HOUSING - LIQUID CRYSTAL POLYMER (LCP)
   GLASS-FILLED, UL94V-0
   TERMINALS - HIGH PERFORMANCE COPPER ALLOY
2. FINISH: 30uIN GOLD IN CONTACT AREA.
   SELECTIVE TIN ON PCB TAILS.
   NICKEL OVERALL.
3. REFER TO MOLEX PRODUCT SPECIFICATION 172740-9990 FOR PERFORMANCE SPECIFICATIONS.
4. THIS PART CONFORMS TO CLASS B REQUIREMENTS OF MOLEX COSMETIC SPEC PS-45499-002.
5. PRODUCT WILL BE PACKAGED PER PK-78888-085.
6. SEE SHEET 2 FOR HOLE PATTERN DIMENSIONS.
7. MATES WITH IMPEL CABLE SERIES 730650.
8. REFER TO IMPEL PLUS PCB ROUTING GUIDE AS-172730-990 FOR ANTIPAD, ROUTING RECOMMENDATIONS, AND ADDITIONAL PCB INFORMATION.
9. WHEN STACKING MODULES END TO END, THE MINIMUM COLUMN TO COLUMN HOLE SPACING IS 10.52 MM.
10. COMPONENT PLACEMENT AND TRACE KEEP OUT ZONE; DO NOT PLACE ANY OTHER COMPONENT OR ROUTE ANY TOP LAYER TRACES IN THIS AREA.
11. CONNECTOR IS SUPPLIED WITH TWO 2-32 THREAD FORMING SCREWS. REFER TO TABLE ON SHEET 2 FOR MOLEX PART NUMBER.
12. WHEN USING THE SUPPLIED 2-32 THREAD FORMING SCREWS, REFER TO TABLE ON SHEET 2 FOR THE APPLICABLE BOARD THICKNESS RANGE.

THIS DESIGN IS BASED ON DESIGN OBJECTIVES AND IS STRICTLY TENTATIVE. IT MAY CHANGE BASED ON RESULTS OF ADDITIONAL DESIGN REVIEWS & VERIFICATIONS.

DIMENSION UNITS	SCALE	
mm	NTS	CURRENT REV DESC: 1. AMEND "QSP" TO "QSFP" ON PIN OUT TABLE, SHEET 2. 2. ADD "PCB EDGE" NOTE ON SHEET 2 C6

GENERAL TOLERANCES (UNLESS SPECIFIED)		
ANGULAR TOL	±	°
4 PLACES	±	
3 PLACES	±	
2 PLACES	±	0.13
1 PLACE	±	0.25
0 PLACES	±	

DRAFT WHERE APPLICABLE MUST REMAIN WITHIN DIMENSIONS

EC NO:	714548	
DRWN:	JLTAN06	2022/07/06
CHK'D:	YLLIM	2022/07/18
APPR:	DWLEE01	2022/07/18
INITIAL REVISION:		
DRWN:	JLTAN06	2022/04/12
APPR:	DWLEE01	2022/06/27

molex

IMPEL PLUS 4PX8C CUSTOM RIGHT GUIDED DC SPI (OPEN19 V2)

PRODUCT CUSTOMER DRAWING

DOCUMENT NUMBER	DOC TYPE	DOC PART	REVISION
2040669001-SD	PSD	000	A1

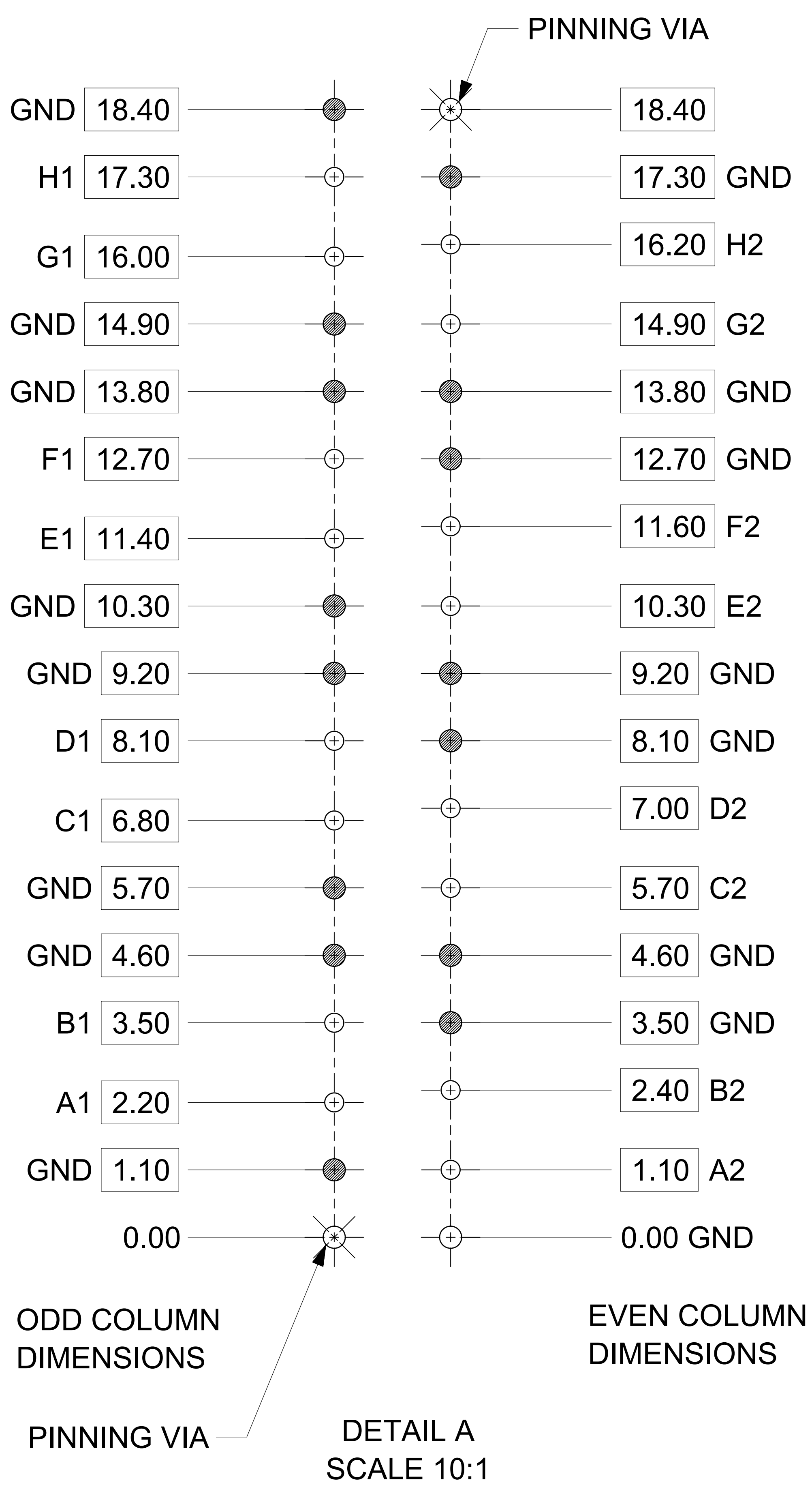
THIRD ANGLE PROJECTION	DRAWING	SERIES	MATERIAL NUMBER	CUSTOMER	SHEET NUMBER
	C-SIZE	204066	SEE TABLE	GENERAL MARKET	1 OF 3

DOCUMENT STATUS	P1	RELEASE DATE	2022/07/18	10:32:07

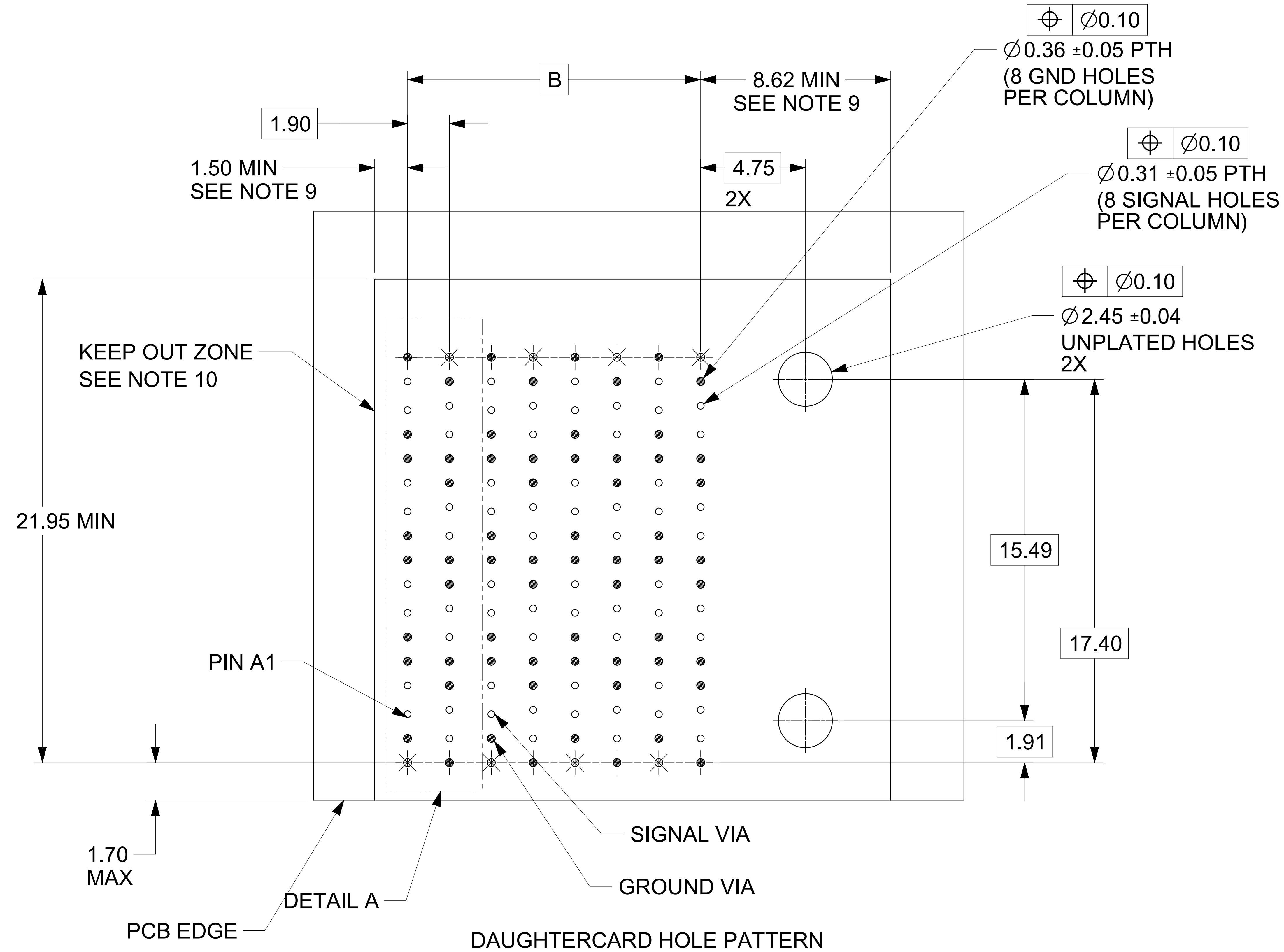
MATERIAL NUMBER	NUMBER OF COLUMNS	NUMBER OF DIFF PAIR	DIM "A"	DIM "B"	SUPPLIED SCREW PN	APPLICABLE BOARD THICKNESS
2040669038	8	32	23.40 MAX	13.30	737263100	2.20 TO 4.10 MM
2040669138	8	32	23.40 MAX	13.30	737260015	1.00 TO 1.90 MM

BRICK (4X8) PIN OUT MAPPING

PIN	COLUMN 1	COLUMN 2	COLUMN 3	COLUMN 4	COLUMN 5	COLUMN 6	COLUMN 7	COLUMN 8
H	BI_DD-	NC	SFP2_LN1_RX_N	SFP4_LN1_RX_N	QSFP1_LN2_RX_N	QSFP1_LN1_RX_N	QSFP2_LN2_RX_N	QSFP2_LN1_RX_N
G	BI_DD+		SFP2_LN1_RX_P	SFP4_LN1_RX_P	QSFP1_LN2_RX_P	QSFP1_LN1_RX_P	QSFP2_LN2_RX_P	QSFP2_LN1_RX_P
F	BI_DC-	NC	SFP1_LN1_RX_N	SFP3_LN1_RX_N	QSFP1_LN4_RX_N	QSFP1_LN3_RX_N	QSFP2_LN4_RX_N	QSFP2_LN3_RX_N
E	BI_DC+		SFP1_LN1_RX_P	SFP3_LN1_RX_P	QSFP1_LN4_RX_P	QSFP1_LN3_RX_P	QSFP2_LN4_RX_P	QSFP2_LN3_RX_P
D	BI_DB-	NC	SFP2_LN1_TX_N	SFP4_LN1_TX_N	QSFP1_LN3_TX_N	QSFP1_LN4_TX_N	QSFP2_LN3_TX_N	QSFP2_LN4_TX_N
C	BI_DB+		SFP2_LN1_TX_P	SFP4_LN1_TX_P	QSFP1_LN3_TX_P	QSFP1_LN4_TX_P	QSFP2_LN3_TX_P	QSFP2_LN4_TX_P
B	BI_DA-	NC	SFP1_LN1_TX_N	SFP3_LN1_TX_N	QSFP1_LN1_TX_N	QSFP1_LN2_TX_N	QSFP2_LN1_TX_N	QSFP2_LN2_TX_N
A	BI_DA+		SFP1_LN1_TX_P	SFP3_LN1_TX_P	QSFP1_LN1_TX_P	QSFP1_LN2_TX_P	QSFP2_LN1_TX_P	QSFP2_LN2_TX_P



DETAIL A
SCALE 10:1



DAUGHTERCARD HOLE PATTERN
(CONNECTOR SIDE)

DIMENSION UNITS: mm | SCALE: NTS

GENERAL TOLERANCES (UNLESS SPECIFIED)
ANGULAR TOL ± °
4 PLACES ±
3 PLACES ±
2 PLACES ± 0.13
1 PLACE ± 0.25
0 PLACES ±

DRAFT WHERE APPLICABLE MUST REMAIN WITHIN DIMENSIONS

CURRENT REV DESC: 1. AMEND "QSP" TO "QSFP" ON PIN OUT TABLE, SHEET 2. 2. ADD "PCB EDGE" NOTE ON SHEET 2 C6

EC NO: 714548
DRWN: JLTAN06 2022/07/06
CHK'D: YLLIM 2022/07/18
APPR: DWLEE01 2022/07/18

INITIAL REVISION:
DRWN: JLTAN06 2022/04/12
APPR: DWLEE01 2022/06/27

THIRD ANGLE PROJECTION | DRAWING: C-SIZE | SERIES: 204066

molex

IMPEL PLUS 4PX8C CUSTOM RIGHT GUIDED DC SPI (OPEN19 V2)

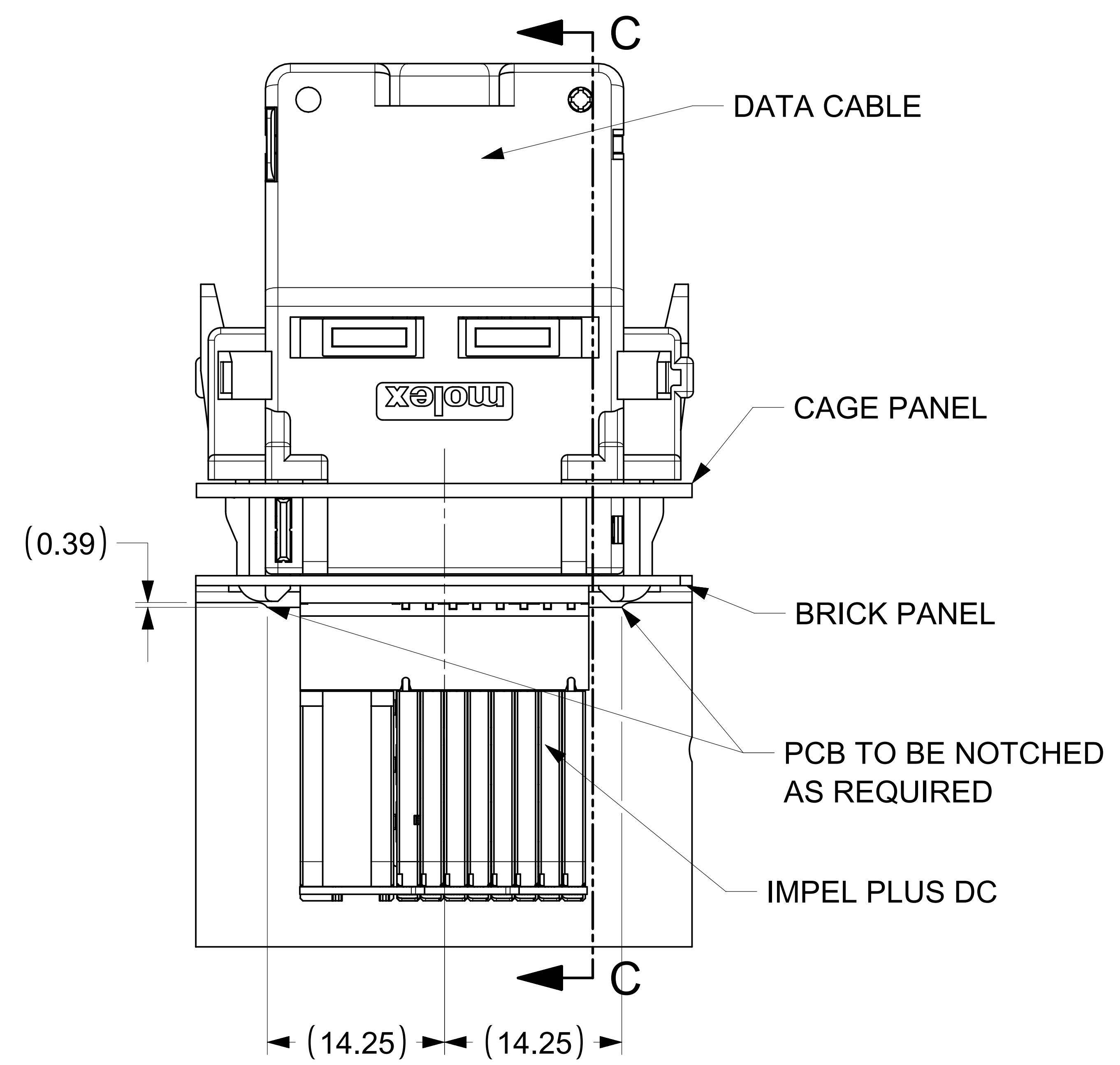
PRODUCT CUSTOMER DRAWING

DOCUMENT NUMBER: 2040669001-SD | DOC TYPE: PSD | DOC PART: 000 | REVISION: A1

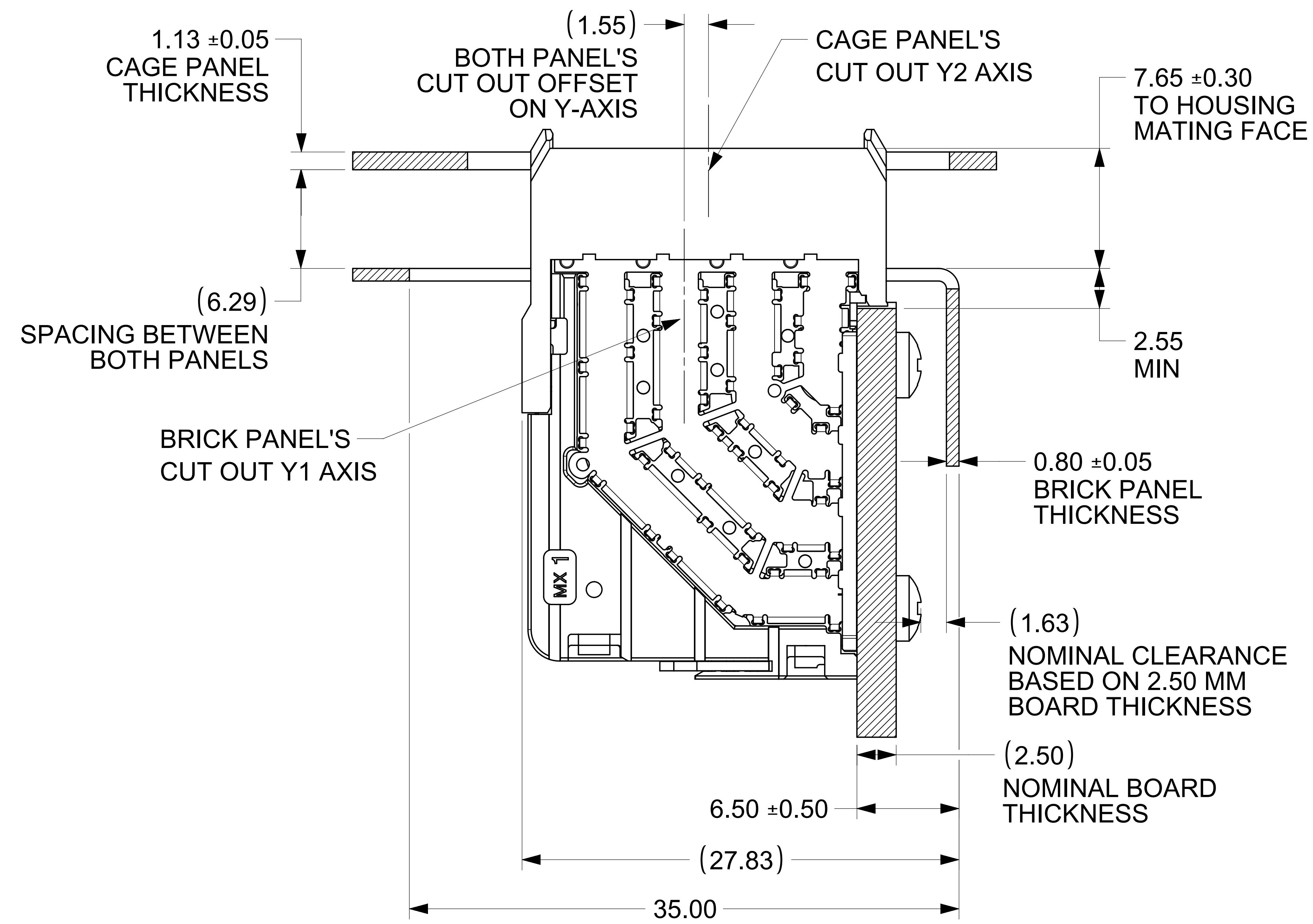
MATERIAL NUMBER: SEE TABLE | CUSTOMER: GENERAL MARKET | SHEET NUMBER: 2 OF 3

DOCUMENT STATUS P1 | RELEASE DATE 2022/07/18 10:32:07

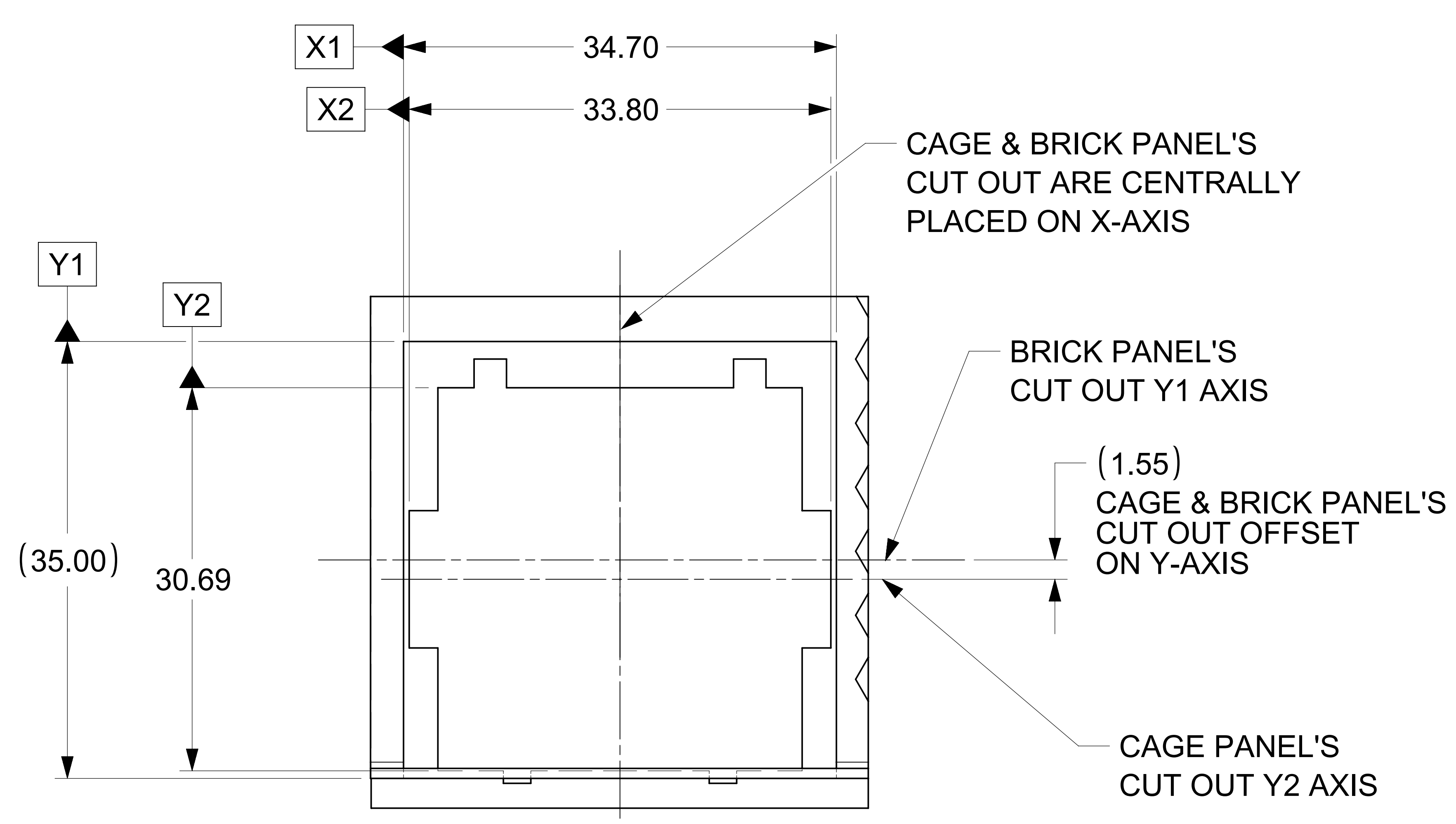
IMPEL PLUS DC MATED TO DATA CABLE



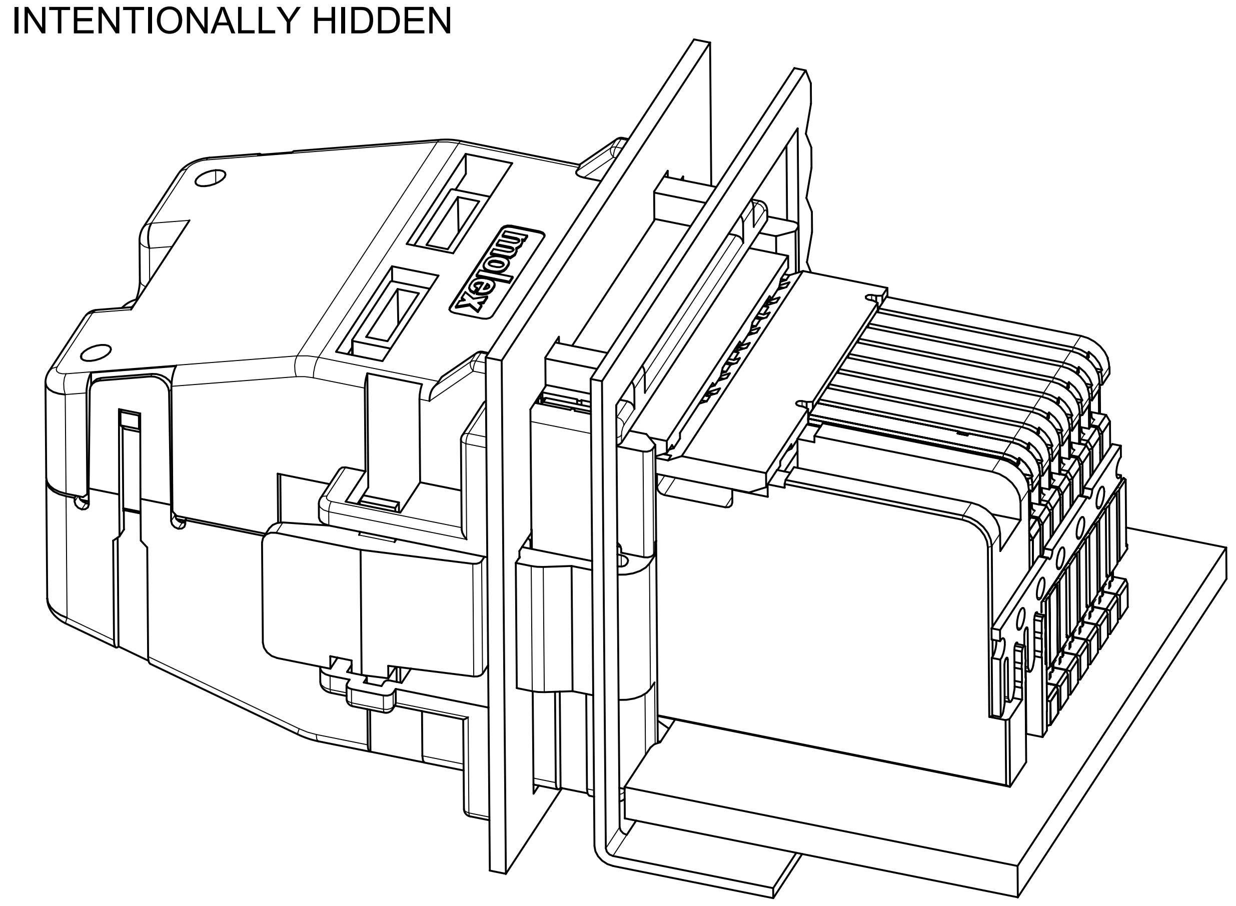
IMPEL PLUS DC MATING PROFILE WITH RESPECT TO PANEL



SECTION C-C
DATA CABLE IS INTENTIONALLY HIDDEN



VIEW D
RECOMMENDED BRICK PANEL
CUT OUT PROFILE



ISOMETRIC VIEW B

DIMENSION UNITS: mm	SCALE: NTS	CURRENT REV DESC: 1. AMEND "QSP" TO "QSFP" ON PIN OUT TABLE, SHEET 2. 2. ADD "PCB EDGE" NOTE ON SHEET 2 C6
GENERAL TOLERANCES (UNLESS SPECIFIED)		
ANGULAR TOL ± °		EC NO: 714548
4 PLACES ±		DRWN: JLTAN06 2022/07/06
3 PLACES ±		CHK'D: YLLIM 2022/07/18
2 PLACES ± 0.13		APPR: DWLEE01 2022/07/18
1 PLACE ± 0.25		INITIAL REVISION: DRWN: JLTAN06 2022/04/12
0 PLACES ±		APPR: DWLEE01 2022/06/27
DRAFT WHERE APPLICABLE MUST REMAIN WITHIN DIMENSIONS	THIRD ANGLE PROJECTION	DRAWING: C-SIZE; SERIES: 204066

molex

IMPEL PLUS 4PX8C CUSTOM RIGHT GUIDED DC SPI (OPEN19 V2)

PRODUCT CUSTOMER DRAWING

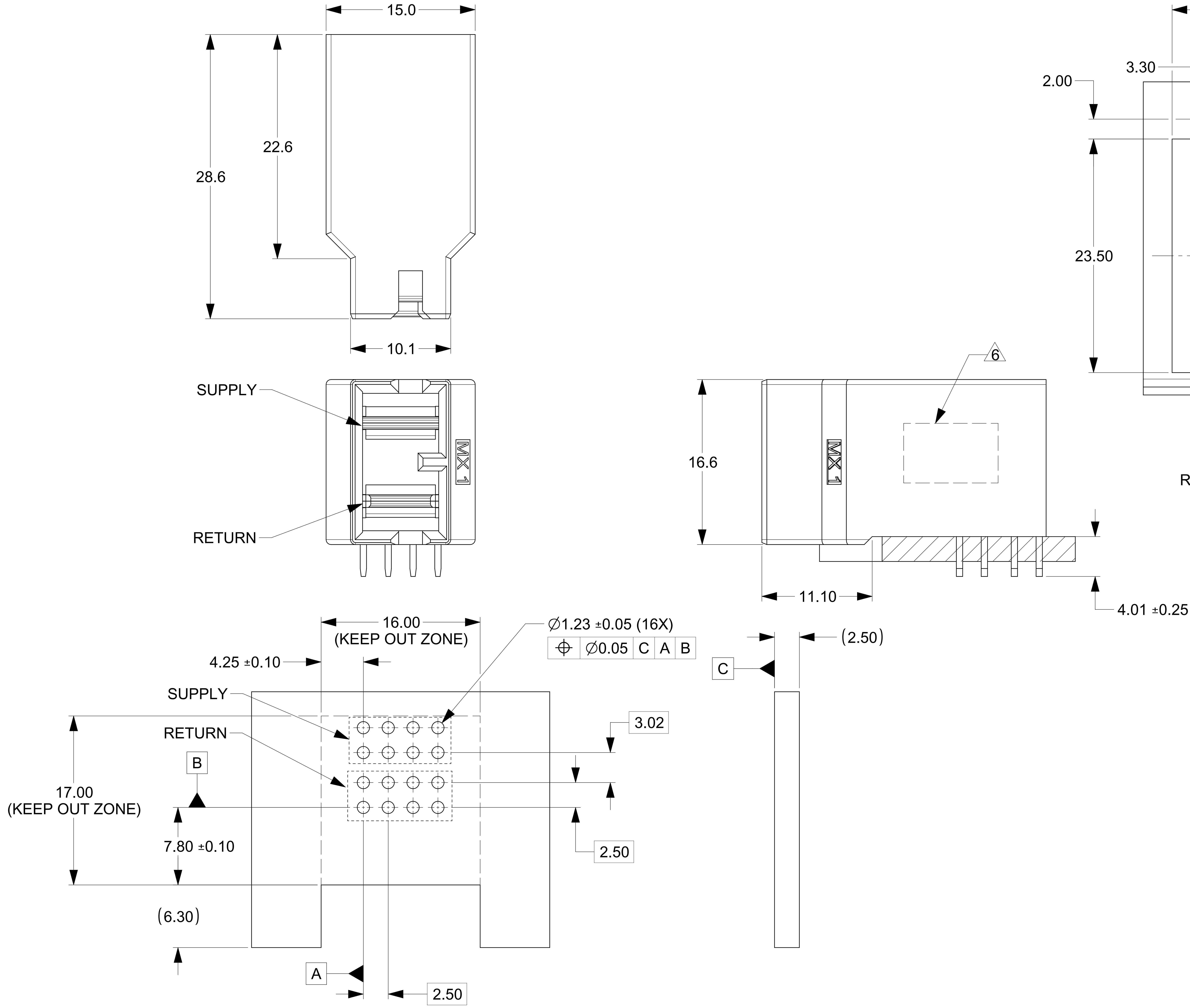
DOCUMENT NUMBER	DOC TYPE	DOC PART	REVISION
2040669001-SD	PSD	000	A1

MATERIAL NUMBER	CUSTOMER	SHEET NUMBER
SEE TABLE	GENERAL MARKET	3 OF 3

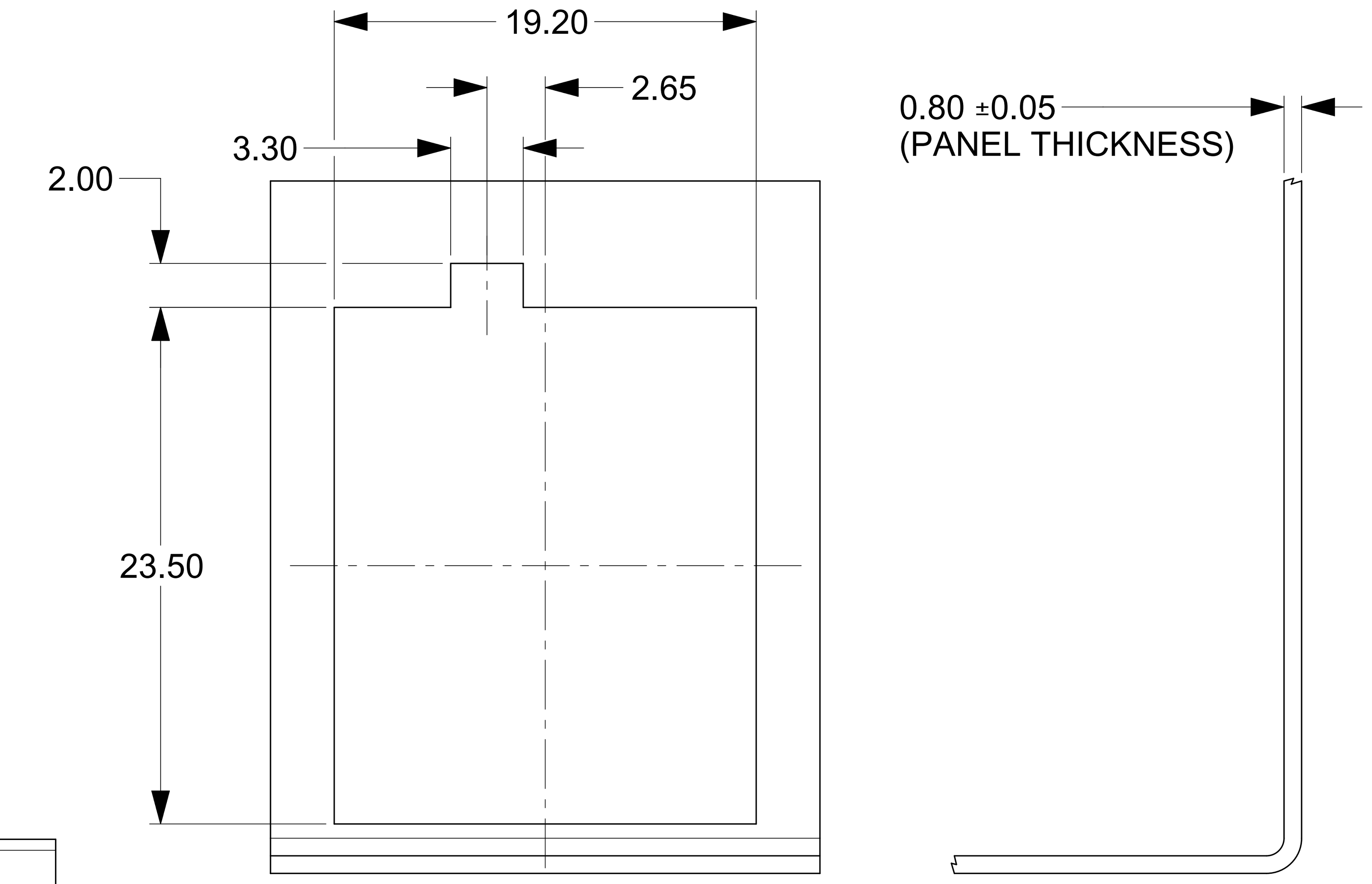
DOCUMENT STATUS	P1	RELEASE DATE	2022/07/18 10:32:07

FORMAT: master-tb-prod-C REVISION: J1 DATE: 2021/05/01

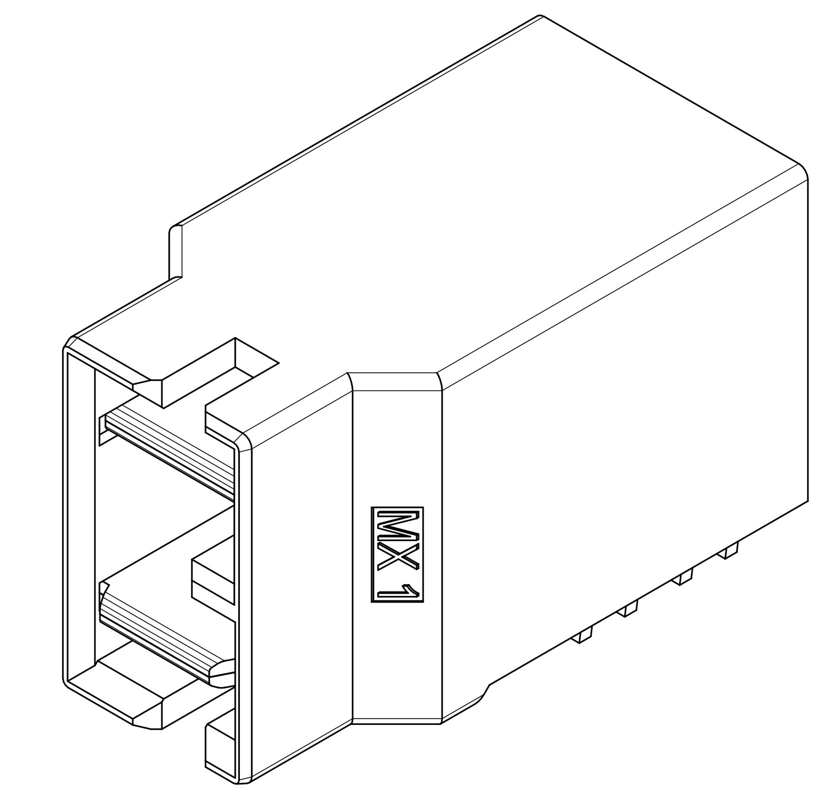
7.2. Power connector sales drawing

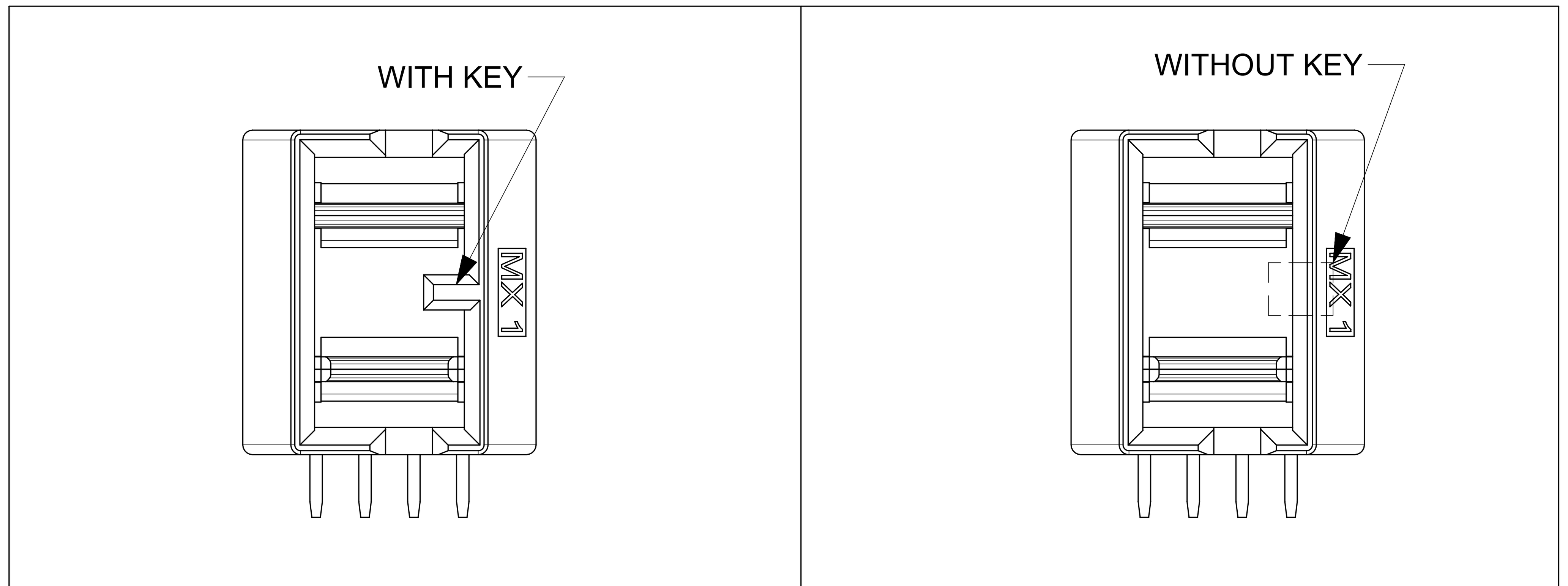


RECOMMENDED PCB LAYOUT
AND KEEP OUT ZONE



RECOMMENDED PANEL
CUTOUT





2203700000 (FOR 70 AMPS)	2203700001 (FOR 35 AMPS)

FOR
INFORMATION
USE ONLY

PART NUMBER	CURRENT RATING	MATES WITH	NOTES
2203700000	70 AMPS	2203650001	WITH KEY
2203700001	35 AMPS	2203650001 2203650002	WITHOUT KEY

NOTES:
1. MATERIAL:
   HOUSING : LCP, GF, UL94V-0, COLOUR: BLACK.
   TERMINAL: COPPER ALLOY.
2. FINISH:
   CONTACT:
   0.76uM MIN GOLD OVER 1.27um MIN NICKEL UNDERPLATE OVERALL.
   SOLDERTAIL:
   2.54uM MIN TIN OVER 1.27uM MIN NICKEL UNDERPLATING OVERALL
3. PRODUCT SPECIFICATION: TBD
4. PACKAGING SPECIFICATIONS: 2203700000-PK
5. MATE WITH MOLEX PART NUMBER : SEE TABLE
6. DATE CODE & PART NUMBER TO BE LASER MARKED AT APPROXIMATE LOCATION AS SHOWN. (TBA)
7. TERMINALS ARE LUBRICATED.

DOCUMENT STATUS	ER	RELEASE DATE	2022/08/30	18:19:45

DIMENSION UNITS	SCALE
mm	4:1

GENERAL TOLERANCES (UNLESS SPECIFIED)

ANGULAR TOL	±	2.0 °
4 PLACES	±	
3 PLACES	±	
2 PLACES	±	0.13
1 PLACE	±	0.25
0 PLACES	±	

DRAFT WHERE APPLICABLE MUST REMAIN WITHIN DIMENSIONS

CURRENT REV DESC:
EC NO:
DRWN: GSHARMA 2022/07/01
CHK'D:
APPR:
INITIAL REVISION:
DRWN: GSHARMA 2022/02/07
APPR:

THIRD ANGLE PROJECTION | DRAWING: D-SIZE | SERIES: 220370

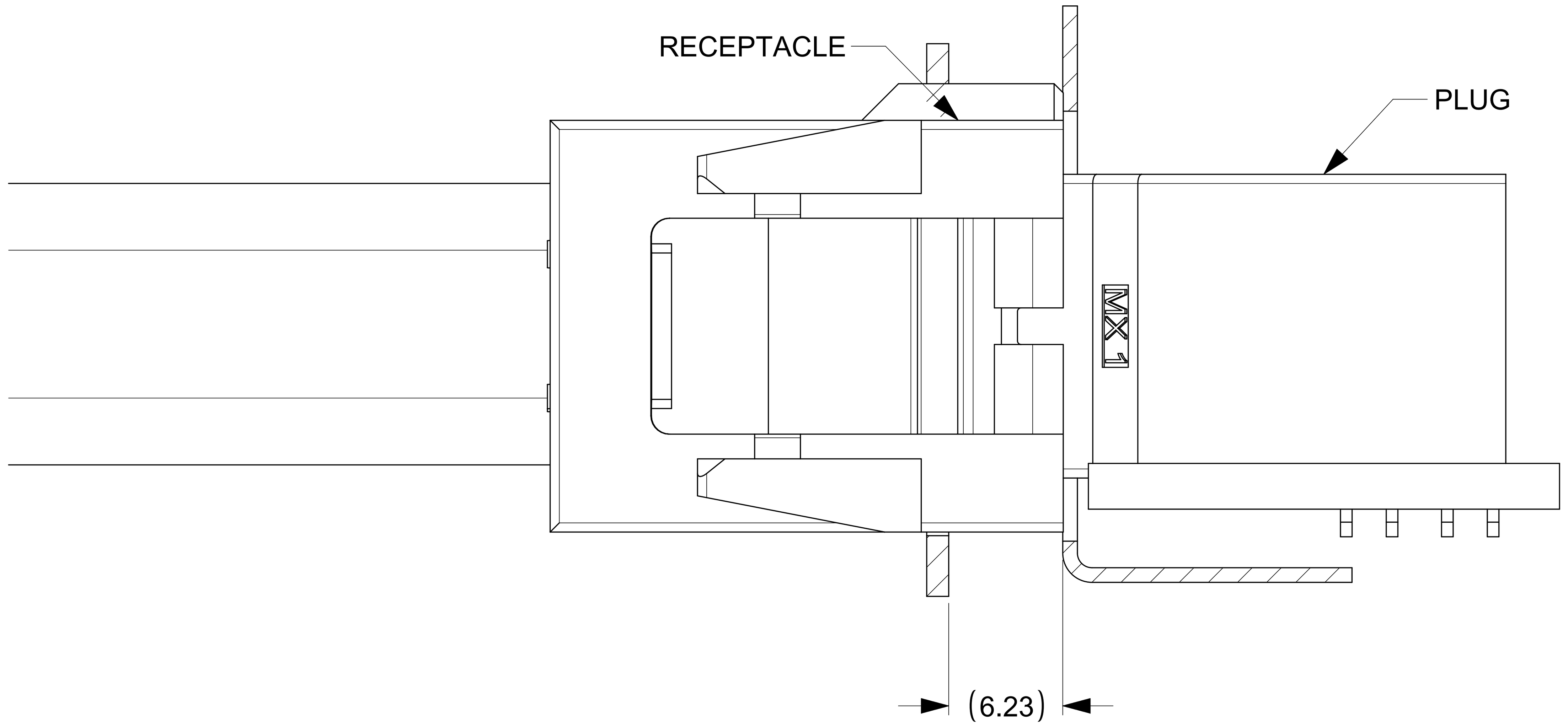
molex

BRICK POWER PLUG CONNECTOR ASSY

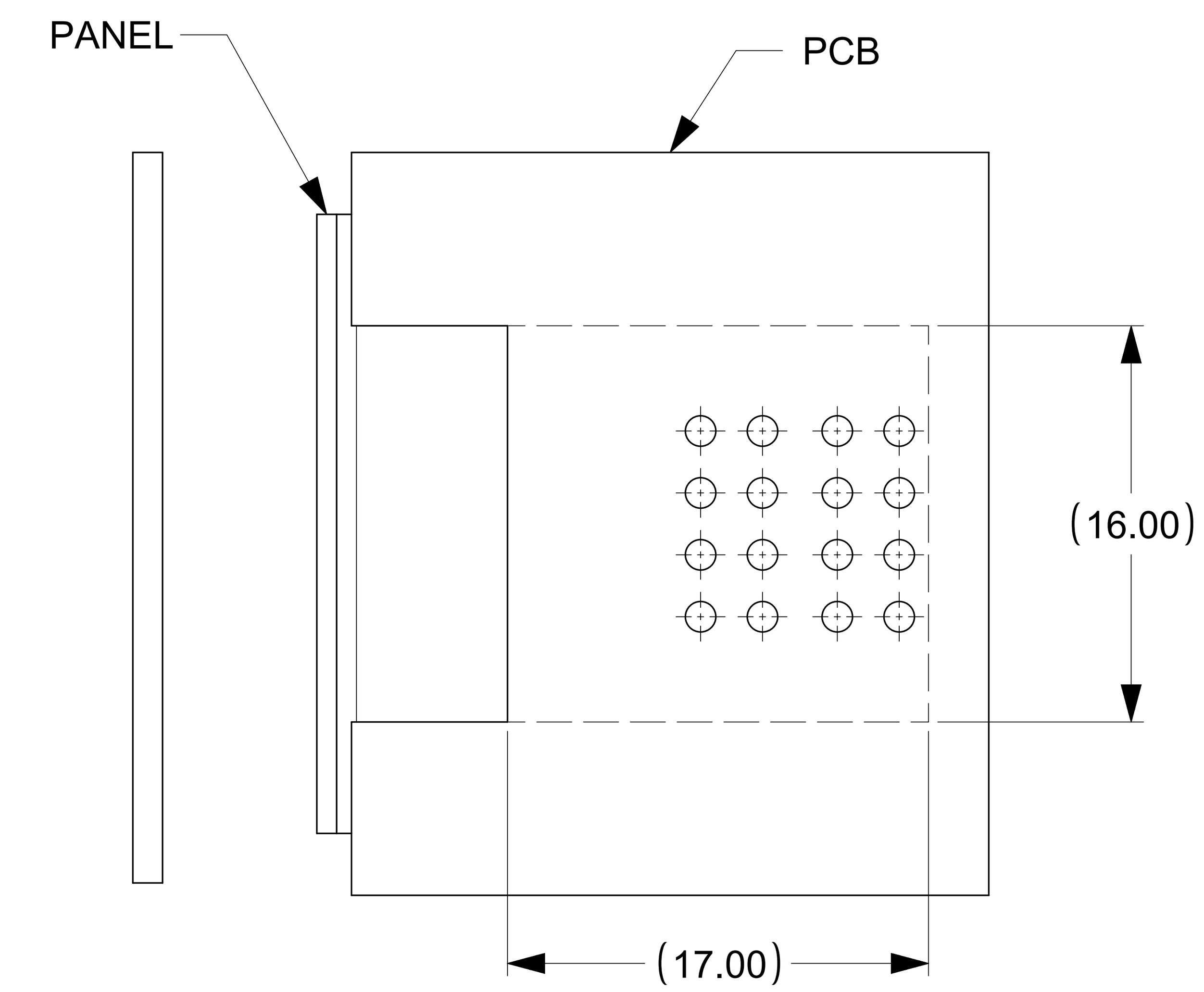
PRODUCT CUSTOMER DRAWING

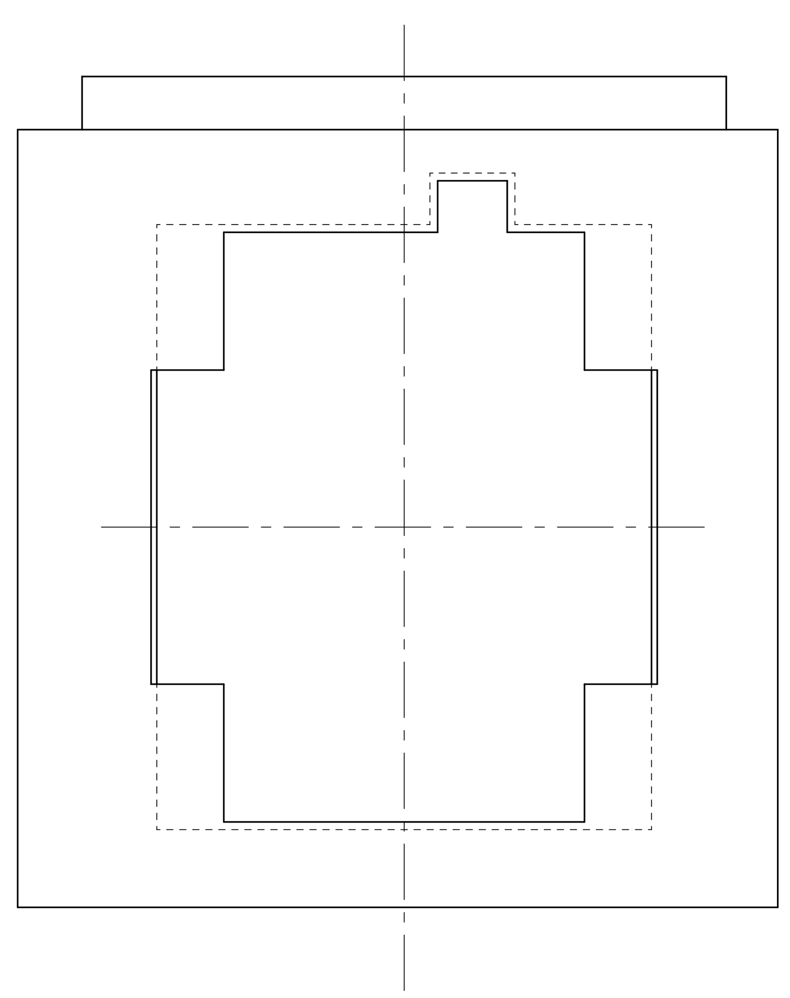
DOCUMENT NUMBER	DOC TYPE	DOC PART	REVISION
2203700000	PSD	000	3

MATERIAL NUMBER	CUSTOMER	SHEET NUMBER
SEE TABLE	GENERAL MARKET	1 OF 2

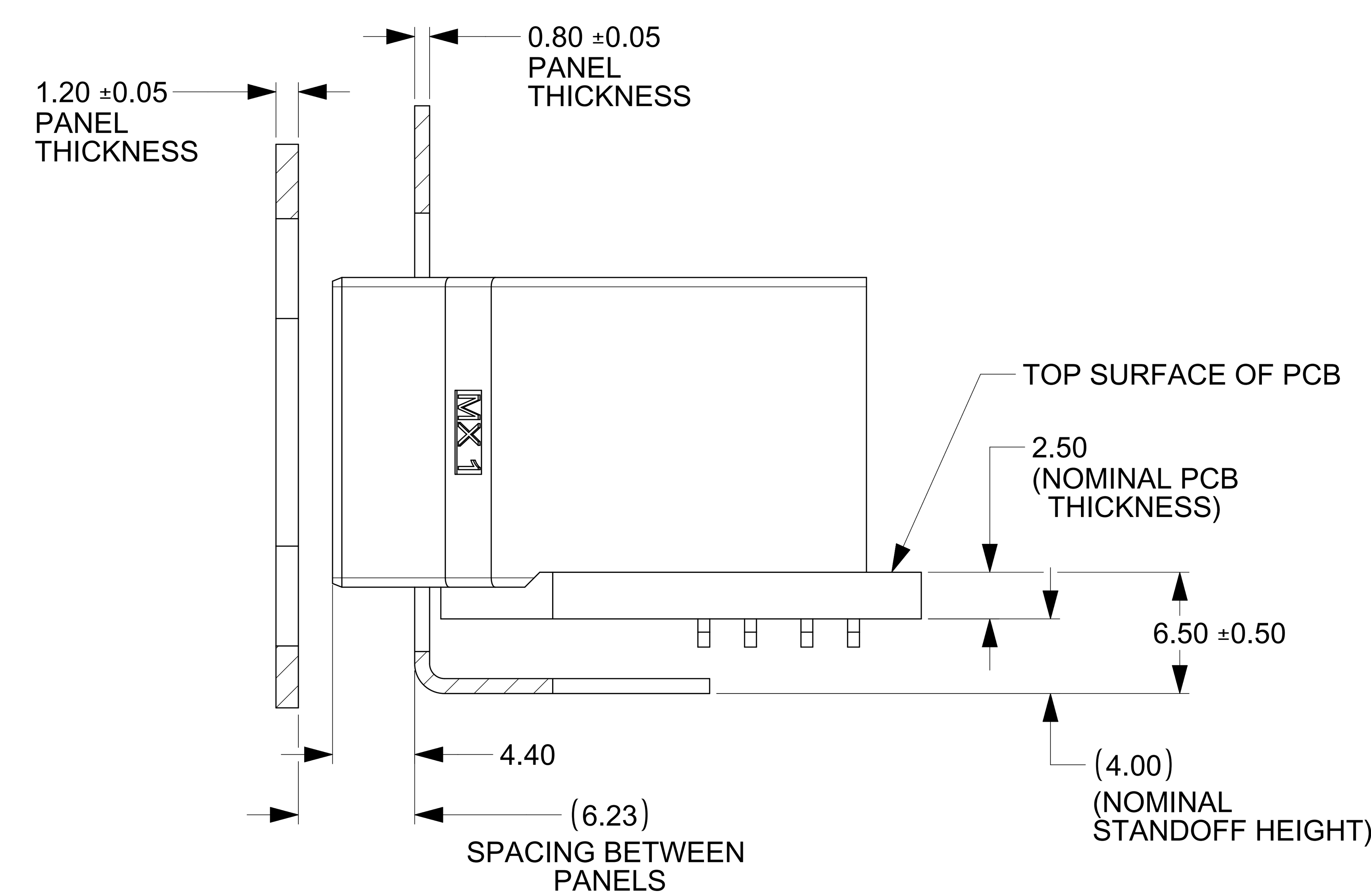


PLUG AND RECEPTACLE MATED PROFILE





BOTH PANELS CUT OUTS ARE CENTRALLY PLACED
ON X-AXIS AS WELL AS Y-AXIS



PLUG MATING PROFILE WITH RESPECT TO PANELS

FOR INFORMATION USE ONLY

DIMENSION UNITS	SCALE
mm	4:1

GENERAL TOLERANCES (UNLESS SPECIFIED)

ANGULAR TOL	±	2.0 °
4 PLACES	±	
3 PLACES	±	
2 PLACES	±	0.13
1 PLACE	±	0.25
0 PLACES	±	

DRAFT WHERE APPLICABLE MUST REMAIN WITHIN DIMENSIONS

CURRENT REV DESC:

EC NO:
DRWN: GSHARMA 2022/07/01
CHK'D:
APPR:

INITIAL REVISION:
DRWN: GSHARMA 2022/02/07
APPR:

THIRD ANGLE PROJECTION	DRAWING	SERIES
	D-SIZE	220370

molex

BRICK POWER PLUG CONNECTOR ASSY

PRODUCT CUSTOMER DRAWING

DOCUMENT NUMBER	DOC TYPE	DOC PART	REVISION
2203700000	PSD	000	3

MATERIAL NUMBER	CUSTOMER	SHEET NUMBER
SEE TABLE	GENERAL MARKET	2 OF 2

DOCUMENT STATUS	ER	RELEASE DATE	2022/08/30	18:19:45

=== Liquid connector spec sheet

EVERIS™ BLQ4 SERIES CONNECTOR

Everis™ BLQ4 Series quick disconnect couplings

provide ultra-reliable, dripless connections and disconnections that protect valuable electronics. Designed specifically for rack mounted liquid cooling applications, Everis BLQ4 utilizes patent-pending valve technology that eliminates drips and is designed for long-term connected use.

FEATURES	BENEFITS
Non-spill design	Disconnect under pressure with no drips
Redundant seals	Extra protection from leaks
Friction-free valve	Enables extended periods in connected state
Rugged construction	Able to withstand long-term, repeated use
Axial engagement tolerance	Allows full flow even when not fully engaged
Single-piece options for body & insert	Space saving

Specifications

PRESSURE:
Vacuum to 120 psi, 8.3 bar

TEMPERATURE:
0°F to 240°F (-17°C to 115°C)

MAXIMUM FLOW AT DISCONNECT:
3.0 gal/min, 11.3L/min

ENGAGEMENT TOLERANCE:
Coupling must be within 1/8" (3mm) of fully engaged to achieve maximum flow.

MATERIALS:
Main Housing Components:
Nickel chrome-plated brass

Valves: Polysulfone

Valve Springs (wetted): Stainless steel

Seals: EPDM

THREAD SIZES:
Insert: G 1/4, G 3/8, SAE-4, SAE-6

Body: G 1/4, G 3/8, SAE-4, SAE-6, SAE-8

LUBRICANTS: Krytox® PFPE

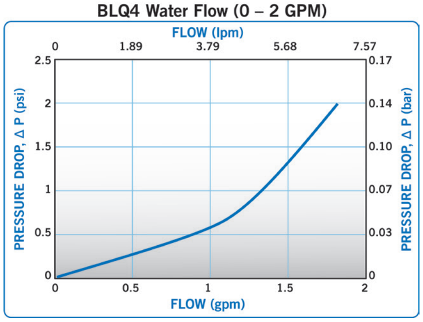
SPILLAGE:
<0.025 cc per disconnect at 0 psi;
<0.055 cc per disconnect at 120 psi

INCLUSION: <0.025 cc per connect

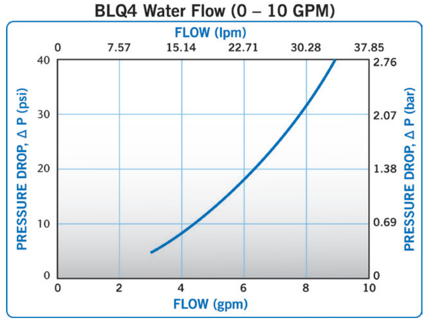
***WARNING**: Pressure, temperature, chemicals, and operating environment can affect the performance of couplings. It is the customer's responsibility to test the suitability of CPC's products in their own application conditions.*

These graphs are intended to give you a general idea of the performance capabilities of each product line. Contact CPC for flow of a particular coupling combination.

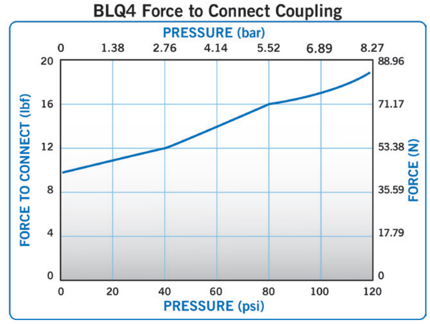
BLQ4 Water Flow (0 – 2 GPM)



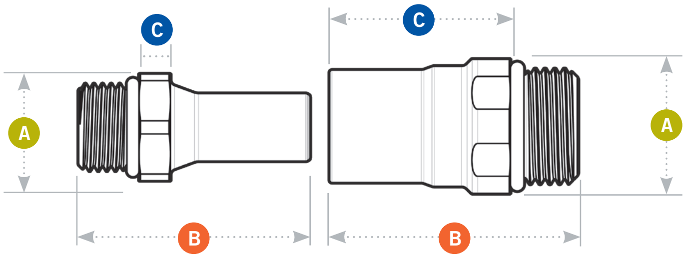
BLQ4 Water Flow (0 – 10 GPM)



BLQ4 Force to Connect Coupling



EVERIS™ BLQ4 SERIES CONNECTOR



All measurements are in inches (millimeters) unless otherwise noted. Tubing must meet stated inside and outside diameters.

A = Height/Diameter
B = Total Length
C = Connected Length

Coupling Bodies • CHROME-PLATED BRASS

TERMINATION	TUBING/THREAD SIZE	METRIC EQ	SHUTOFF	A	B	C
IN-LINE STRAIGHT THREAD PORT SAE	1/4 SAE-4: 7/16-20[1]		BLQ4D30004	0.93 (23.6mm)	2.05 (52.1mm)	1.64 (41.7mm)
	3/8 SAE-6: 9/16-18[1]		BLQ4D30006	0.93 (23.6mm)	2.05 (52.1mm)	1.64 (41.7mm)
	1/2 SAE-8: 3/4-16[1,3]		BLQ4D30008	0.90 (22.9mm)	1.60 (40.6mm)	1.16 (29.5mm)
	G 1/4 (BSPP)[2]		BLQ4D31004	0.93 (23.6mm)	2.11 (53.6mm)	1.70 (43.2mm)
	G 3/8 (BSPP)[2]		BLQ4D31006	0.93 (23.6mm)	2.11 (53.6mm)	1.70 (43.2mm)

Coupling Inserts • CHROME-PLATED BRASS

TERMINATION	TUBING/THREAD SIZE	METRIC EQ	SHUTOFF	A	B	C
IN-LINE STRAIGHT THREAD PORT SAE	1/4 SAE-4: 7/16-20[1]		BLQ4D46004	0.93 (23.6mm)	1.99 (50.5mm)	0.69 (17.5mm)
	3/8 SAE-6: 9/16-18[1,3]		BLQ4D46006	0.75 (19.1mm)	1.49 (37.8mm)	0.19 (4.8mm)
	G 1/4 (BSPP)[2]		BLQ4D47004	0.93 (23.6mm)	2.05 (52.1mm)	0.75 (19.1mm)
	G 3/8 (BSPP)[2,3]		BLQ4D47006	0.93 (23.6mm)	1.49 (37.8mm)	0.19 (4.8mm)

Note:
[1]All SAE terminations are compatible with SAE J1926-1 ports.
[2]All G (BSPP) terminations are compatible with ISO 1179-1 ports.
[3]One-piece design.

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Germany
PHONE +49 6026 9973-0

COLDER PRODUCTS COMPANY LIMITED
Shanghai, China
PHONE +86 21 2411 2666
TOLL FREE +86 400 990 1978

Confidence at every poinmt of connection.

cpcworldwide.com/ContactUs

EVERIS™ BLQ6
SERIES CONNECTOR

Everis™ BLQ6 Series quick disconnect couplings Ultra-reliable, no-drip connections for thermal management to help protect valuable electronic systems. Designed specifically for blind mate liquid cooling applications, the BLQ6 Series uses patented technology that eliminates drips and is specifically designed to withstand long-term connection. An optional accessory kit is available for panel mount connections.

SPECIFICATIONS

PRESSURE: Vacuum to 120 psi, 8.3 bar

TEMPERATURE:
Operating: 0°F to 240°F (-17°C to 115°C)
Storage/Shipping:
-40°F to 240°F (-40°C to 115°C)

MAXIMUM FLOW AT DISCONNECT:
3.0 gal/min, 11.3L/min

ENGAGEMENT TOLERANCE:
Coupling must be within 1/8" of fully engaged to achieve maximum flow.

MATERIALS:
Main Components: Anodized Aluminum
Valves: Polysulfone
Valve Springs (wetted): Stainless steel
Seals: EPDM
Panel Mount Kit: Stainless steel
Compliance: RoHS, REACH

THREAD SIZES:
Insert: SAE-6, G 1/2
Body: SAE-6, G 1/2

LUBRICANTS: Krytox® PFPE

SPILLAGE:
<0.03 cc per disconnect at 0 psi;
<0.03 cc per disconnect at 120 psi

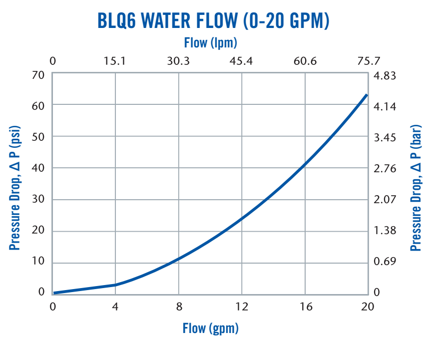
AIR INCLUSION: <0.022 cc per connect

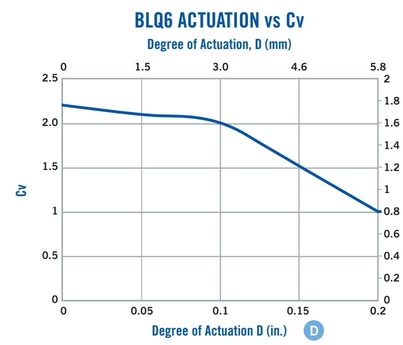
FLOW COEFFICIENT: Cv ~ 2.2 (1.90 Kv)

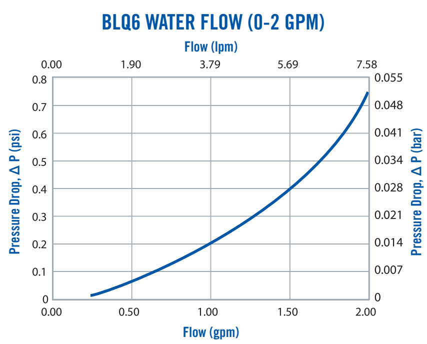
AXIAL MISALIGNMENT: 1 mm max

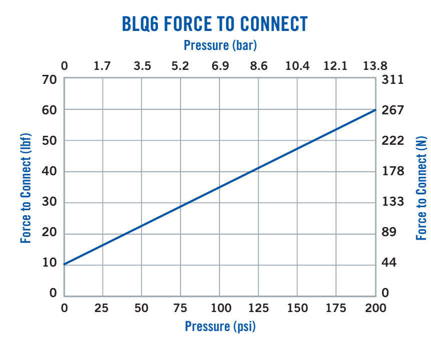
WARNING: Pressure, temperature, chemicals and operating environment can affect the performance of couplings. It is the customer's responsibility to test the suitability of CPC)'s products in their own application conditions.

FEATURES	BENEFITS
Non-spill valve	Disconnect under pressure with no spills
Redundant seals	Extra protection from leak-causing contaminants and debris
Innovative valve design	Provides reliability for extended periods of operation
Rugged anodized aluminum	Able to withstand long-term, ongoing and repeated use
Axial engagement tolerance	Allows full flow even when not fully engaged
Optional panel mount kit	Enables either the body, insert or both to be panel mounted









These graphs are intended to give you a general idea of the performance capabilities of each product line. Contact CPC for flow of a particular coupling combination.

CPC A DOVER COMPANY

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E-MAIL: asiapacific@cpcworldwide.com

EVERIS™ BLQ6 SERIES DIMENSIONS

COUPLING BODIES - Nickel-chrome plated brass

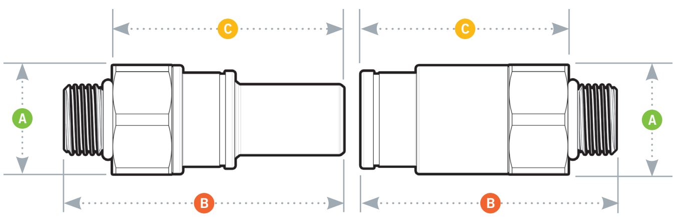
TERMINATION	TUBING/THREAD SIZE	METRIC EQ	SHUTOFF	HEX	A	B	C
IN-LINE STRAIGHT THREAD SAE	3/8 SAE-6: 9/16-18[1]	N/A	BLQ6D30006	7/8"	0.96 (24.4)	2.08 (52.7)	1.68 (42.8)
IN-LINE STRAIGHT THREAD G / BSPP	G 1/2[2]	N/A	BLQ6D31008	26mm	1.12 (28.4)	2.26 (57.5)	1.76 (44.7)

COUPLING INSERTS - Nickel-chrome plated brass

TERMINATION	TUBING/THREAD SIZE	METRIC EQ	SHUTOFF	HEX	A	B	C
IN-LINE STRAIGHT THREAD PORT SAE	3/8 SAE-6: 9/16-18[1]	N/A	BLQ6D46006	7/8"	0.96 (24.4)	2.27 (57.7)	0.99 (25.2)
IN-LINE STRAIGHT THREAD G / BSPP	G 1/2[2,3]	N/A	BLQ6D47008	26mm	1.12 (28.4)	2.00 (50.9)	0.70 (17.8)

All measurements are in inches (millimeters) unless otherwise noted.
[1]All SAE terminations are compatible with SAE J1926-1 ports.
[2]All G (BSPP) terminations are compatible with ISO 1179-1 ports.
[3]One-piece design

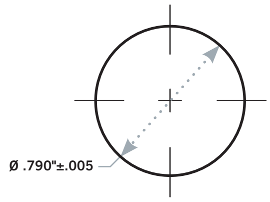
PRODUCT DIMENSIONS



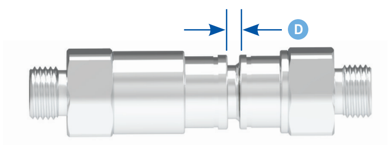
A = Height/Diameter B = Total Length C = Connected Length

PANEL DIMENSIONS

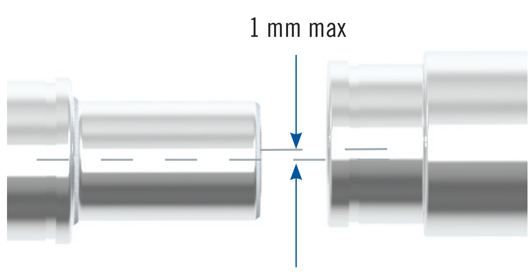
	PANEL OPENING	PANEL THICKNESS MAX.–MIN.
	see drawing	0.075 -0.175"



ACTUATION LENGTH

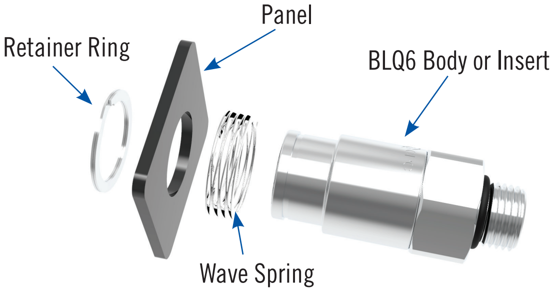


RADIAL TOLERANCE



PANEL MOUNT KIT (P/N BLQ6PMKIT)

Kit includes wave spring and retainer ring only.



SCAN

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